

1. DC Circuits

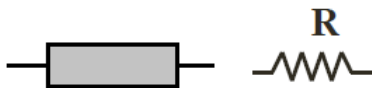
1.1 Introduction

- DC Circuit stands for direct current circuits. DC current flow one direction.
- We can get DC supply by Batteries, Solar Panel, DC Generator, Rectifier etc.
- A simple DC circuit consists of a current source (e.g. battery) and one or more "loads" (circuit elements). Each "load" absorbs electrical energy, converting it to some other form of energy, e.g. a light bulb emits heat and light energy, an electric motor performs mechanical work and emits heat.

1.2 Electrical circuit elements (R, L and C)

RESISTANCE (R)

- A Resistor is a device that provides resistance in an electric circuit and control the current in a circuit.
- Resistance is the property of a circuit element which offers opposition to the flow of current. It is denoted by R and its unit is the ohm (Ω). Symbol



RESISTIVITY

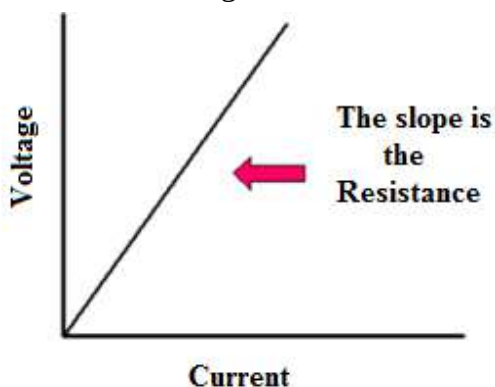
- Resistivity or Specific resistance is the resistance per unit length.
- It is denoted by the symbol by ' ρ ' (Roh)

1.2.1 Ohms Law

The relationship between voltage, current and resistance of a circuit was derived by the scientist ohm is known as ohms law.

Statement:

At a constant temperature, the ratio of potential difference between any two points in a conductor to the current flowing between them is constant.



Explanation:

$$\text{Hence, } \frac{V}{I} = \text{Constant} = R$$

$$\text{or, } V = IR$$

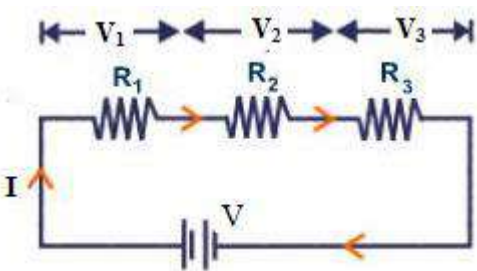
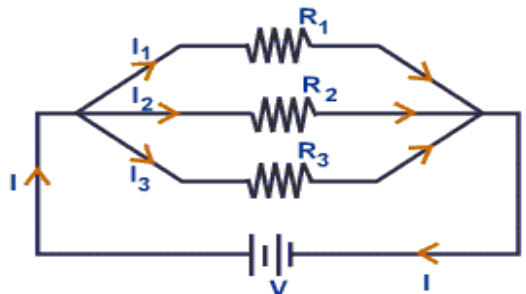
Current and voltage are linearly proportional.

Limitation of ohms law:

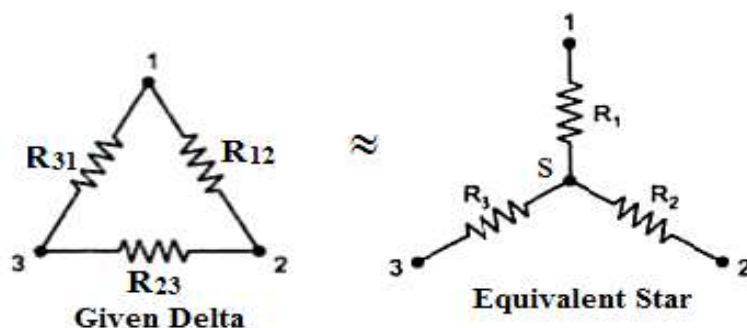
- This law is applicable only to circuits whose physical conditions remain constant. Also, it is not applicable to circuits involving mutual induction.

- It is applicable for linear circuits only.

1.2.2 Series Parallel Connection of Resistance

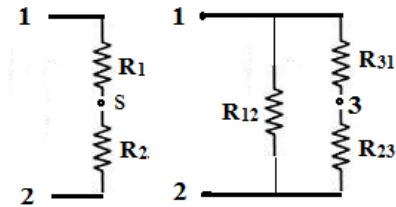
Resistances in series	Resistances in parallel
	
Fig. Three-resistances are connected in series. The DC voltage is applied across the series combination of resistance. As per ohms law, $R = \frac{V}{I}$ In series circuit, voltage across each resistance will be different and it will, $V = IR_1 + IR_2 + IR_3$ while current will remain same. $\therefore R = \frac{IR_1 + IR_2 + IR_3}{I}$ Now, if R_{eq} is the equivalent resistance when no. of resistances are connected in series then, $\boxed{R_{eq} = R_1 + R_2 + R_3}$	Fig. Three-resistances are connected in parallel. The DC voltage is applied to the parallel resistance. As per ohms law, $R = \frac{V}{I}$ In parallel circuit, voltage across resistances will remain same but current passing through each resistance will be different and it will be, $I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$ $\therefore R = \frac{V}{\frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$ Now, if R_{eq} is the equivalent resistance when no. of resistances are connected in parallel then, $\boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$

1.2.3 Star to delta conversion of resistance



Here, we find the value of R_1 , R_2 and R_3 in terms of R_{12} , R_{23} and R_{31} of delta.

In star the resistance between 1-2 terminal R_1 and R_2 is in series. But In delta, resistance between 1-2 terminal R_{12} is parallel with $(R_{31} + R_{23})$, see in fig,



So, equivalent resistance between 1-2 terminal of star = equivalent resistance between 1-2 terminal of delta is given by,

$$R_1 + R_2 = \frac{R_{12} \times (R_{23} + R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{---(1)}$$

Similarly;

$$R_2 + R_3 = \frac{R_{23} \times (R_{31} + R_{12})}{R_{12} + R_{23} + R_{31}} \quad \text{---(2) and}$$

$$R_3 + R_1 = \frac{R_{31} \times (R_{12} + R_{23})}{R_{12} + R_{23} + R_{31}} \quad \text{---(3)}$$

Now adding equation (1) (2) and (3) we get,

$$2(R_1 + R_2 + R_3) = \frac{2(R_{12}R_{23} + R_{23}R_{31} + R_{31}R_{12})}{(R_{12} + R_{23} + R_{31})}$$

$$(R_1 + R_2 + R_3) = \frac{(R_{12}R_{23} + R_{23}R_{31} + R_{31}R_{12})}{(R_{12} + R_{23} + R_{31})} \quad \text{---(4)}$$

Now to find R_1 , subtract equation (4) – (2)

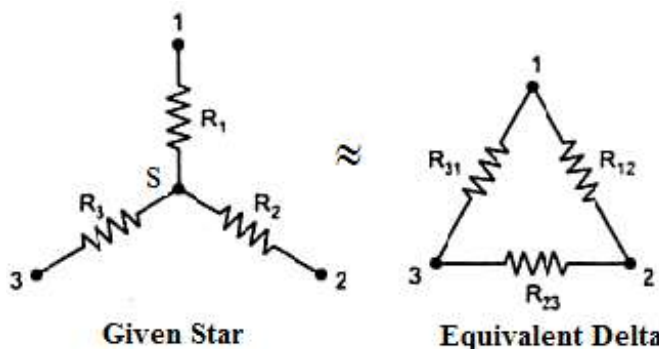
So, we get,
$$R_1 = \frac{R_{12} \times R_{31}}{R_{12} + R_{23} + R_{31}}$$

Similarly; to find R_2 , subtract eqⁿ (4) – (3)
$$R_2 = \frac{R_{23} \times R_{12}}{R_{12} + R_{23} + R_{31}}$$

And to find R_3 , subtract equation (4) – (1)
$$R_3 = \frac{R_{31} \times R_{23}}{R_{12} + R_{23} + R_{31}}$$

1.2.4 Delta to Star conversion of resistance

Here, we find the value of R_{12} , R_{23} and R_{31} in terms of R_1 , R_2 and R_3 of star.



Now, we have already below three equations from delta to star transformation theory.

$$R_1 = \frac{R_{12}R_{31}}{R_{12} + R_{23} + R_{31}} \quad \text{---(5),}$$

$$R_2 = \frac{R_{23}R_{12}}{R_{12} + R_{23} + R_{31}} \quad \text{---(6)and}$$

$$R_3 = \frac{R_{31}R_{23}}{R_{12} + R_{23} + R_{31}} \quad \text{---(7)}$$

Now, multiplying eqⁿ (5) and (6), eqⁿ (6) and (7), eqⁿ (7) and (5) we get,

$$R_1 R_2 = \frac{R_{12}^2 R_{31} R_{23}}{(R_{12} + R_{23} + R_{31})^2} \quad \text{---(8)}$$

Similarly;

$$R_2 R_3 = \frac{R_{23}^2 R_{12} R_{31}}{(R_{12} + R_{23} + R_{31})^2} \quad \text{---(9) and}$$

$$R_3 R_1 = \frac{R_{31}^2 R_{12} R_{23}}{(R_{12} + R_{23} + R_{31})^2} \quad \text{---(10)}$$

Now adding equation (8), (9) and (10) we get,

$$R_1 R_2 + R_2 R_3 + R_3 R_1 = \frac{R_{12}^2 R_{31} R_{23} + R_{23}^2 R_{12} R_{31} + R_{31}^2 R_{12} R_{23}}{(R_{12} + R_{23} + R_{31})^2} \quad \text{---(4)}$$

Now to find R_{12} , divide eqⁿ (4) to eqⁿ (7)

$$\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = \frac{R_{12}^2 + R_{23} R_{12} + R_{31} R_{12}}{(R_{12} + R_{23} + R_{31})}$$

$$\frac{R_1 R_2}{R_3} + R_2 + R_1 = \frac{R_{12}(R_{12} + R_{23} + R_{31})}{(R_{12} + R_{23} + R_{31})}$$

Now, rearranging the term; we get,

$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

Similarly to find value of R_{23} , divide eqⁿ (4) to eqⁿ (5)

$$R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

and to find R_{31} , divide eqⁿ (4) to eqⁿ (6)

$$R_{31} = R_3 + R_1 + \frac{R_3 R_1}{R_2}$$

1.2.5 Inductor (L)

An inductor is a passive component which designed to resist change in current. Inductors are formed from coils of wire, often around a steel or ferrite core.

Inductor stores energy in the form of a magnetic field.

→ When an Inductor is connected to a circuit with Direct Current (DC) source, two processes, which are called “storing” and “decaying” energy, will happen in specific conditions.

→ A unit of Inductance is Henry (H)

→ **Symbol:** 

→ If there is a changing magnetic field in an inductor, current will be induced in the circuit.

1.2.6 Capacitor (C)

- The capacitor is important devices which used electrical circuits. It is used to store electrical energy (Capacitance) to a particular time and discharged it when it needed.
- It consists of two conductors which are separated by a dielectric medium. The dielectric medium can be air, wax coated paper, mica or oil etc.

→ **Symbol:** 

- The Charge storing capability of a capacitor is called capacitance of the capacitor. The unit of capacitance is Coulomb per Volt, which is called the Farad (F). .
- The charge stored in a capacitor is given by

$$Q = CV$$

Where a Q - charge stored by the capacitor

C - Capacitance value of the capacitor

V - Voltage applied across the capacitor

- **Capacitors** are widely used in a variety of **applications** of electronic circuits such as;

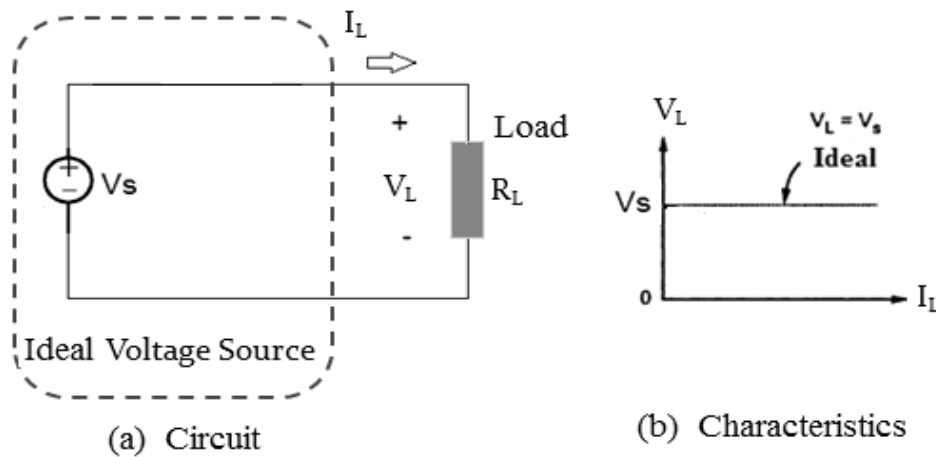
- Store charges such as in a camera flash circuit
- Smoothing the output of power supply circuits
- Coupling of two stages of a circuit (coupling of an audio stage with a loud speaker)
- Tuning radios to particular frequencies

1.3 Voltage and Current Source

The most important active elements are voltage and current sources that generally deliver power to the circuit connected to them. Generally, these sources are classified as (a) ideal source and (b) practical source.

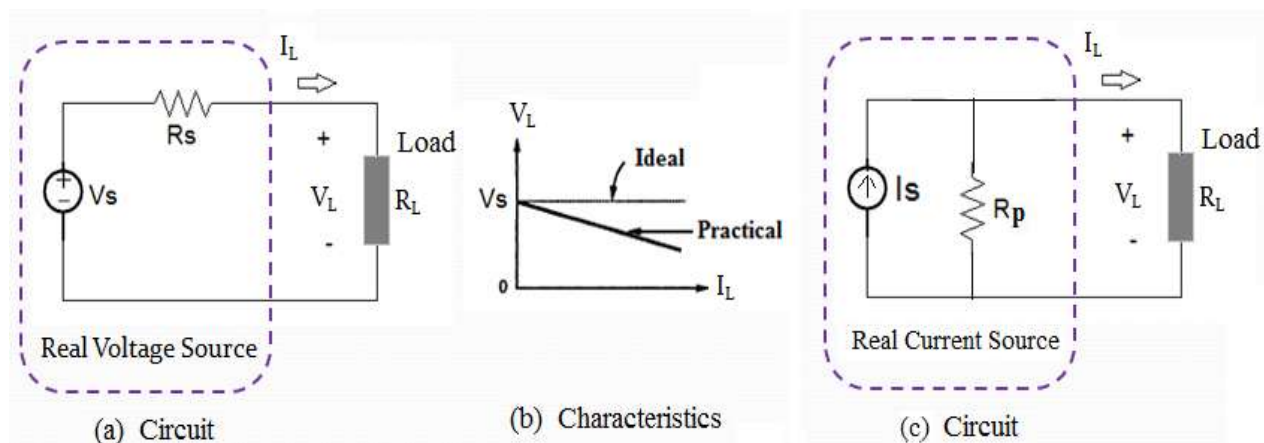
(a) Ideal Source

- An *ideal sources* are classified as; (i) Ideal voltage sources and (2) ideal current source
- **An ideal voltage source has zero internal resistance** and terminal voltage is independent of the current through it.
- An ideal voltage source that is symbolized by a model as shown in figure below is a device which generates a constant voltage across its terminals ($V_L = V_S$) no matter what current is drawn from it (i.e., terminal voltage is independent of load resistance linked across the terminals).
- An ideal voltage source has the following characteristics that allow it to act as a 100% efficient source of voltage. See a circuit and characteristics of the ideal voltage source as shown in Figure (a) and (b).
- **An ideal current source has infinite internal resistance means zero conductance.**



(b) Practical Source:

- A **practical or real voltage source** has an **internal resistance** which, to be accounted for, It is also two types of practical source (i) voltage and (ii) current source
- Practical or real voltage source represented with an external resistance R_s in series with it. As shown in Figure (a) and its characteristic in fig. (b).
- Similarly a **real current source** has an internal resistance R_p in parallel with it. As shown in Figure (c)



1.4 Kirchhoffs Current and Voltage Law

In 1845, German Physicist Gustav Kirchhoff first stipulated two laws that become fundamental laws to understanding of electrical **circuits**; these laws are known as; (1) KCL and (2) KVL

Kirchhoff's Current Law (KCL):

Statement: "The algebraic sum of all currents meeting at the junction is equal to zero".

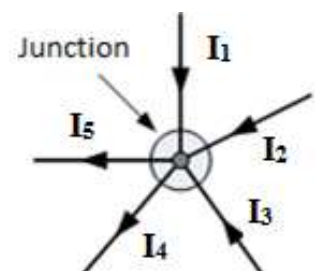
Explanation:

See a fig. Here a junction having five currents. Current I_1 , I_2 , and I_3 coming towards the junction while I_4 and I_5 leaving the junction.

Assume incoming currents are positive signs (+) and outgoing currents are negative Sign (-)

$$I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

$$\therefore I_1 + I_2 + I_3 = I_4 + I_5$$



Incoming current = outgoing current

Hence, this law can be stated as total incoming currents are equal to total outgoing currents at junction.

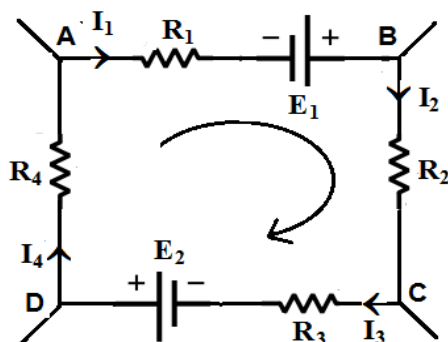
Then; $\sum I$ (Incoming) = $\sum I$ (Outgoing)

Kirchhoff's Voltage Law, (KVL)

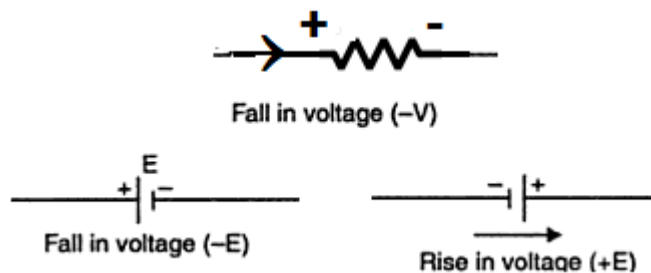
Statement: "The algebraic sum of all the voltages in a closed loop circuit is equal to zero."

Explanation:

See a fig. As per mention above statement, all the voltages mean here, in ABCD closed loop circuit, two possible voltages, (1) battery voltage E and (2) voltage drop in resistances = IR .



For calculation, let we go to clockwise direction A-B-C-D-A. So note following sign.



Similarly, apply above sign for other element so, we can write,

$$-I_1 R_1 + E_1 - I_2 R_2 - I_3 R_3 + E_2 - I_4 R_4 = 0$$

$$\therefore E_1 + E_2 = I_1 R_1 + I_2 R_2 + I_3 R_3 + I_4 R_4$$

Hence, this law can be stated as "The algebraic sum of the all the emf's in the path is equal to the algebraic sum of the voltage drops across the circuit element."

Then; $\sum E$ (emf's) = $\sum IR$ (Voltage drop)

1.5 Superposition Theorem

Statement:

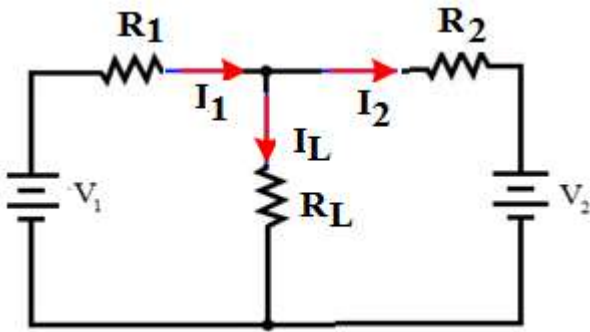
The superposition theorem states that in a linear circuit with several sources, the current and voltage for any element in the circuit is the algebraic sum of the currents and voltages produced by each source acting independently.

For the Calculation consider;

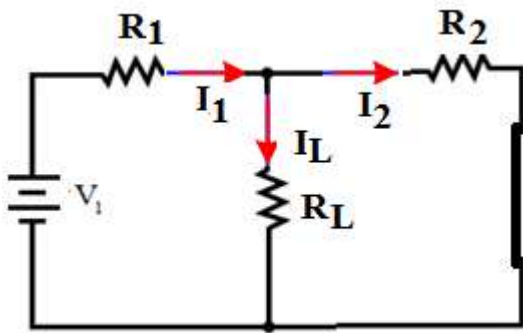
Voltage Source $\rightarrow$ **Short circuited**

Current Source $\rightarrow$ **Open Circuited**

Consider Below Circuit

**Step:1-**

At first, find the current through load resistance R_L with V_1 source acting alone. Hence replace the V_2 source by a short circuit.



To find the load current, find the total current supplied by the source (I_1) with its total resistance. Then apply current division rule and find the current through R_L resistor with V_1 source.

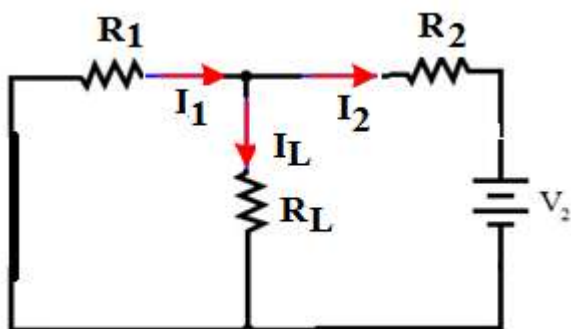
$$R_{eq} = R_1 + \frac{R_2 \times R_L}{R_2 + R_L} = \Omega$$

$$\text{Now, } I_{total} = \frac{V_1}{R_{eq}} = \text{Amp.}$$

$$\text{So, } I_{L(V_1)} = I_{total} \times \frac{R_2}{R_2 + R_L} = \text{Amp}$$

Step:-2

Now consider the V_2 source alone and replace V_1 source by a short circuit.



$$R_{eq} = R_2 + \frac{R_1 \times R_L}{R_1 + R_L} = \Omega$$

$$\text{Now, } I_{total} = \frac{V_2}{R_{eq}} = \text{Amp.}$$

$$\text{So, } I_{L(V_2)} = I_{total} \times \frac{R_1}{R_1 + R_L} = \text{Amp}$$

Step:- 3

Finally, Current passing through load resistance R_L is the algebraic sum of $I_{L(V_1)}$ and $I_{L(V_2)}$.

$$I_L = I_{L(V_1)} + I_{L(V_2)}$$

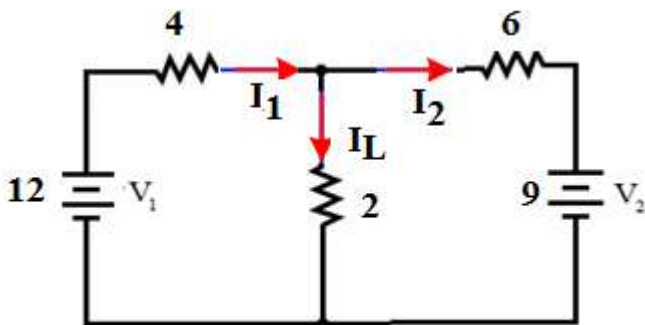
Application of superposition theorem

This Theorem can employ only for linear circuits as well as the circuit which has more power supplies.

From the above superposition theorem examples, this theorem cannot be used for non-linear circuits, but applicable for linear circuits.

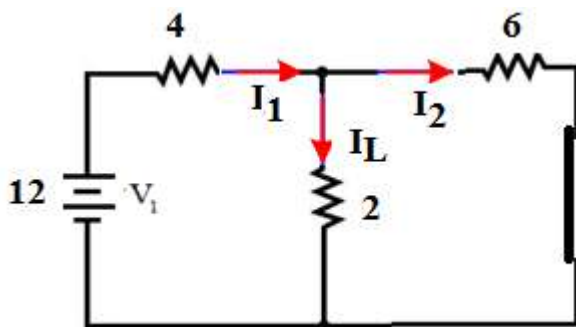
Example On Superposition Theorem:

Find Current passing through $2\ \Omega$ load resistance.



Step:1-

First, we find current through $2\ \Omega$ resistance with $12\ \text{V}$ source alone by V_2 short circuit.



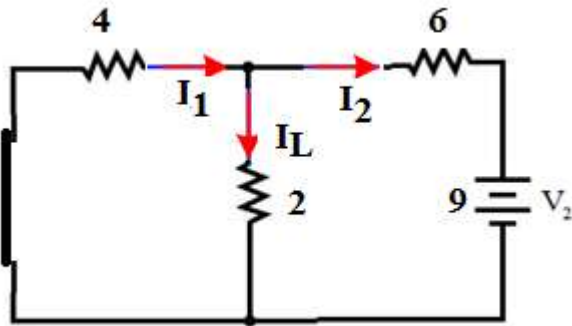
$$R_{eq} = 4 + \frac{6 \times 2}{6 + 2} = 5.5 \, \Omega$$

$$\text{Now, } I_{total} = \frac{V_1}{R_{eq}} = \frac{12}{5.5} = 2.18 \, \text{Amp.}$$

$$\text{So, } I_{L(12)} = I_{total} \times \frac{R_2}{R_2 + R_L} = 2.18 \times \frac{6}{6 + 2} = 1.635 \, \text{Amp}$$

Step:2-

First, we find current through $2 \, \Omega$ resistance with $12 \, \text{V}$ source alone by V_2 short circuit.



$$R_{eq} = R_2 + \frac{R_1 \times R_L}{R_1 + R_L} = 6 + \frac{4 \times 2}{4 + 2} = 7.33 \, \Omega$$

$$\text{Now, } I_{total} = \frac{V_2}{R_{eq}} = \frac{9}{7.33} = 1.23 \, \text{Amp.}$$

$$\text{So, } I_{L(9)} = I_{total} \times \frac{R_1}{R_1 + R_L} = 1.23 \times \frac{4}{4 + 2} = 0.82 \, \text{Amp}$$

Step:- 3

Finally, Current passing through $2 \, \Omega$ load resistance is the algebraic sum of $I_L(V_1)$ and $I_L(V_2)$.

$$I_L = I_{L(12)} + I_{L(9)}$$

$$I_L = 1.635 + 0.82$$

$$I_L = 2.45 \, \text{Amp}$$

1.6 Thevenin Theorem

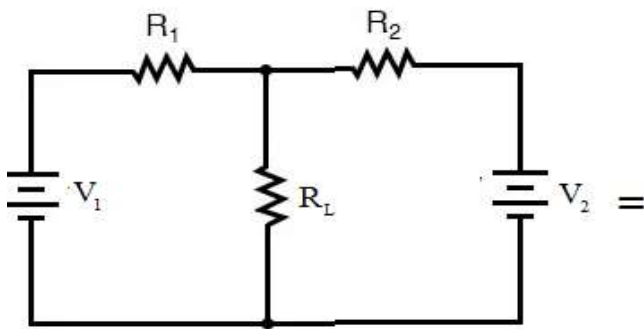
Statement: Thevenin's theorem states that any linear complex electrical circuit can be reduced into a simple electric circuit with one voltage and resistance connected in series.

The process for analyzing a DC circuit using Thevenin's Theorem requires the following steps:

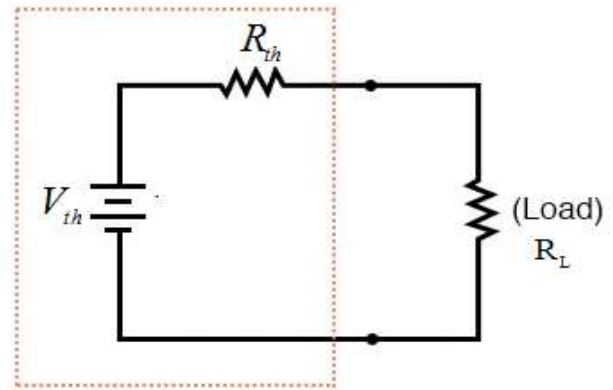
1. Find the Thevenin Resistance (R_{th}) by removing all voltage sources and load resistor.
2. Find the Thevenin Voltage (V_{th}) by Connecting voltmeter.

3. Find the load current by using Thevenin Resistance (R_{th}) and Voltage (V_{th}).

Consider below complex circuit:



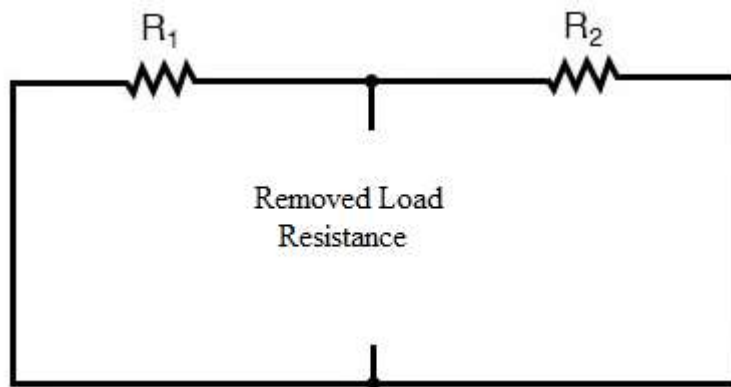
A Linear network contain several voltage sources and resistance



Equivalent Thevenin Circuit

Step-1: Thevenin Resistance

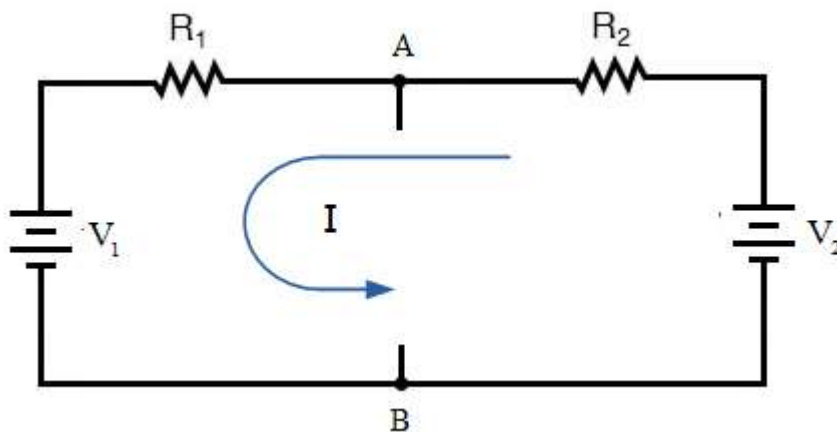
We first need to remove the load resistor connecting terminals A and B, along with all voltage sources.



$$R_{th} = \frac{R_1 \times R_2}{R_1 + R_2}$$

Step-2: Thevenin Voltage

Connect the voltages and find thevenin equivalent Voltage



As per Ohms Law, $I = \frac{V}{R} = \frac{V_2 - V_1}{R_1 + R_2}$

or $\frac{V_{\text{Higher}} - V_{\text{Lower}}}{R_1 + R_2}$

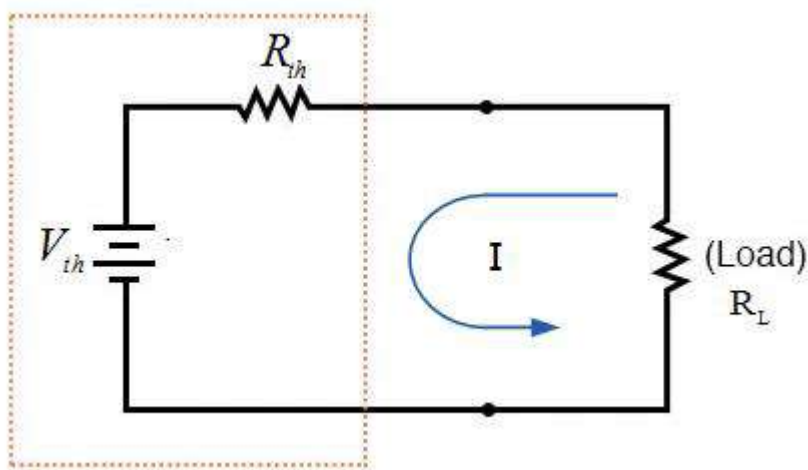
$V_{AB} = V_{th} = V_2 - (IR_2)$

OR

$V_{AB} = V_{th} = V_1 + (IR_1)$

Step-3: Load Current

Now connect R_L in circuit and find load current by V_{th} and R_{th} .



$$I = \frac{V}{R} = \frac{V_{th}}{R_{th} + R_L} = \text{Amp}$$

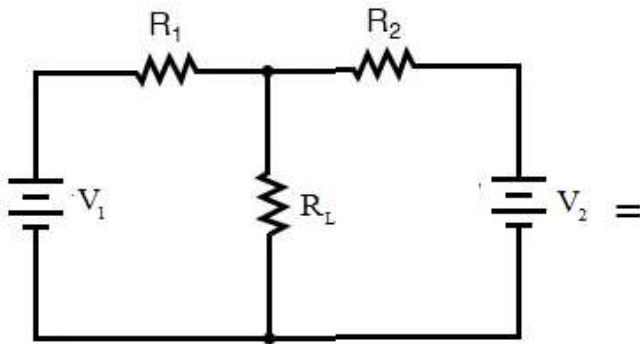
1.7 Norton Theorem

Statement: Norton's theorem states that any linear complex electrical circuit can be reduced into a simple electric circuit with one current and resistance connected in parallel.

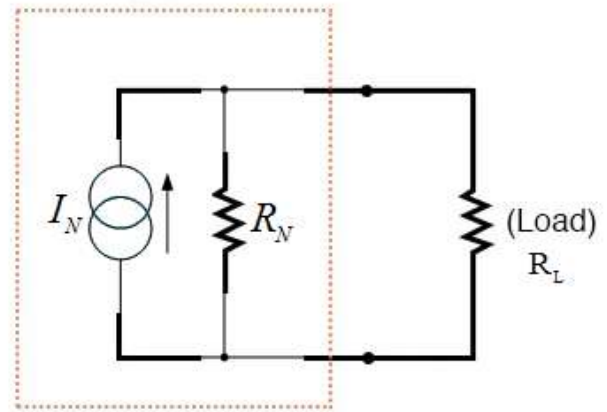
The process for analyzing a DC circuit using Norton's Theorem requires the following steps:

1. Find the Norton's Resistance (R_N) by shorting all voltage sources or by open circuiting all the current sources.
2. Find the Norton's Current (I_N) by placing a shorting link on the output terminals A and B.
3. Find the load current R_L

Consider below complex circuit:

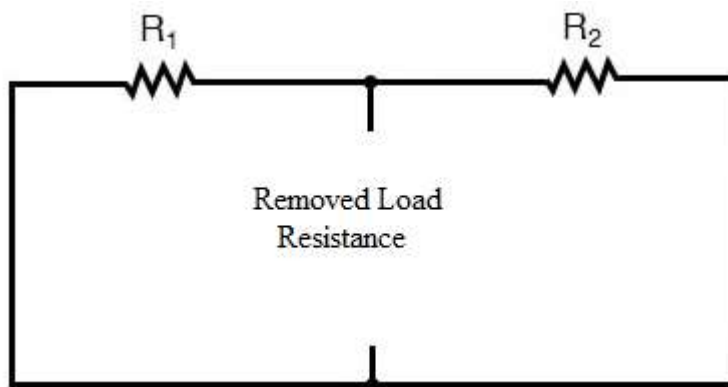


A Linear network contain several voltage sources and resistance



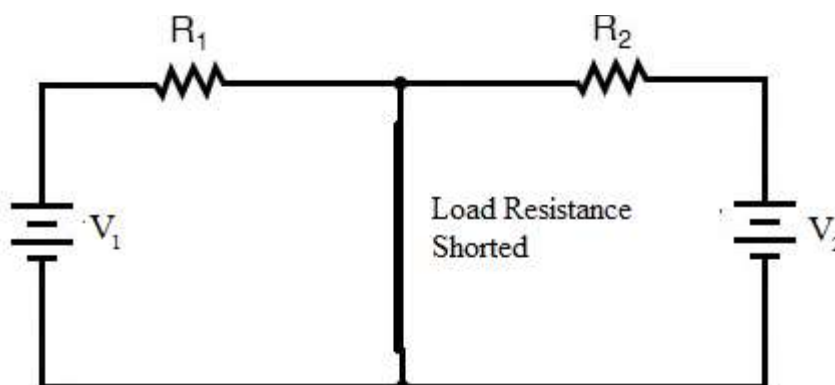
Equivalent Nortons Circuit

Step:1 Find the Norton's Resistance (R_N) remove the centre load resistor.



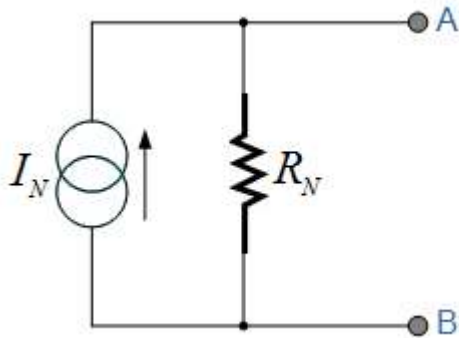
$$R_N = \frac{R_1 \times R_2}{R_1 + R_2} = \Omega$$

Step:2 Find the Norton's Current (I_N) by placing a shorting link on the output terminals A and B.

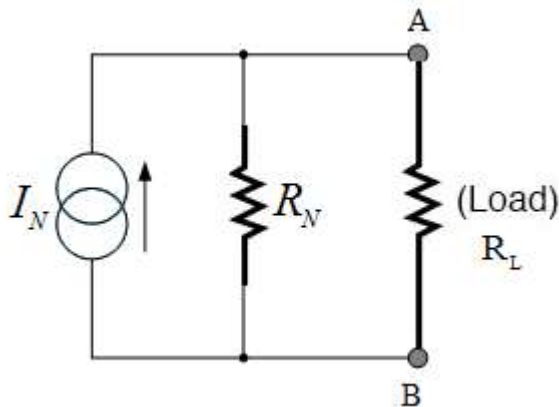


$$I_1 = \frac{V_1}{R_1} \text{ and } I_2 = \frac{V_2}{R_2}$$

$$\therefore I_N = I_1 + I_2$$



Step:3 Find the load current R_L



$$I_L = \frac{V_{AB}}{R_L}$$

$$V_{AB} = I_N \times R_{eq.}$$

$$R_{eq.} = \frac{R_N \times R_L}{R_N + R_L}$$

1.8 Time domain Analysis of first order RL and RC Circuit

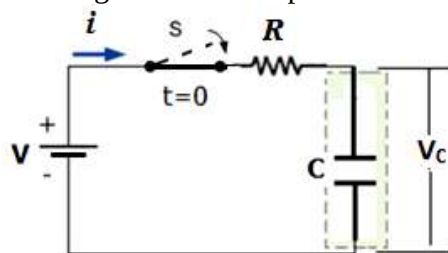
- Analysis of response of circuit consisting of R, L, C voltage source, current source & switches to sudden application of voltage or current is called as Time domain Analysis & Transient Response.
- It is also assumed that circuit elements R, L, C are constant and source is very strong to absorb any disturbances.
- Amongst basic circuit elements Resistor is energy dissipating component & Inductor, Capacitor are energy storing elements.
- Response of these elements to nature of source and disturbance varies from source to source.

- Transients (current or voltage lasting for short duration) in circuit is due to energy storing elements.
- For source free circuit transients response is called as Natural Response .
- For circuit with source transient response is called as Forced Response.

1.8.1 Charging of capacitor

When a capacitor is charged through a resistor, then the voltage across the capacitor and the charge on the capacitor will increase exponentially in time, when the voltage across the capacitor reaches the battery voltage, then it will stop charging.

Fig, shows a series RC circuit connected across Voltage V , **at $t=0$ switch is closed, the capacitor starts charging.** At any time, let V_c is the voltage across the capacitor.



According to KVL,

$$V = V_c + iR \quad \text{-----(1)}$$

$$\text{Now, } i = C \frac{dV_c}{dt}$$

$$\therefore V = V_c + C \frac{dV_c}{dt} R,$$

$$\therefore V - V_c = RC \frac{dV_c}{dt}$$

$$\frac{dV_c}{V - V_c} = \frac{dt}{RC}$$

Integrating and Multiplying both side -1 we get,

$$\int \frac{-dV_c}{V - V_c} = \int \frac{-dt}{RC}$$

$$\text{Log}_e(V - V_c) = \frac{-t}{RC} + k \dots \dots (2)$$

Where k = constant of integration

Now, initially when switch is closed at $t = 0$ sec, at

this time $V_c = 0$, put this value in equation(2),

$$\log_e V = 0 + k, \text{ Hence } k = \log_e V,$$

put k again in equation(2),

$$\therefore \log_e(V - V_c) = \frac{-t}{RC} + \log_e V$$

$$\log_e \frac{(V - V_c)}{V} = \frac{-t}{RC}$$

Where $RC = \lambda$ = time constant

$$\text{So, } \log_e \frac{(V - V_C)}{V} = \frac{-t}{\lambda}$$

$$\therefore \frac{(V - V_C)}{V} = e^{\frac{-t}{\lambda}}$$

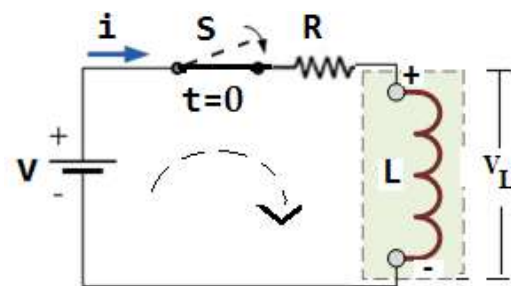
$$V - V_C = Ve^{\frac{-t}{\lambda}}, \therefore V_C = V - Ve^{\frac{-t}{\lambda}}$$

$$\text{Hence, voltage across capacitor, } V_C = V(1 - e^{\frac{-t}{\lambda}})$$

$$\text{or } V_C = V(1 - e^{\frac{-t}{RC}})$$

1.8.2 Rise of current in inductive circuit

Fig, shows a series RL circuit connected across voltage V , at $t=0$ switch is closed, current starts to rise in an inductor. At any time, let V_L is the voltage across an inductor.



According to KVL,

$$V = V_L + iR \quad \text{-----(1)}$$

Note : as per defination; $e = -L \frac{di}{dt}$,

But here; $e = -V_L$ bcz we going to clockwise direction

$$\therefore V_L = L \frac{di}{dt}$$

$$\text{So, } V = L \frac{di}{dt} + iR$$

$$V - iR = L \frac{di}{dt}$$

$$\frac{di}{V - iR} = \frac{dt}{L}$$

Intigrating and Multiplying of $-R$ on both side we get,

$$\int \frac{-R di}{V - iR} = \int \frac{-R dt}{L}$$

$$\therefore \log_e (V - iR) = \frac{-Rt}{L} + k \quad \text{-----(2)}$$

Where, k = constant of intigration

Now initially when switch is closed at $t = 0$ sec,
 at this time $i = 0$, put this value in equation(2),
 $\log_e V - 0 = 0 + k$, Hence $k = \log_e V$, put k in equation(2),

$$\log_e (V - iR) = \frac{-Rt}{L} + \log_e V$$

$$\therefore \log_e \frac{(V - iR)}{V} = \frac{-Rt}{L}$$

Where $\frac{L}{R} = \lambda = \text{time constant}$

$$\text{So, } \log_e \frac{(V - iR)}{V} = \frac{-t}{\lambda}$$

$$\frac{(V - iR)}{V} = e^{\frac{-t}{\lambda}}$$

$$V - iR = Ve^{\frac{-t}{\lambda}}$$

$$\therefore iR = V(1 - e^{\frac{-t}{\lambda}})$$

$$\therefore i = \frac{V}{R}(1 - e^{\frac{-t}{\lambda}})$$

Now, $\frac{V}{R}$ represents the maximum steady value of current I_m ,

$$\therefore \text{Rise of current in an inductive circuit, } i = I_m(1 - e^{\frac{-t}{\lambda}})$$

$$\text{or, } i = I_m(1 - e^{\frac{-Rt}{L}})$$

LIST OF EQUATIONS:

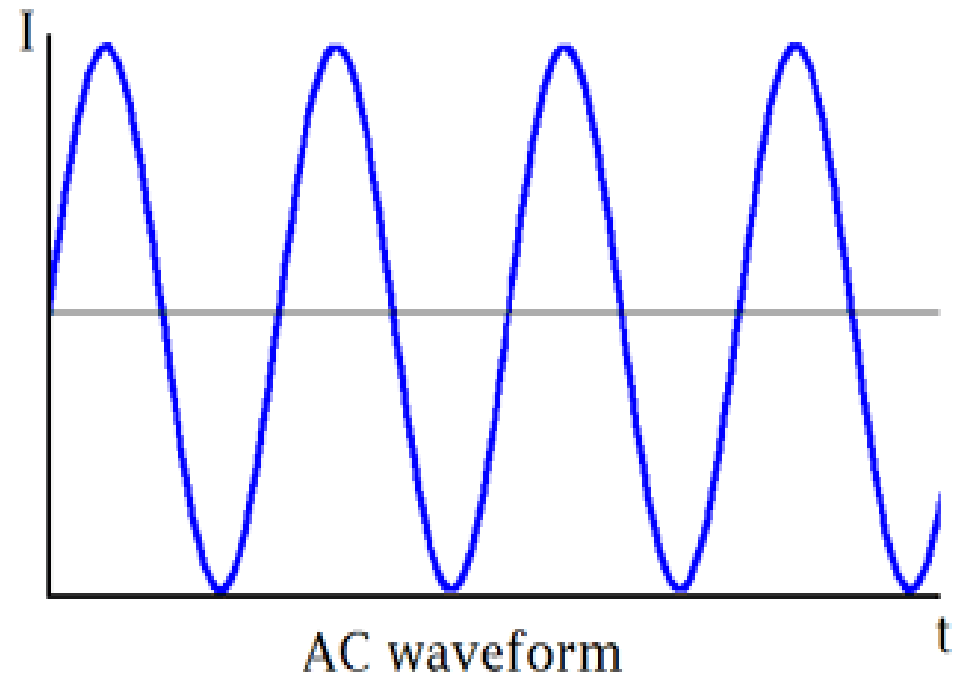
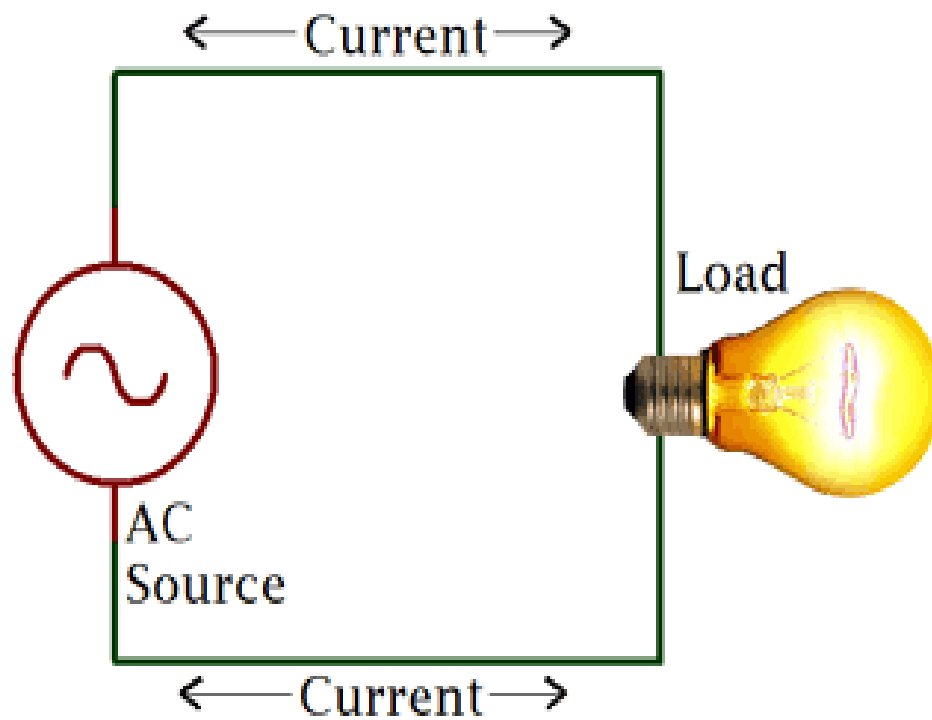
Ohm's Law for DC Circuits	
Current:	$I = \frac{V}{R}, \frac{P}{V}, \sqrt{\frac{P}{R}}$
Voltage:	$V = I \times R, \frac{P}{I}, \sqrt{P \times R}$
Resistance:	$R = \frac{V}{I}, \frac{P}{I^2}, \frac{V^2}{P}$
Power:	$P = I^2 R, V \times I, \frac{V^2}{R}$
Factors affection on resistance	
Resistance of material	$R = \frac{\rho l}{a}$ Where, $a = \frac{\pi d^2}{4}$
Series – Parallel connection of resistance	
Series: Equivalent resistance	$R_{eq} = R_1 + R_2 + R_3 + \dots$
Voltage divider rule	$V_1 = \frac{V}{R_1 + R_2} \times R_1, \quad V_2 = \frac{V}{R_1 + R_2} \times R_2$

Parallel: Equivalent resistance	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
Star –delta and delta – star transformation.	
Delta to star transformation of resistance	$R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}}, R_2 = \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{31}}$ $R_3 = \frac{R_{31} R_{23}}{R_{12} + R_{23} + R_{31}}$
Star to delta transformation of resistance	$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}, R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1},$ $R_{31} = R_3 + R_1 + \frac{R_3 R_1}{R_2}$

IMPORTANT QUESTIONS

1. State ohms law
2. Explain ideal and practical current and voltage sources
3. State and explain Kirchhoff's voltage and current laws.
4. Explain Superposition theorem
5. State and explain Thevenin's theorem
6. Explain Norton's Theorems
7. Derive an expression for equivalent resistances of a star connected network to Delta connected network.
8. Derive an expression for equivalent resistances of a Delta connected network to Star connected network.

BEE CH-2 : AC Circuit



Dr. S. G. Prajapati
(D.E, B.E, M.E, Ph.D. - Electrical Engineering)

Monday, August 18, 2025

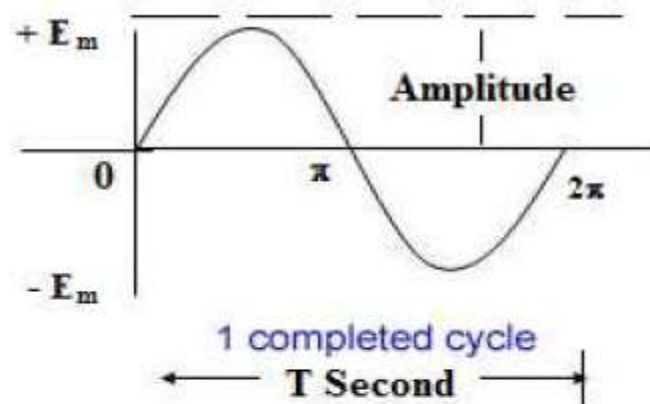
AC Circuit-10 Hours, 25% Weightage

1. Representation of sinusoidal waveforms,
2. Peak and RMS values,
3. Phasor representation of AC quantities,
4. Real power, reactive power, apparent power,
5. power factor.
6. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel),
7. Series and parallel resonance.
8. Three phase balanced circuits,
9. Voltage and current relations in star and delta connections
10. Power measurement in three phase circuits.

1	Define Amplitude, Frequency and Time period for alternating quantities.	03	GTU-Jan-2019
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(1) Cycle:

One complete set of positive and negative values of an alternating quantity is known as cycle.



(2) Frequency:

- The number of cycles completed in one second is known as frequency.
- It is denoted by f and its unit is Hertz (H_z)
- In India, the adopted standard frequency is $50 H_z$

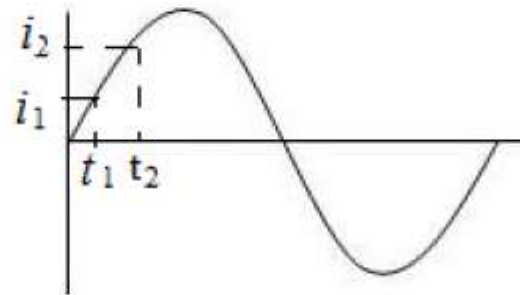
(3) Time period:

- The time required to complete one cycle is known as the time period.
- It is denoted by T and measure in terms of seconds.
- It is inversely proportional to frequency, $T = \frac{1}{f}$

(4) Instantaneous value:

- It is the value of an alternating quantity at any instant of time; see in fig.
- They are represented by small letters, i , v , e , etc.
- Here the value of current i_1 at t_1 time and i_2 at t_2 time.

Thus the relation between instantaneous current and time is expressed as $i = I_m \sin \omega t$



(5) Maximum, Peak, Amplitude or Crest value:

- The maximum positive or negative values of an alternating quantity are known as peak value.
- They are represented by I_m , V_m , E_m , etc
- See in fig: $+E_m$ and $-E_m$ are the peak value of voltages.

(6) R.M.S value:

- The RMS (root mean square) value of the ac current is called effective values of an AC.
- The RMS value of an AC voltage or current is defined as the equivalent DC value, which would have the same heating effect. So, heat produce due to AC = heat produce due to DC.
- They are represented by I_{rms} , V_{rms} , or V , I , etc
- For a sine wave, $RMS\ value = \frac{1}{\sqrt{2}} \times \text{maximum value}$. Hence its value is 0.707 of the maximum value.

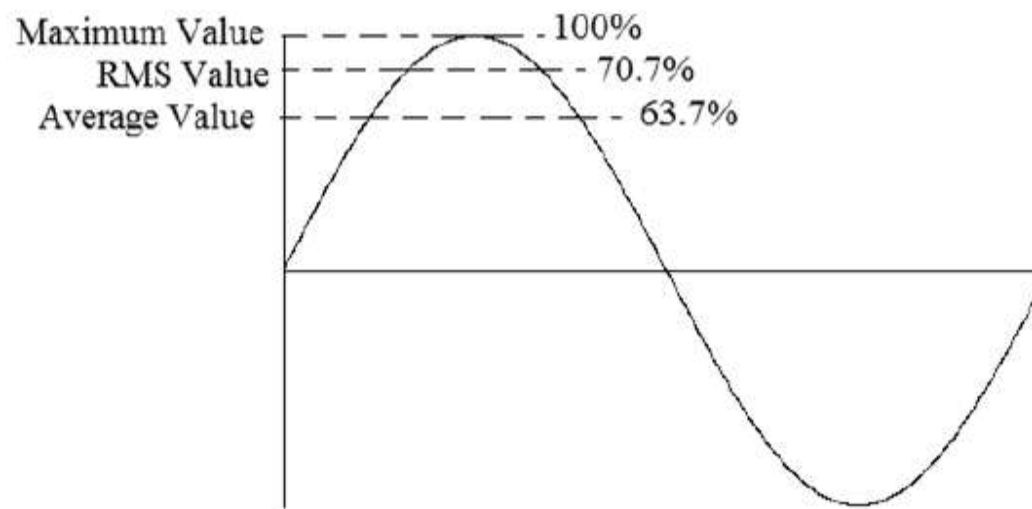
(7) Average or mean value:

- The average value is defined with reference to same charge transfer by DC and AC circuit. So, charge transferred by DC circuit = charge transferred by AC circuit.
- They are represented by I_{avg} , V_{avg} , etc
- For a sine wave, average value = $\frac{2}{\pi} \times \text{maximum value}$. Hence its value is 0.637 of the maximum value.

(8) Peak factor or crest factor or amplitude factor: It is the ratio of maximum value to RMS value. It is indicated by K_p .

$$\text{Hence, } K_p = \frac{\text{Maximum Value}}{\text{RMS Value}} = \frac{I_m}{I_m / \sqrt{2}} = 1.414$$

$$\therefore K_f = \frac{\text{RMS Value}}{\text{Average Value}} = \frac{I_m / \sqrt{2}}{2I_m / \pi} = 1.11$$



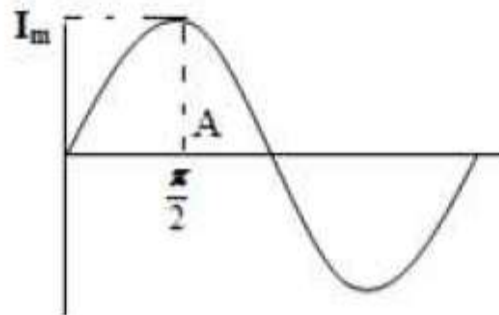
(9) Form factor: It is the ratio of RMS value to average value. It is indicated by K_f .

$$\therefore K_f = \frac{\text{RMS Value}}{\text{Average Value}} = \frac{I_m / \sqrt{2}}{2I_m / \pi} = 1.11$$

Hence its value is 1.11

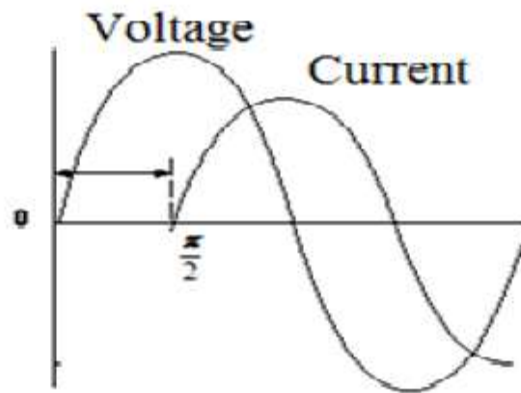
(10) Phase: The phase of an alternating quantity is the fraction of a cycle (in degree, second or radians) that has gone by since a voltage or current has passed through a given reference. (Usually zero) value. See in fig.

(10) Phase: The phase of an alternating quantity is the fraction of a cycle (in degree, second or radians) that has gone by since a voltage or current has passed through a given reference. (Usually zero) value. See in fig.



Phase at point A = $\frac{\pi}{2}$ radian or $\frac{T}{4}$ Second or 90 degree.

(11) Phase difference: When two alternating quantities are not attaining maximum value or zero value at the same time then they have phase difference. See fig.



Here the phase difference between voltage and current is 90° or $\frac{\pi}{2}$ radian.

R.M.S means root mean(average) square

$$\text{Means, } I_{rms} = \sqrt{i_{avg}^2}$$

First, square of current;

$$i = I_m \sin \theta, \text{ so, } i^2 = I_m^2 \sin^2 \theta$$

$$i^2 = I_m^2 \left(\frac{1 - \cos 2\theta}{2} \right)$$

$$= \frac{I_m^2}{2} (1 - \cos 2\theta)$$

Now, Average of square of current of one complete cycle,

$$i_{avg}^2 = \frac{1}{2\pi} \int_0^{2\pi} i^2 d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{I_m^2}{2} (1 - \cos 2\theta) d\theta$$

$$= \frac{I_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{I_m^2}{4\pi} \left[2\pi - \frac{\sin 4\pi}{2} - 0 + \frac{\sin 2\theta}{2} \right]$$

$$= \frac{I_m^2}{4\pi} (2\pi) = \frac{I_m^2}{2}$$

Finally, root of average of square of current;

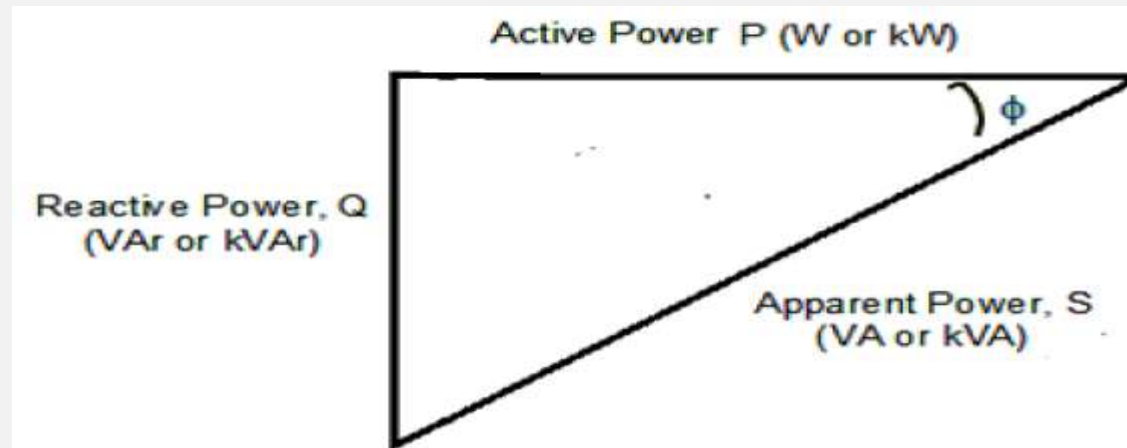
$$I_{rms} = \sqrt{i_{avg}^2} = \sqrt{\frac{I_m^2}{2}}$$

$$\therefore I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

RMS and Average value

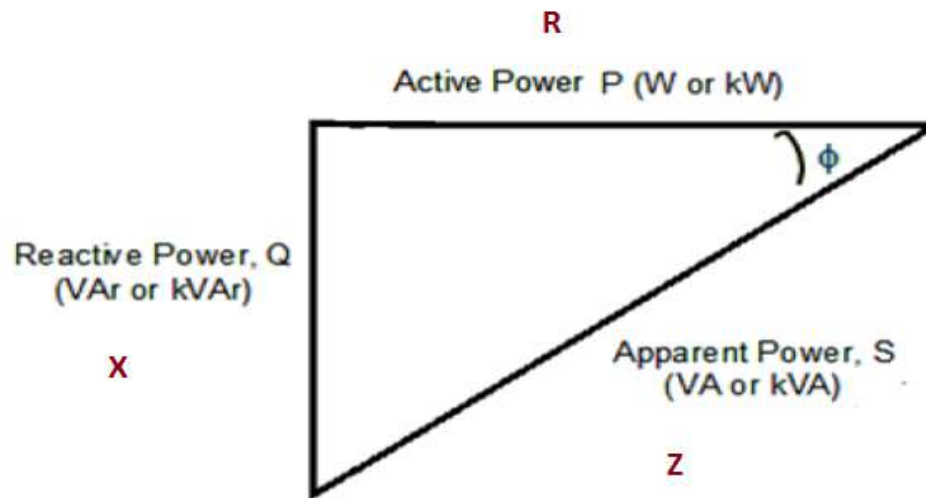
Determination of RMS Value	Determination of Average Value
R.M.S means root mean(average) square <i>Means, $I_{rms} = \sqrt{i_{avg}^2}$</i> <i>First, square of current;</i> $i = I_m \sin \theta$, so, $i^2 = I_m^2 \sin^2 \theta$ $i^2 = I_m^2 \left(\frac{1 - \cos 2\theta}{2} \right)$ $= \frac{I_m^2}{2} (1 - \cos 2\theta)$ <i>Now, Average of square of current of one complete cycle;</i> $i_{avg}^2 = \frac{1}{2\pi} \int_0^{2\pi} i^2 d\theta$ $= \frac{1}{2\pi} \int_0^{2\pi} \frac{I_m^2}{2} (1 - \cos 2\theta) d\theta$ $= \frac{I_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi}$ $= \frac{I_m^2}{4\pi} \left[2\pi - \frac{\sin 4\pi}{2} - 0 + \frac{\sin 2\theta}{2} \right]$ $= \frac{I_m^2}{4\pi} (2\pi) = \frac{I_m^2}{2}$ <i>Finally, root of average of square of current;</i> $I_{rms} = \sqrt{i_{avg}^2} = \sqrt{\frac{I_m^2}{2}}$ $\therefore I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$	$i_{avg} = \frac{1}{\pi} \int_0^{\pi} i d\theta$ $= \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta d\theta$ $= \frac{I_m}{\pi} \int_0^{\pi} \sin \theta d\theta$ $= \frac{I_m}{\pi} \left[-\cos \theta \right]_0^{\pi}$ $= \frac{I_m}{\pi} \left[-\cos \pi + \cos 0 \right]$ $i_{avg} = \frac{I_m}{\pi} [1 + 1]$ $\therefore I_{avg} = \frac{2I_m}{\pi} = 0.637 I_m$

Real power, reactive power, apparent power



<i>Active/Real Power</i>	<i>Reactive Power</i>	<i>Apparent Power</i>
Real or true power is the power that is used to do the work on the load. It is denoted by P. $P = VI \cos \phi$ It is the power that is consumed by the resistance R. The unit of active power is Watt (W).	Reactive power is the power that is wasted and not used to do work on the load. It is denoted by Q. $Q = VI \sin \phi$ It is the circulating power in the L and C components. The unit of reactive power is Volt Amperes Reactive (VAr).	The apparent power is the total power in the circuit. It is denoted by S. $S = \sqrt{P^2 + Q^2}$ $= \sqrt{(VI \cos \phi)^2 + (VI \sin \phi)^2}$ $= VI \sqrt{(\cos \phi)^2 + (\sin \phi)^2}$ $\therefore S = VI$ The unit of apparent power is Volt Amperes (VA).

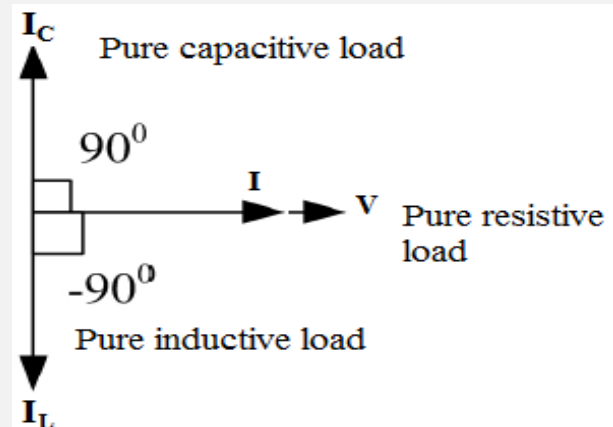
Power factor



From power triangle, $\cos\phi = \frac{\text{Active Power}}{\text{Apparent Power}}$

From impedance triangle, Power Factor, $\cos\phi = \frac{R}{Z}$

From phasor diagram it is angle between V and I

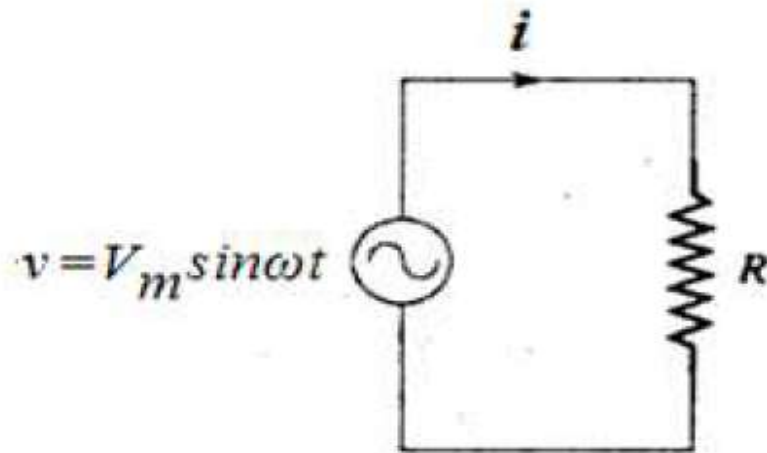


- *Power factor for pure R: $\cos\phi = \cos 0 = 1$*
- *Power factor for pure L: $\cos\phi = \cos 90 = 0$ (Lagging)*
- *Power factor for pure C: $\cos\phi = \cos 90 = 0$ (Leading)*

Analysis of single-phase ac circuits consisting of R

4.5.2 AC through pure resistance

Consider a circuit consisting of an AC source and a pure resistor (zero inductance).



Let, instantaneous value of alternating voltage is,

$$v = V_m \sin \omega t \quad \text{--- (1)}$$

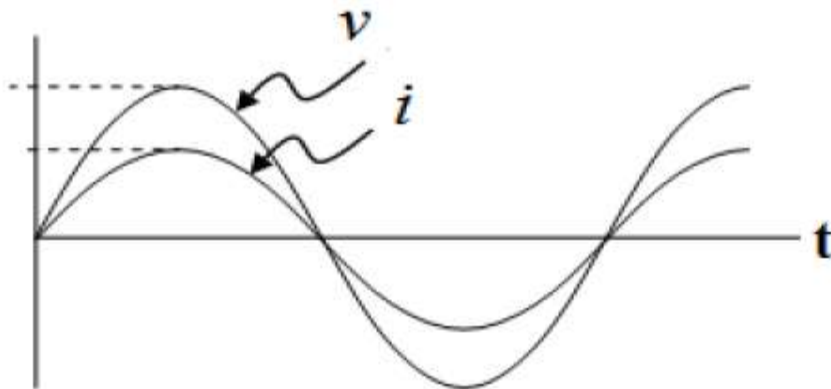
⇒ Instantaneous value of current :

$$\text{As we know, } i = \frac{v}{R} = \frac{V_m \sin \omega t}{R}$$

$$\text{But, } \frac{V_m}{R} = I_m$$

$$\therefore i = I_m \sin \omega t \quad \text{--- (2)}$$

⇒ Voltage, current waveform and phasor diagram:



Voltage and current waveform



Phasor Diagram

⇒ Instantaneous value of power :

$$\begin{aligned} p &= v \times i \\ &= V_m \sin \omega t \times I_m \sin \omega t \\ &= V_m I_m \sin^2 \omega t \\ &= V_m I_m \left(\frac{1 - \cos 2\omega t}{2} \right) \\ p &= \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\omega t \end{aligned}$$

Here, $\frac{V_m I_m}{2}$ is constant part while $\frac{V_m I_m}{2} \cos 2\omega t$ is variable part.

For one complete cycle, the average value of $\frac{V_m I_m}{2} \cos 2\omega t$ becomes zero because the frequency of the fluctuating power is twice the applied voltage.

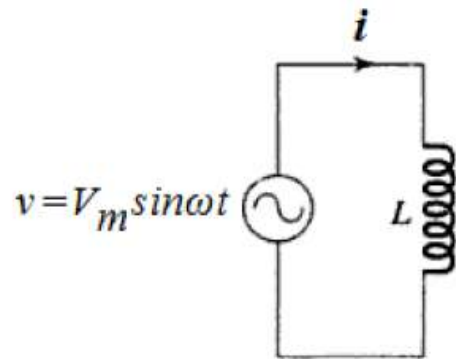
$$\begin{aligned} \text{Hence, } P_{avg} &= \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} \\ P_{avg} &= V \times I \quad \text{--- (3)} \end{aligned}$$

Thus, the active power consumption in purely resistive circuit is VI .

Analysis of single-phase ac circuits consisting of L & C

(a) AC through a pure inductor

Consider a circuit consisting of an AC source and a pure inductance (zero resistance).



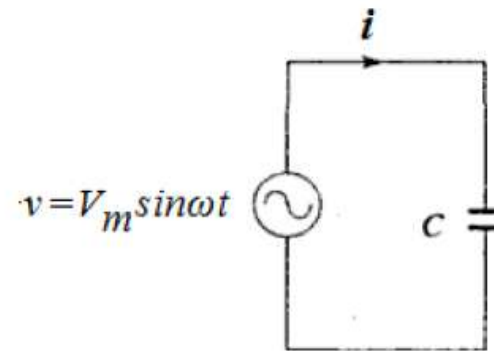
Let, instantaneous value of alternating voltage is,

$$v = V_m \sin \omega t \quad \text{--- (1)}$$

⇒ Instantaneous value of current :

(b) AC through a pure capacitor

Consider a circuit consisting of an AC source and a pure capacitor.



Let instantaneous value of alternating voltage is,

$$v = V_m \sin \omega t \quad \text{--- (1)}$$

⇒ Instantaneous value of current :

⇒ Instantaneous value of current :

As we know, $v = L \frac{di}{dt}$

$$\therefore \frac{di}{dt} = \frac{v}{L} = \frac{V_m \sin \omega t}{L}$$

from this,

$$\begin{aligned} i &= \int \frac{V_m \sin \omega t}{L} dt \\ &= \frac{V_m}{L} \int \sin \omega t dt \\ &= \frac{V_m}{L} \left(\frac{-\cos \omega t}{\omega} \right) \\ &= \frac{V_m}{\omega L} (-\cos \omega t) \end{aligned}$$

Now, $-\cos \omega t = \sin(\omega t - \frac{\pi}{2})$ and $\omega = 2\pi f$

So, $i = \frac{V_m}{2\pi f L} \sin(\omega t - \frac{\pi}{2})$

Now, $2\pi f L = X_L$

Hence, $i = \frac{V_m}{X_L} \sin(\omega t - \frac{\pi}{2})$

But, $\frac{V_m}{X_L} = I_m$

$$\therefore i = I_m \sin(\omega t - \frac{\pi}{2}) \quad \text{--- (2)}$$

→ Instantaneous value of current .

As we know, $i = \frac{dq}{dt}$

$$q = Cv$$

$$i = \frac{dCv}{dt}$$

$$= \frac{d}{dt} (CV_m \sin \omega t)$$

$$= CV_m \frac{d}{dt} \sin \omega t$$

$$= CV_m (\cos \omega t \times \omega)$$

$$\therefore i = \omega CV_m \cos \omega t$$

Now, $\cos \omega t = \sin(\omega t + \frac{\pi}{2})$ and $\omega = 2\pi f$

So, $i = 2\pi f C V_m \sin(\omega t + \frac{\pi}{2})$

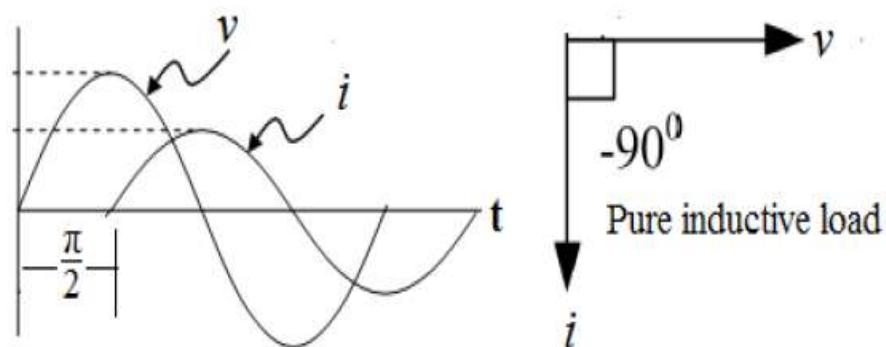
Now, $2\pi f C = \frac{1}{X_c}$

$$i = \frac{V_m}{X_c} \sin(\omega t + \frac{\pi}{2})$$

But $\frac{V_m}{X_c} = I_m$

$$\text{Hence, } i = I_m \sin(\omega t + \frac{\pi}{2}) \quad \text{--- (2)}$$

⇒ **Voltage, current waveform and phasor diagram**



Voltage and current waveform

Phasor Diagram

⇒ **Instantaneous value of power :**

$$\text{Power, } p = v \times i$$

$$= V_m \sin \omega t \times I_m \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\text{Now, again } \sin \left(\omega t - \frac{\pi}{2} \right) = -\cos \omega t$$

$$\text{Hence, } p = V_m \sin \omega t \times I_m (-\cos \omega t)$$

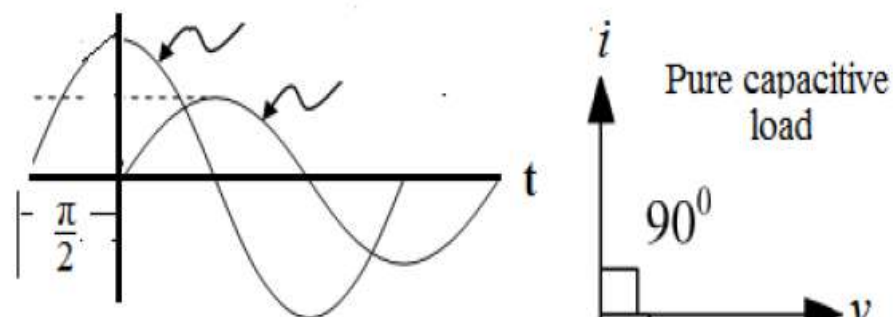
$$= -V_m I_m \sin \omega t \cos \omega t$$

$$\text{Now, we have } \sin \omega t \cos \omega t = \frac{\sin 2\omega t}{2}$$

$$\text{Put, } \omega t = \theta$$

$$\therefore p = -V_m I_m \frac{\sin 2\theta}{2}$$

⇒ **Voltage, current waveform and phasor diagram**



Voltage and current waveform

Phasor Diagram

⇒ **Instantaneous value of power :**

$$\text{Power, } p = v \times i$$

$$= V_m \sin \omega t \times I_m \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$\text{Now, again } \sin \left(\omega t + \frac{\pi}{2} \right) = \cos \omega t$$

$$\text{Hence, } p = V_m \sin \omega t \times I_m (\cos \omega t)$$

$$= V_m I_m \sin \omega t \cos \omega t$$

$$\text{Now, we have } \sin \omega t \cos \omega t = \frac{\sin 2\omega t}{2}$$

$$\text{Put, } \omega t = \theta$$

$$\therefore p = V_m I_m \frac{\sin 2\theta}{2}$$

Now, Average power for complete one cycle;

$$\begin{aligned}
 P_{\text{avg}} &= \frac{1}{2\pi} \int_0^{2\pi} p d\theta \\
 &= \frac{1}{2\pi} \int_0^{2\pi} -V_m I_m \frac{\sin 2\theta}{2} d\theta \\
 &= \frac{-V_m I_m}{4\pi} \int_0^{2\pi} \sin 2\theta d\theta \\
 &= \frac{-V_m I_m}{4\pi} \left| \frac{-\cos 2\theta}{2} \right|_0^{2\pi} \\
 &= \frac{V_m I_m}{8\pi} \left| \cos 2\theta \right|_0^{2\pi} \\
 &= \frac{V_m I_m}{8\pi} (\cos 4\pi - \cos 2\pi)
 \end{aligned}$$

$$= \frac{V_m I_m}{8\pi} (1 - 1)$$

$$P_{\text{avg}} = 0 \quad \text{---(3)}$$

Thus, the active power consumption in purely inductive circuit is zero.

Now, Average power for complete one cycle;

$$\begin{aligned}
 P_{\text{avg}} &= \frac{1}{2\pi} \int_0^{2\pi} p d\theta \\
 &= \frac{1}{2\pi} \int_0^{2\pi} V_m I_m \frac{\sin 2\theta}{2} d\theta \\
 &= \frac{V_m I_m}{4\pi} \int_0^{2\pi} \sin 2\theta d\theta \\
 &= \frac{V_m I_m}{4\pi} \left| \frac{-\cos 2\theta}{2} \right|_0^{2\pi} \\
 &= \frac{V_m I_m}{8\pi} \left| -\cos 2\theta \right|_0^{2\pi} \\
 &= \frac{V_m I_m}{8\pi} |-\cos 4\pi + \cos 2\pi|
 \end{aligned}$$

$$= \frac{V_m I_m}{8\pi} (-1 + 1)$$

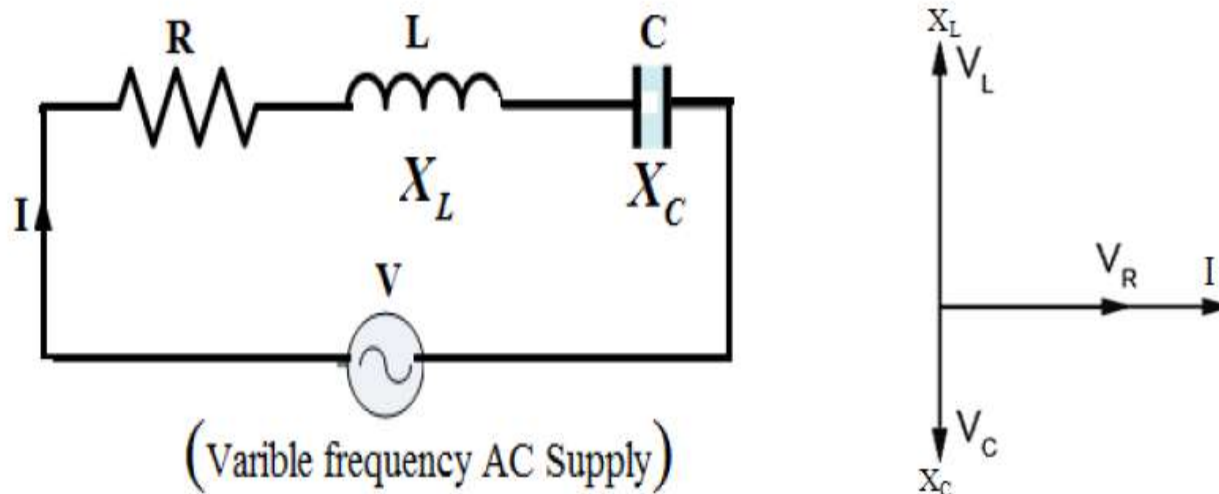
$$P_{\text{avg}} = 0 \quad \text{---(3)}$$

Thus, the active power consumption in purely capacitive circuit is zero.

Series RLC Circuit resonance

4.5.7.1 Series Resonance

A series resonant circuit consists of a resistor, an inductor and a capacitor in series. The variable frequency, constant voltage AC supply is applied to the circuit to get resonance condition. See fig.



Condition for resonance: A series RLC circuit is said to be resonant when net reactance of the circuit becomes zero. Hence, $X = X_L - X_C = 0$ or $X_L = X_C$

Explanation

See phasor diagram for net reactance.

As we know that, $X_L = 2\pi fL$ and $X_C = \frac{1}{2\pi fC}$

Both above are the functions of frequency. When frequency increase, X_L increase and X_C decreases. So, at certain values of frequency, both X_L and X_C becomes exactly equal. This value of frequency is known as resonant frequency. It denoted by f_r .

Expression of resonant frequency(f_r):

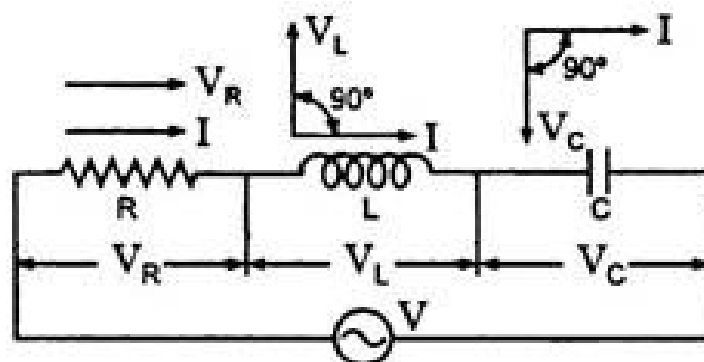
As per condition, $X_L = X_C$

$$2\pi f_r L = \frac{1}{2\pi f_r C}$$

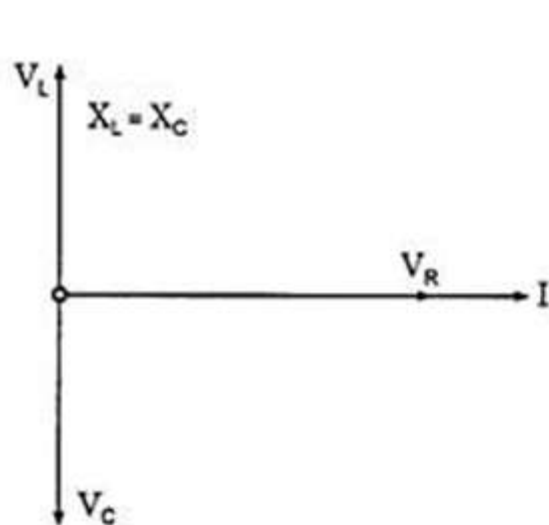
$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$\therefore \mathbf{f_r = \frac{1}{2\pi\sqrt{LC}}}$$

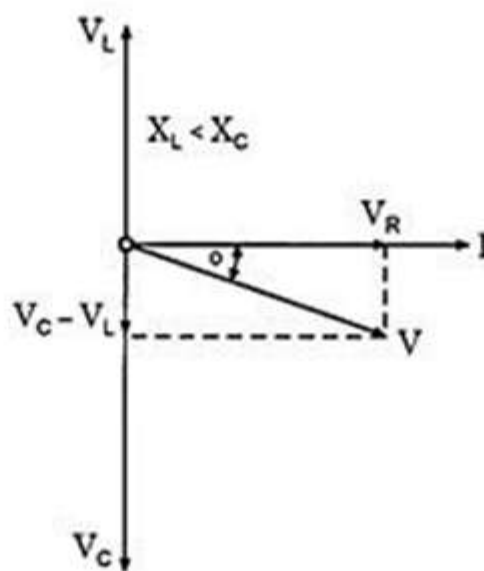
3	For series resonant circuit with brief description draw the phasor diagrams for following conditions (i) At resonant (ii) Below resonant (iii) Above resonant.	03	GTU-JUNE -2019
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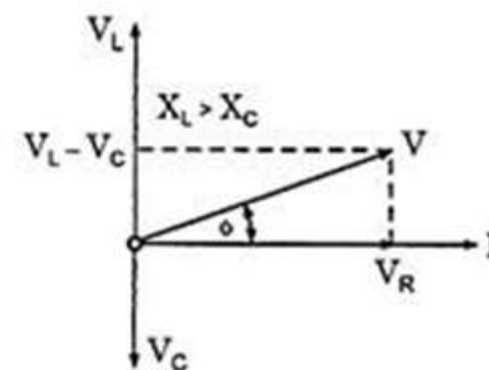
Phasor diagrams for the series R-L-C circuit shown in Fig. 6.1 at three different frequencies i.e. (a) $f = f_r$ (b) $f < f_r$ and (c) $f > f_r$



(a) At resonance



(b) Below resonance



(c) Above resonance

Voltage and current relations in star connections

5.6.1 RELATIONSHIP BETWEEN LINE AND PHASE QUANTITIES IN BALANCED STAR CONNECTED SYSTEM

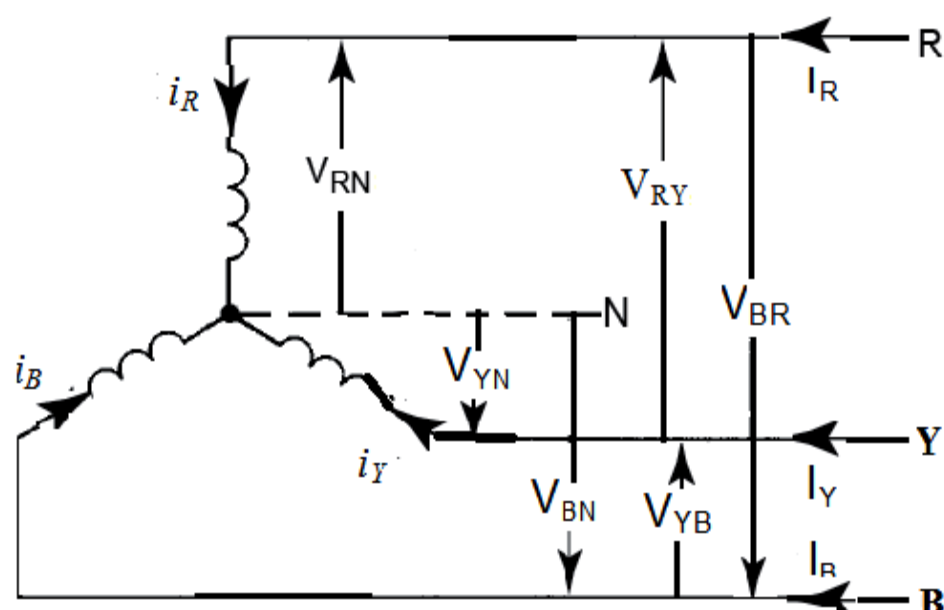


Figure 5.6 Balanced star connected system

Here V_{RY} is the phasor sum of V_{RN} and $-V_{YN}$ and geometrically it can be obtained by completing the parallelogram as shown in fig.

Consider a star connected, 3-phase balanced load.
See fig. 5.6

As per explain earlier; here,

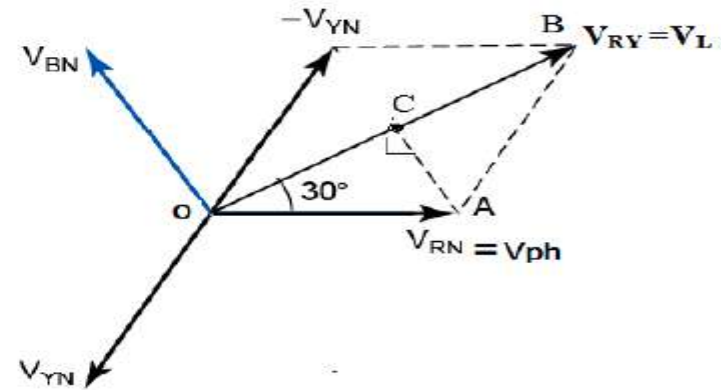
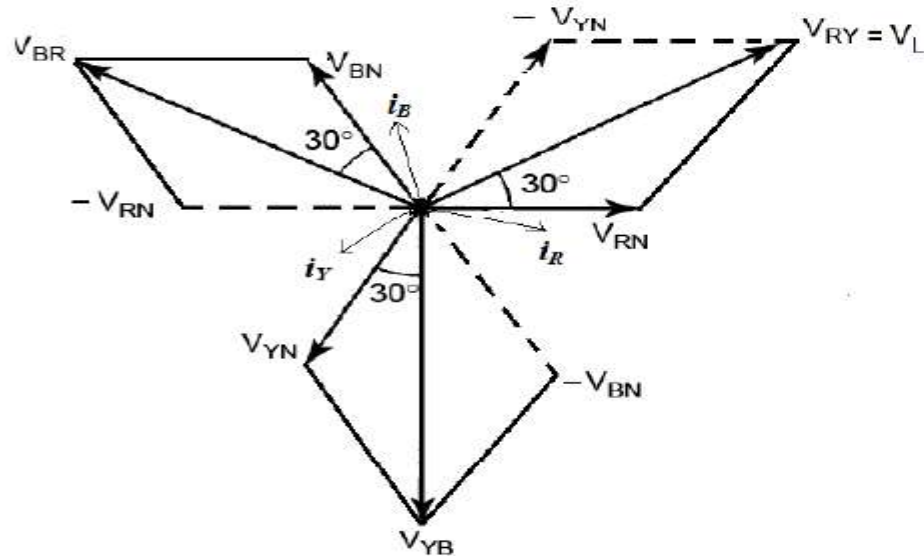
$$V_{RN} = V_{YN} = V_{BN} = V_{ph} \text{ (Phase voltage)}$$

$$V_{RY} = V_{YB} = V_{BR} = V_L \text{ (Line voltage)}$$

$$I_R = I_Y = I_B = I_L \text{ (Line current) and}$$

$$i_R = i_Y = i_B = I_{ph} \text{ (Phase current)}$$

The phase voltages, V_{RN} , V_{YN} , and V_{BN} are placed by 120° apart from each other.



Expression of Line voltage and phase voltage:

Draw the perpendicular line from point A on vector OB. So the line AC bisects the vector OB.

Now, in triangle, OAC,

$$\cos 30^\circ = \frac{OC}{OA} = \frac{\frac{OB}{2}}{OA} = \frac{\frac{V_L}{2}}{V_{ph}}$$

$$\therefore \frac{\sqrt{3}}{2} = \frac{V_L/2}{V_{ph}}$$

So, in star connection, $V_L = \sqrt{3}V_{ph}$ and $\therefore V_{ph} = \frac{V_L}{\sqrt{3}}$

Expression of Line current and phase current:

See fig. It can be seen that each winding and line are in the series, so the current flowing through them is same. So, the line current is equal to phase current.

Hence, in star connection, $I_L = I_{ph}$

Expression of Power:

Power for 3 phases is given by, $P = 3V_{ph}I_{ph} \cos \phi$

$$\therefore P = 3 \frac{V_L}{\sqrt{3}} I_L \cos \phi$$

$$\therefore P = \sqrt{3} V_L I_L \cos \phi$$

Voltage and current relations in DELTA connections

5.6.2 RELATIONSHIP BETWEEN LINE AND PHASE QUANTITIES IN BALANCED DELTA CONNECTED SYSEM

Consider Fig 5.7, for a delta connected, 3-phase balanced load.

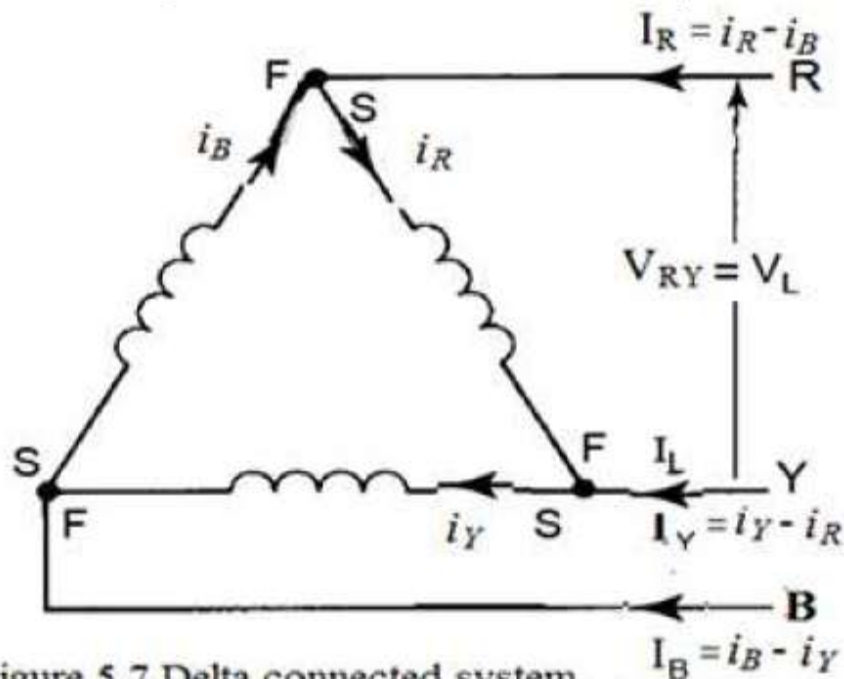


Figure 5.7 Delta connected system

As per explain earlier; here,

$$V_{RY} = V_{YB} = V_{BR} = V_L = V_{ph}$$

$$I_R = I_Y = I_B = I_L \text{ (Line current) and}$$

$$i_R = i_Y = i_B = I_{ph} \text{ (Phase current)}$$

Expression of Line current and phase current:

In fig it can be seen that the current along each line is the phasor difference of a pair of phase currents. The three phase currents are mutually displaced by 120° . See the phasor diagram.

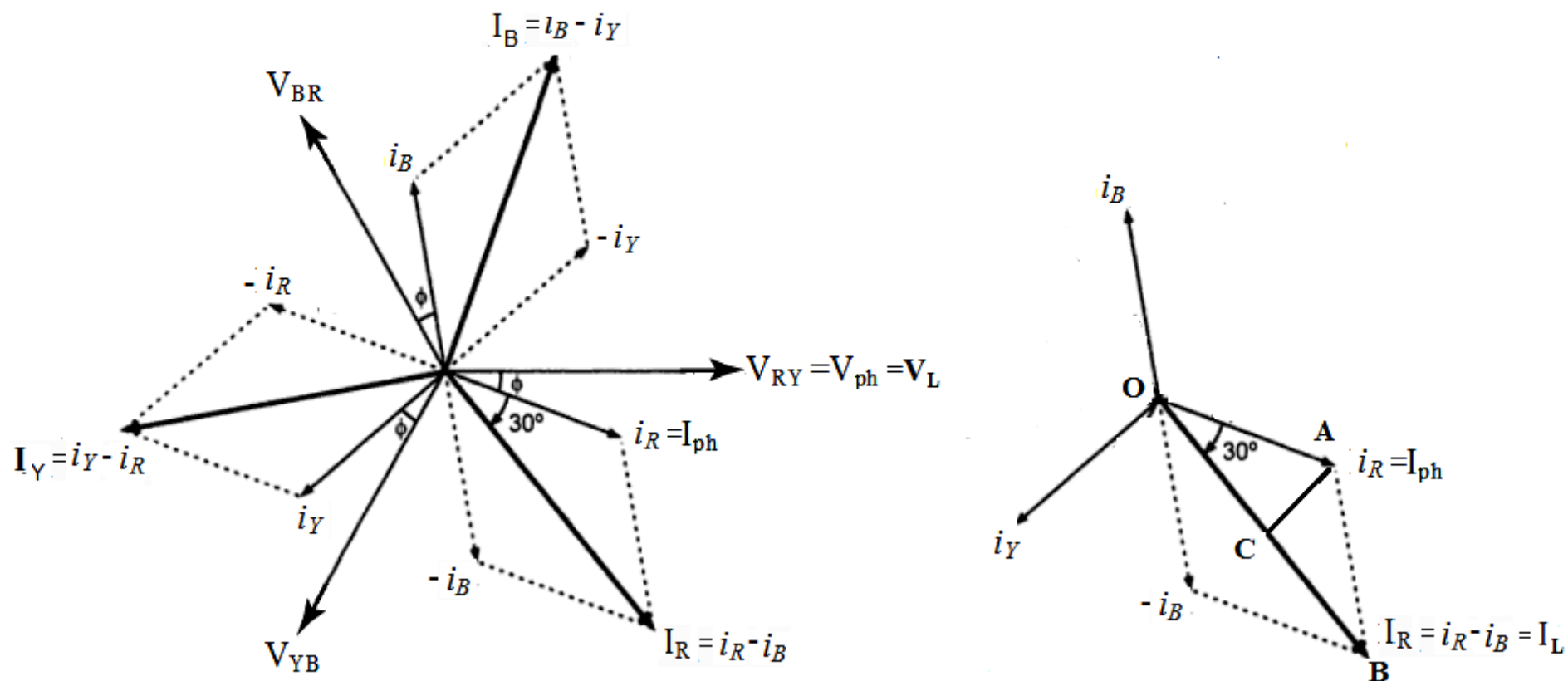


Figure 5.8 Phasor diagram of delta connected system

Draw the perpendicular line from point A on vector OB. So the line AC bisects the vector OB.

Now, in triangle, OAC,

$$\cos 30^\circ = \frac{OC}{OA} = \frac{\frac{OB}{2}}{I_{ph}} = \frac{\frac{I_L}{2}}{I_{ph}}$$
$$\therefore \frac{\sqrt{3}}{2} = \frac{I_L/2}{I_{ph}}$$

So, in Delta connection, $I_L = \sqrt{3}I_{ph}$ and $I_{ph} = \frac{I_L}{\sqrt{3}}$

Expression of Line voltage and phase voltage:

See in fig. It can be clearly seen that; each pair of lines is connected across a phase winding.

Thus for delta connection, $V_L = V_{ph}$

Expression of Power:

Power for 3-phases is given by, $P = 3V_{ph}I_{ph} \cos \phi$

$$\therefore P = 3V_L \frac{I_L}{\sqrt{3}} \cos \phi$$

$$\therefore \boxed{P = \sqrt{3}V_L I_L \cos \phi}$$

Power measurement in three phase circuits

There are three methods to measure 3-phase power.

- 1) One wattmeter method
- 2) Two wattmeter method and
- 3) Three wattmeter method

(1) One wattmeter method:

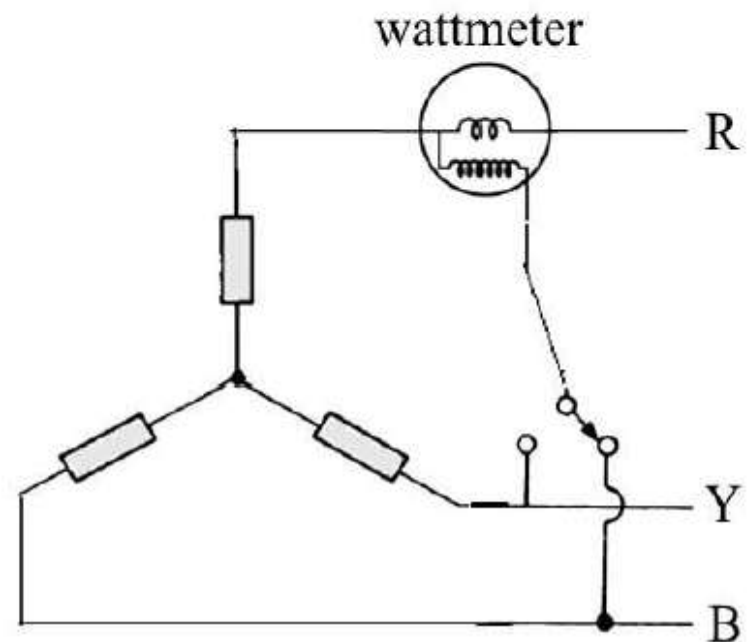
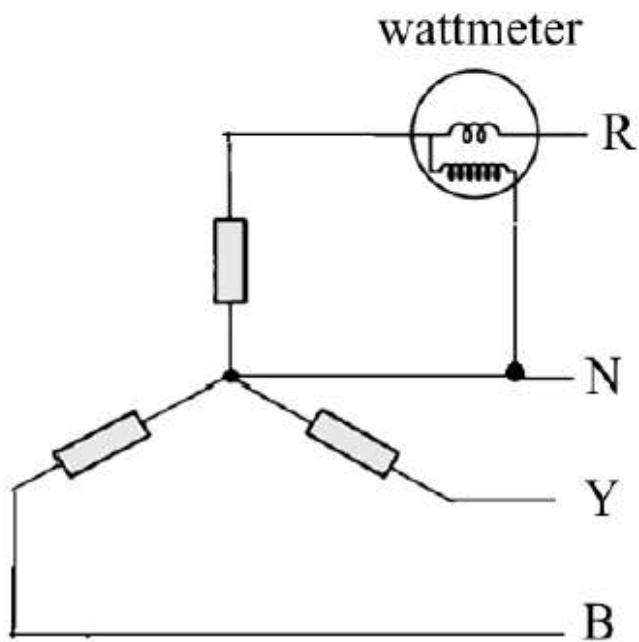
This method is used for balanced load only.

When the load is balanced; the total power is given by $P=3V_{ph}I_{ph}\cos\phi$

Hence, one wattmeter is used to measure power in one phase, then to measure total power in 3-phase circuit wattmeter reading is multiplied by 3.

Total power, $P=3\times\text{Wattmeter reading}$

See circuit diagram.



(2) Two wattmeter method:

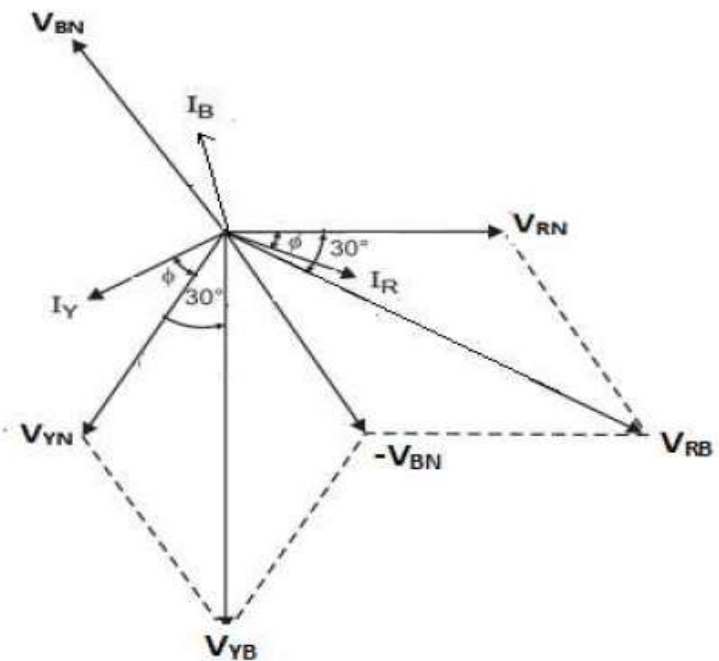
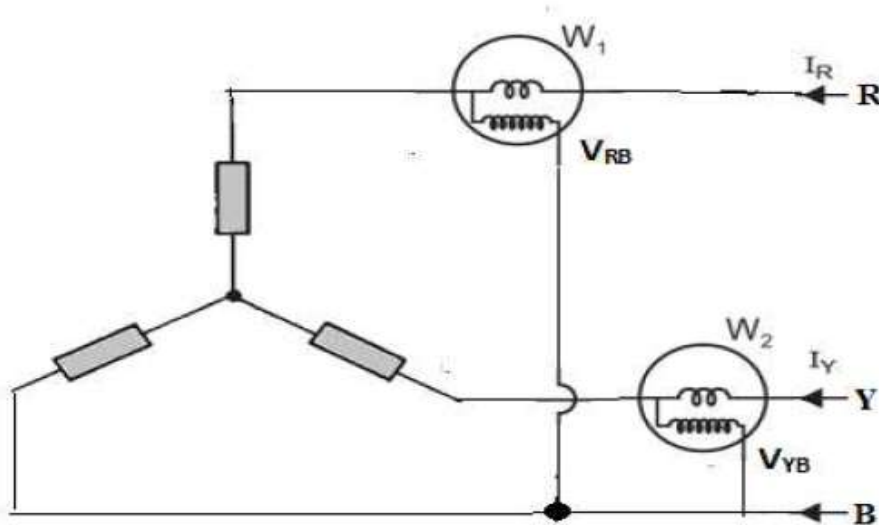
- This method is used for balanced as well as unbalanced load.
- Two-single phase wattmeter's are used to measure the three-phase power.
- The sum of the two wattmeter's readings gives the total power.

$$\text{Total power, } P = W_1 + W_2$$

Consider a balanced star connected load; the power is usually measured by using two wattmeter simultaneously, as shown in fig.

Wattmeter W_1 connected with R phase and W_2 is connected with Y phase.

Now, consider the phasor diagram for an inductive balanced load,



From the phasor diagram it is clear that the phase angle between V_{RB} and I_R is $(30^\circ - \phi)$ and between V_{YB} and I_Y is $(30^\circ + \phi)$.

$$W_1 = V_{RY} I_R \cos(30^\circ - \phi)$$

As per explain earlier, $V_{RB} = V_L$ and $I_R = I_L$

$$\text{Hence, } W_1 = V_L I_L \cos(30^\circ - \phi) \quad \text{--- (1)}$$

$$\text{Simillarly, } W_2 = V_{YB} I_Y \cos(30^\circ + \phi)$$

Here also, $V_{BY} = V_L$ and $I_B = I_L$

$$\text{So, } W_2 = V_L I_L \cos(30^\circ + \phi) \quad \text{--- (2)}$$

Now, the total power, $P = W_1 + W_2$

$$P = V_L I_L \cos(30^\circ - \phi) + V_L I_L \cos(30^\circ + \phi)$$

$$\therefore P = V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)]$$

Now, $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ and $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$$\therefore P = V_L I_L (\cos 30^\circ \cos \phi + \sin 30^\circ \sin \phi + \cos 30^\circ \cos \phi - \sin 30^\circ \sin \phi)$$

$$= V_L I_L (2 \cos 30^\circ \cos \phi)$$

$$= V_L I_L \left(2 \times \frac{\sqrt{3}}{2} \times \cos \phi\right)$$

$$\therefore \text{Total power, } P = \sqrt{3} V_L I_L \cos \phi$$

Thus, total power in three phase circuit is measured by counting the sum of reading of two wattmeter.

GTU Paper

Sr	Question	Mark	GTU Exam
1	Define Q factor of series RLC circuit. What is the importance of it?	03	Jun-025
2	Draw impedance triangle, voltage triangle and power triangle for single phase R-L circuit.	04	June-2025
3	Prove that current in purely capacitive circuit current leads its voltage by 90 and average consumption in pure capacitive circuit is 0.	07	Jun-2025, Aug-2022
4	Explain the concept of active, reactive and apparent power. OR Distinguish between (i) Apparent power (ii) Active power and (iii) Reactive power in ac circuits.	03 04	Jan-2025 Aug-2022
5	Analyze voltage and current relations in three phase star connection.	04	Jan-2025
6	Compare series and parallel resonance OR Write the comparison between Series resonance and Parallel resonance condition in AC circuit	07 03	Jan-2025, Aug-2022
7	Illustrate the Phasor diagram of RC series circuit and explain its development	07	Jan-2025
8	Obtain the relationship between line and phase values of current in a three phase, balanced, delta connected system.	07	Aug-2022

GTU PAPER ANALYSIS

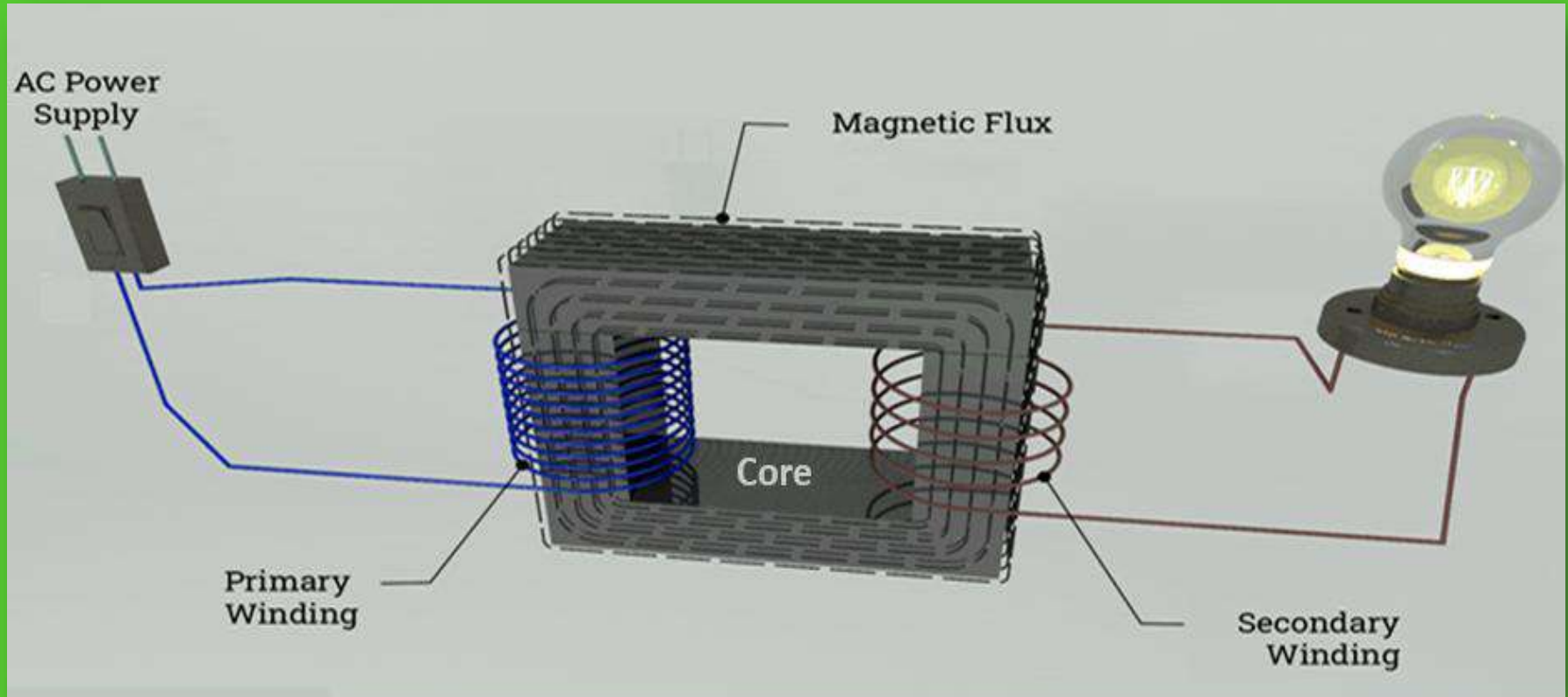
Sr.	Question	Mark	Exam
1	Define Amplitude, Frequency and Time period for alternating quantities.	03	GTU-June-2019
2	For A.C. sinusoidal current prove that $I_{rms} = 0.707 I_m$.	04	GTU-Jan - 2019
3	For series resonant circuit with brief description draw the phasor diagrams for following conditions (i) At resonant (ii) Below resonant (iii) Above resonant.	03	GTU-JUNE - 2019
4	Prove that the sum of readings of two watt meters connected to measure power in three phase circuit gives total power consumed by the circuit.	03	GTU-June-2019

THE END

S.G. Prajapati



Ch-3: Magnetic Circuit & Transformer



Prepared By: Dr. S. G. Prajapati
(D.E, B.E, M.E, Ph.D. - Electrical Engineering)

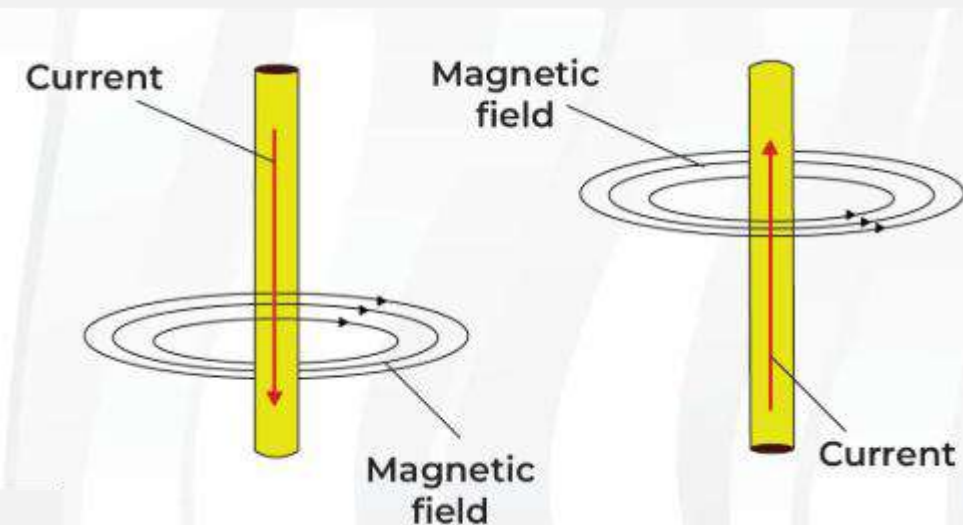
Monday, August 11, 2025

Contents

1. Magnetic effect of electrical current, Cross and dot convention,
2. Right hand thumb rule and cork screw rule,
3. M.M.F, flux, flux density, reluctance, permeability and field strength,
4. Analogy between electric and magnetic circuits,
5. Magnetic Circuits, B-H Curve, Hysteresis Loop,
6. Hysteresis and Eddy current losses,
7. Construction and working principle of single-phase transformers:
8. Voltage and current ratios, Losses,
9. Definition of regulation and efficiency
10. Ideal and practical transformer.



1. Magnetic effect of electrical current



A Danish scientist, Oersted in the early 19th century discovered that there was a magnetic field around a current-carrying conductor

The magnetic effect of electric current is the creation of a magnetic field around a wire that is carrying an electric current.

This magnetic field is invisible, but it can be detected with a magnetic compass.

How it works

When an electric current passes through a wire, the wire behaves like a magnet.

The magnetic field lines around a straight conductor are concentric circles, with the center at the conductor's axis.

How it can be observed

You can observe the magnetic effect of electric current by using a magnetic compass.

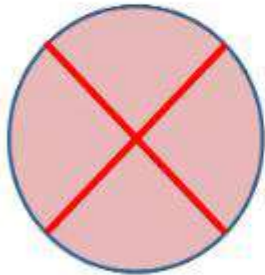
When a current passes through a conductor, the magnetic compass needle will deflect.

Applications

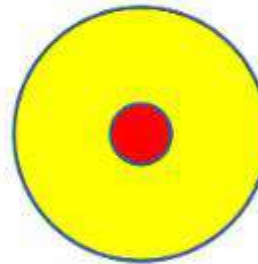
The magnetic effect of electric current is used in motors.

Cross and dot convention

Current INTO the plane of paper

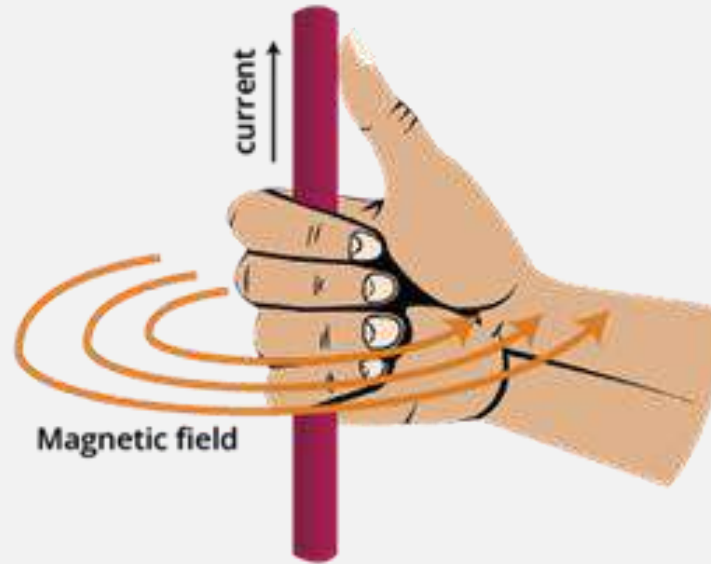


Current Coming Out of the plane of paper



*Current carrying conductor is represented by small circle

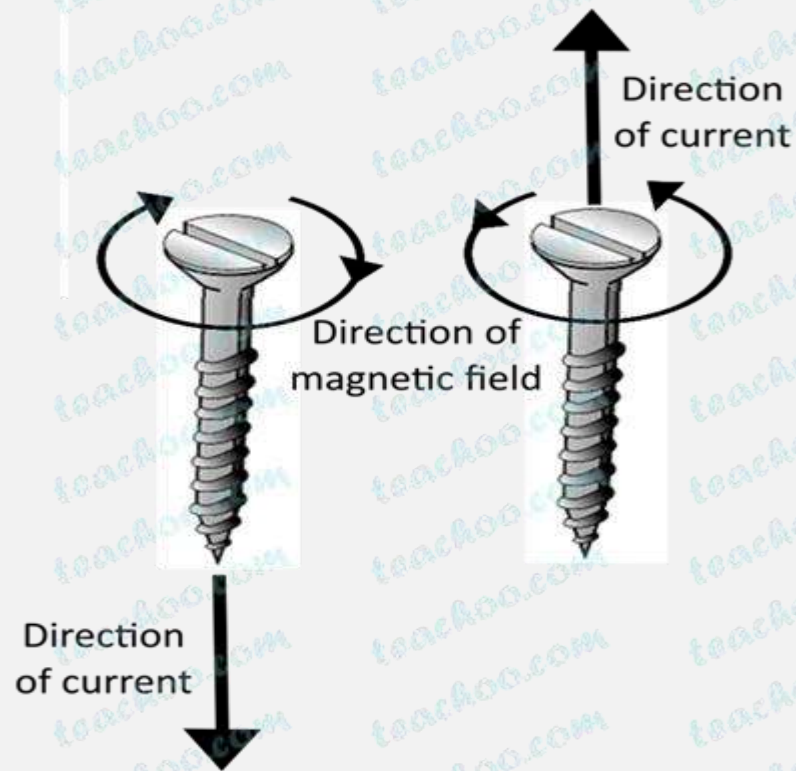
2. Right hand thumb rule and cork screw rule



The right hand rule is used to determine the direction of the magnetic field and current around a straight current carrying conductor or coil.

The right hand thumb shows if we hold the current carrying conductor in our right hand so that the thumb Direction shows current direction and the curly fingers shows the direction of magnetic field.

Maxwells corkscrew rule

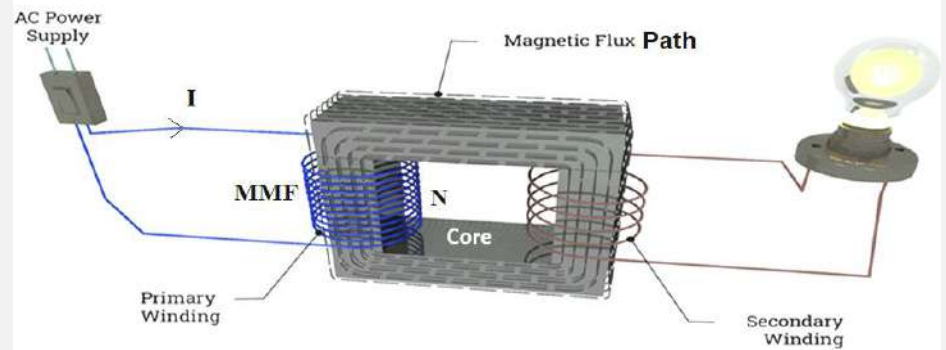


If turning a **screw** into a material:

- The direction of **screw moves forward** = **Direction of Electric Current**
- The direction of **screw rotate** = **Direction of Magnetic Field Lines**

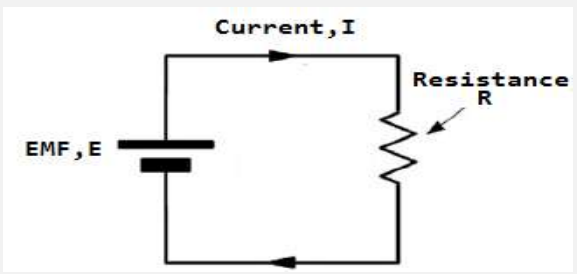
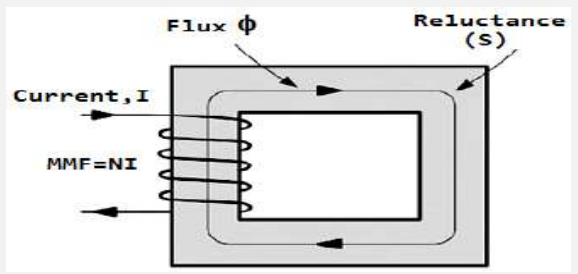
Definitions Magnetic Circuit

on



Sr. no.	Name	Definition	Ind. By	Equation	Unit
1	MMF (Magnetomotive Force)	→ It is the force that creates a magnetic field.	F_m	$N \times I$	AT (Ampere-Turns)
2	Magnetic Field Intensity OR Magnetic field Strength OR Magnetizing Force	→ MMF per unit length of the material. → It shows the strength of the magnetic field.	H	$H = \frac{N \times I}{l}$	AT/m
3	Magnetic Flux	→ Number of magnetic field lines passing through a material. → 1 Weber = 10^8 magnetic Field lines	Φ		Wb
4	Magnetic Flux Density	→ Magnetic flux per unit area.	B	$B = \frac{\Phi}{A}$	Wb/m² OR Tesla
5	Reluctance	→ Magnetic circuit's opposition to the flow of flux. → Depends on material property, area, and length. → Longer length → more reluctance.	S	$\frac{l}{\mu \times A}$	AT/Wb
6	Permeance	→ Magnetic circuit's ability to pass flux easily. → Inverse of reluctance. → Depends on material property, area, and length. → Shorter length → more permeance.	ρ	$\frac{\mu \times A}{l}$	Wb/AT
7	Permeability	→ Property of a material that allows flux to pass through it. → Depends on material property (e.g., silicon steel has high permeability).	μ	B/H	H/m
8	Reluctivity	→ Property of a material that opposes the passage of flux. It is Inverse of permeability. → Depends on material property (e.g., ferrite has low permeability)	ν	$1/\mu$	1/H/m

4. Analogy between electric and magnetic circuits

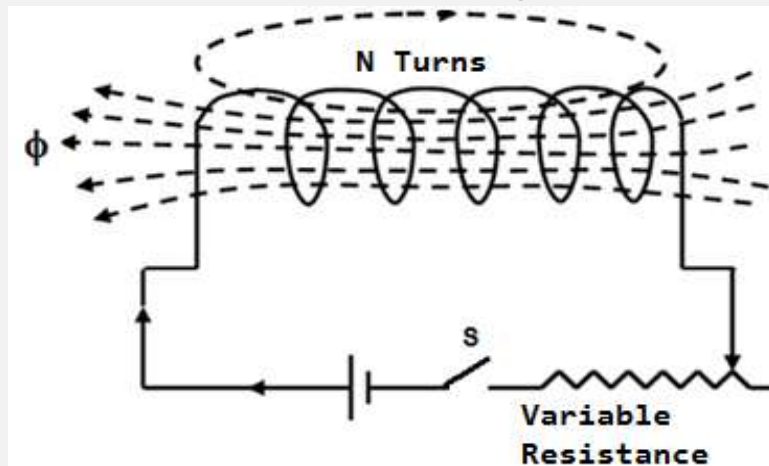
SIMILARITIES	
<u>Electric Circuit</u>	<u>Magnetic Circuit</u>
	
1. The close path of an electric current is known as electric circuit.	1. The close path of magnetic flux is known as magnetic circuit.
2. $EMF = IR$	2. $MMF = NI$ OR ϕS
3. Current, $I = \frac{emf}{R}$	3. Flux, $\phi = \frac{mmf}{S}$
4. Resistance, $R = \frac{emf}{I}$ OR $R = \frac{\rho l}{a}$	4. Reluctance, $S = \frac{mmf}{\phi}$ OR $S = \frac{l}{\mu_0 \mu_r A}$
5. Voltage drop, $V = IR$	5. MMF drop, $= \phi \times S$
6. Current density, $\delta = \frac{I}{A}$ in $\frac{Amp.}{m^2}$	6. Flux density, $B = \frac{\phi}{A}$, $\frac{Weber}{m^2}$
7. Conductance, $G = \frac{1}{R}$	7. Permeance, $p = \frac{1}{S}$
DISIMILARITIES	
1. Current can flow 2. Current cannot flow through air 3. Current is reduced to zero after removal of supply 4. Resistance depends on temperature	1. Flux does not flow 2. Flux can pass through air 3. Some flux is present after removal of supply 4. Reluctance does not depend on temperature

B-H Curve, Hysteresis Loop

Hysteresis loop:

The graphical relation between magnetic flux density B and magnetizing force, H is known as hysteresis curve or hysteresis loop.

Consider a coil having N number of turns and length l .



$$\text{So, Magnetising force, } H = \frac{NI}{l} \propto I$$

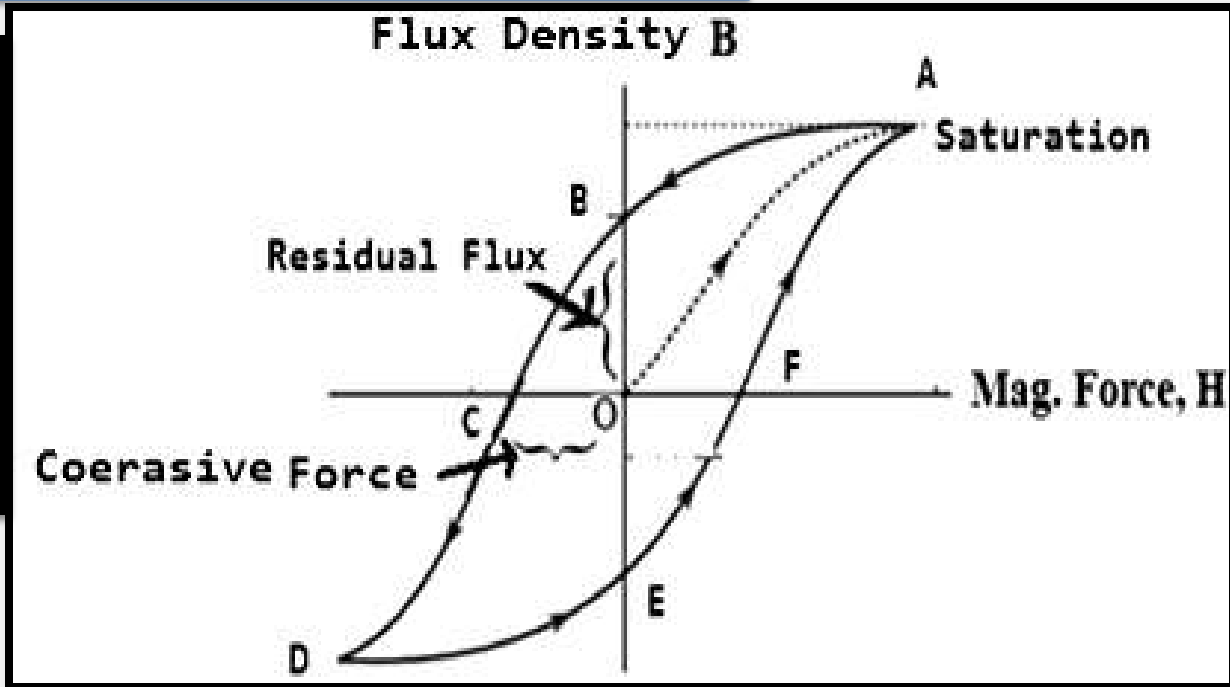
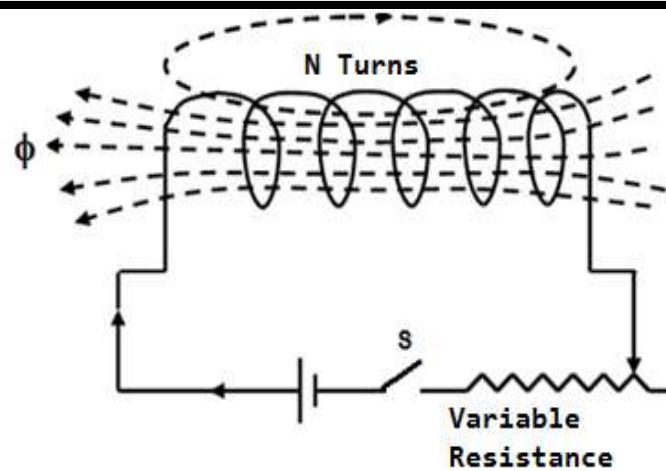
Because N and l remain constant and

$$\text{Flux density, } B = \frac{\Phi}{A} = \frac{\left(\frac{NI}{S}\right)}{A} \propto I$$

Because N, S and A remain constant

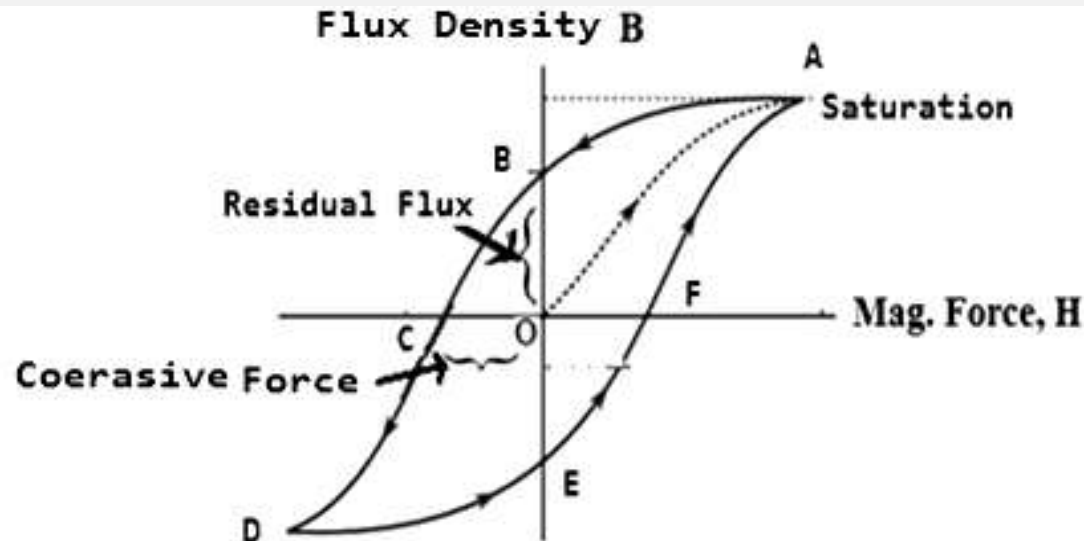
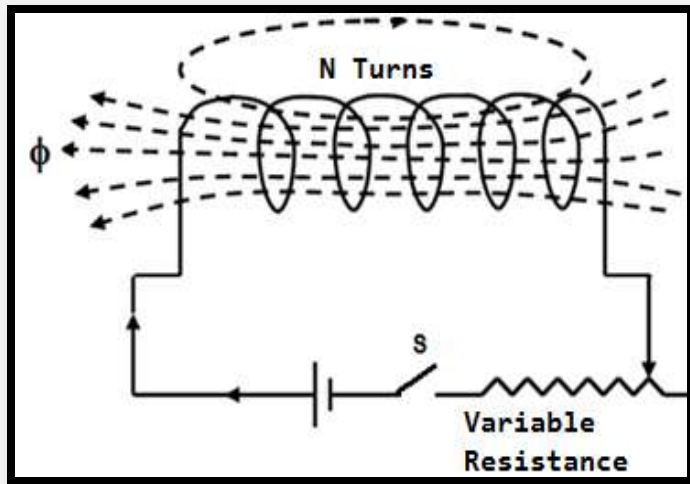
From the above equation, we can see that; both B and H proportional to current I , so both increase as current increase and decrease as current decrease. So following different points found during one cycle of current.

B-H characteristics



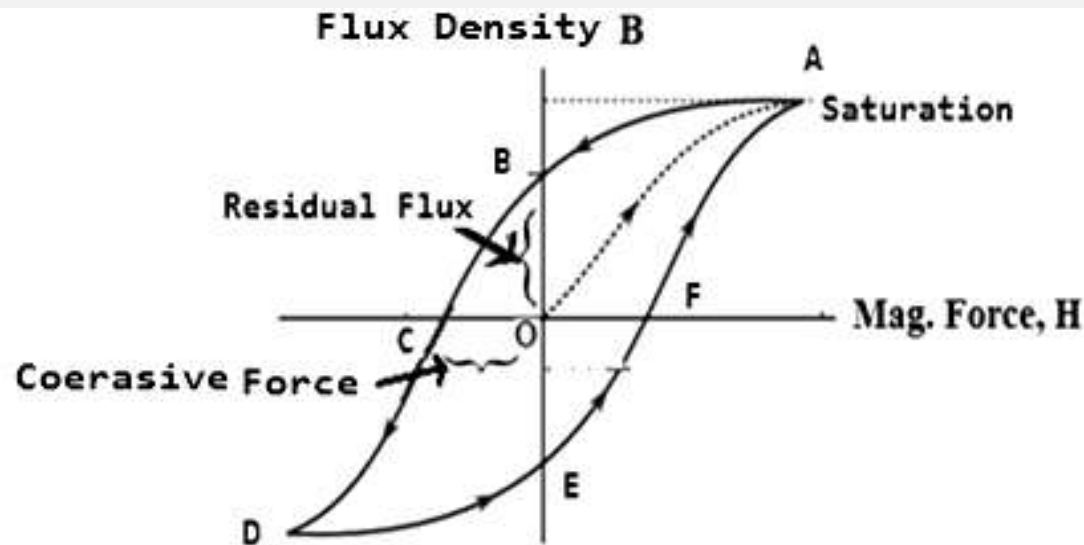
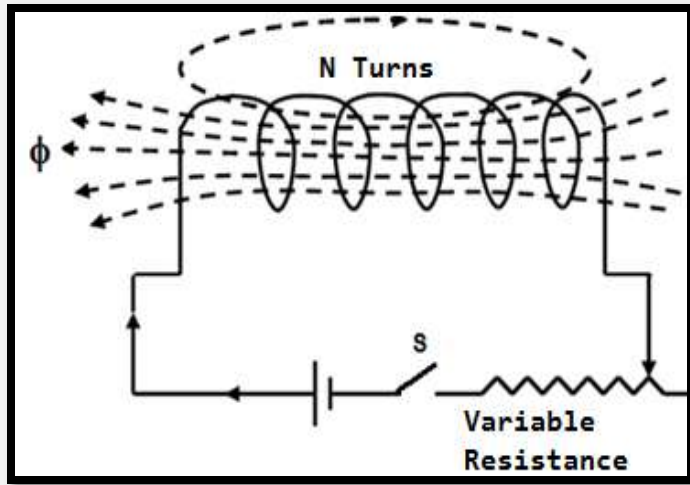
Point O-A: See in fig. When we increase current then both; B and H increase from point O to A but not linearly because it is found that B is lagging behind H . Now, at maximum current, coil goes to saturation at point A . So, at point,

$A: I = I_{\text{max. (Fwd.)}}, B = B_{\text{max. (Fwd.)}}$ and $H = H_{\text{max. (Fwd.)}}$.



Point A-B: Now, we decrease current by variable resistance so, both B and H decreases, but not the same way of AO because there will be some residual magnetic flux and thereby flux density B, so when current $I=0$, $H=0$ but $B=B_{\text{(Residual)}}$ and its value is equal to OB.

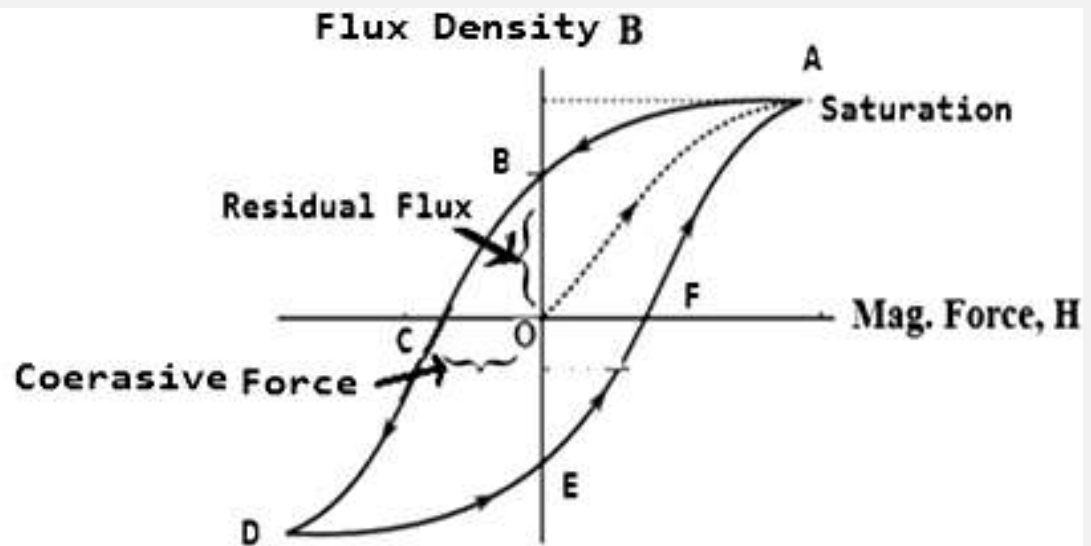
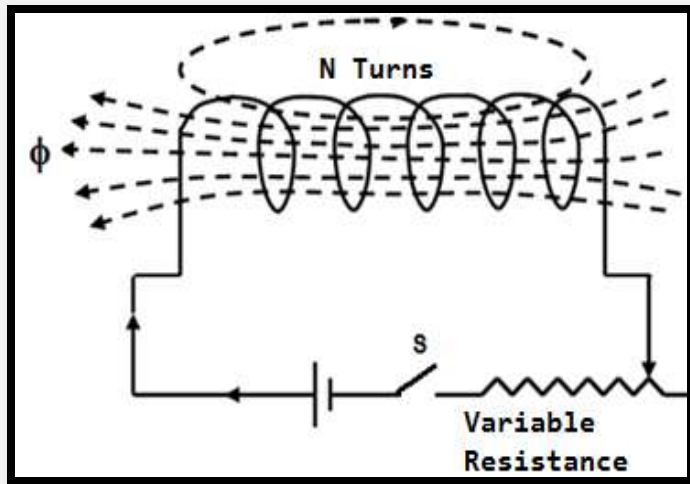
Point B-C: To bring the flux density to zero, the magnetizing force needed in the opposite direction is called the coercive force, for this reverse the battery terminal and increase current in the reverse direction, so flux density, $B=0$ but $H= OC = \text{Coercive force}$.



Point C-D: Further, we increase current in the reverse direction so both B and H increase in the reverse direction so, at the point D coil goes to saturation and at this point $I = I_{\text{max. (Rev)}}$ and $B = B_{\text{max. (Rev)}}$ and $H = H_{\text{max. (Rev)}}$.

Point D-E: Now, decrease current in the reverse direction so both B and H decrease in the reverse direction so, at point E, $I = 0_{\text{(Rev.)}}$ $H = 0$ but $B = B_{\text{Residual (Rev.)}}$ and its value is equal to EO.

Point E-F: Once again, to nullify this reverse residual magnetic flux, some extra energy required to opposite direction, for this change the battery terminal and increase current in the forward direction, so B becomes $B = 0$ and $H = OF$ is known as a coercive force.

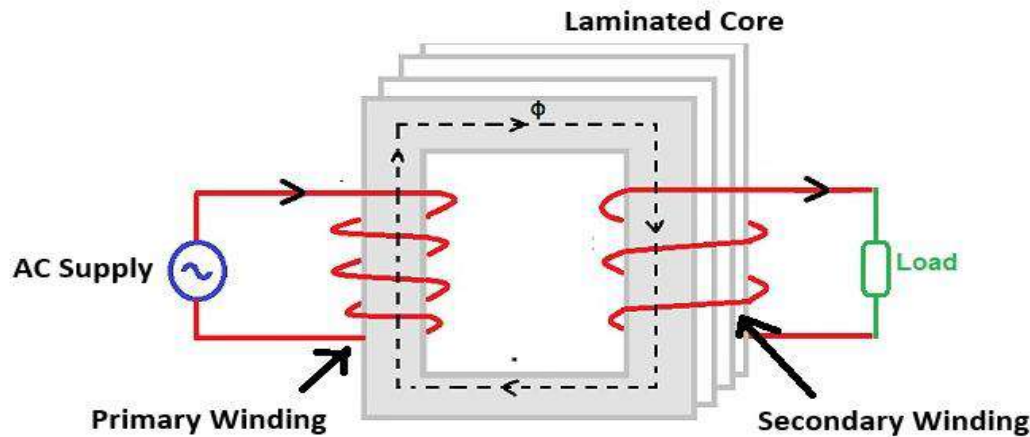


Point F-A: Further, we increase current in the forward direction so both B and H increase in forward direction and at point A once again coil goes to saturation. Hence the cycle repeated.

Hysteresis Loop:

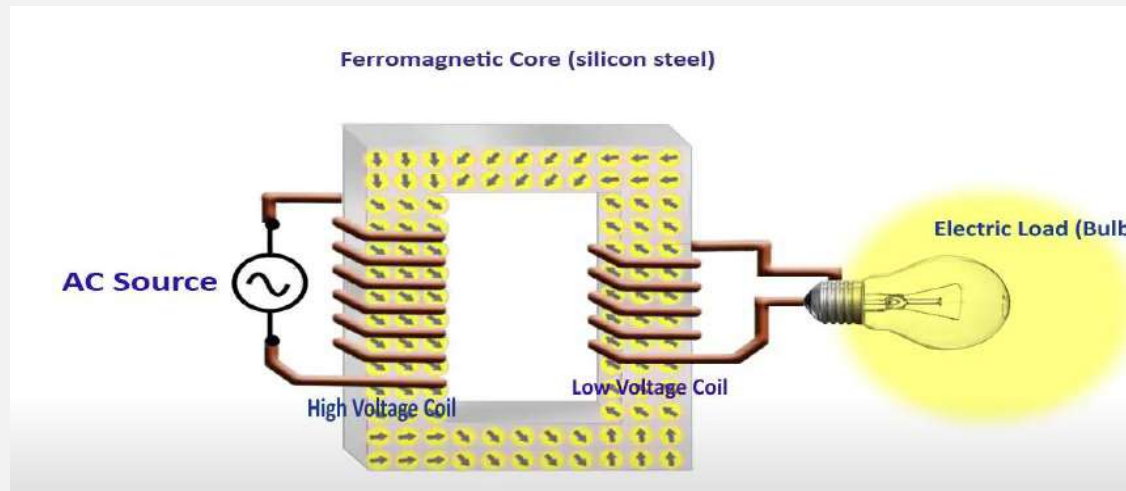
The curve A B C D E F A is called a hysteresis loop. The area of the loop represents energy waste per unit volume in the specimen due to the changing magnetic condition in one cycle.

Short note on transformer



1. Transformer is Static Electric Device.
2. It work Only With the AC Supply.
3. It use to Step up or step down the **voltage**
4. Transfers electric **power** from one circuit to another.
5. It works on the principle of faraday's electromagnetic induction (mutual induction)

Hysteresis Loss



- The transformer core material is made of tiny magnetic dipoles. A dipole means a small magnet inside the material that has a north pole and a south pole.
- When AC supply is given, these dipoles keep changing their alignment. This repeated alignment movement causes friction inside the material, producing heat in the core and energy is wasted which is called hysteresis loss.
- It can be reduce by selecting high permeability material like CRGO, Silicon steel etc.

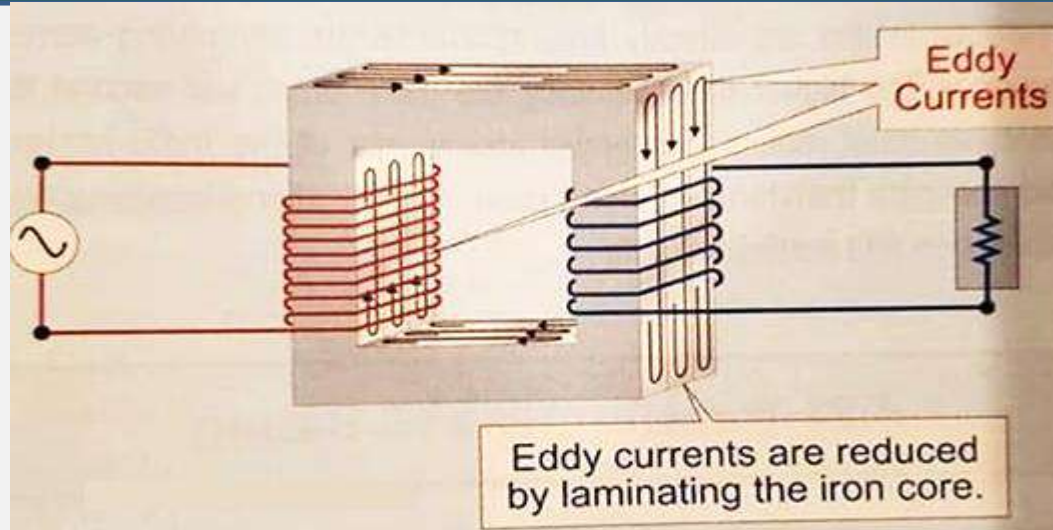
Formula:

$$P_h = \eta B_{max}^{1.6} f V$$

Where

- η = Steinmetz constant (depends on material)
- B_{max} = maximum flux density
- f = frequency of AC
- V = volume of the core

Eddy current losses



When AC Supply given to transformer, Alternating magnetic flux produced in core, causes emf induction inside the core, This EMF produces small loops of current inside the core – called eddy currents. This eddy currents produce heat in the core and energy is wasted which is called eddy current loss.

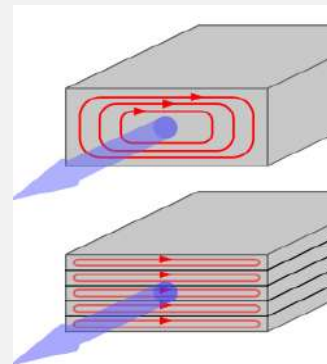
Formula:

$$P_e = K_e B_{max}^2 f^2 t^2 V$$

Where

- K_e = constant
- B_{max} = maximum flux density
- f = frequency
- t = thickness of core lamination
- V = volume of the core

- It can be reduce by Laminating core, Instead of solid,
- Laminations are insulated from each other to stop large loops of current.



Solid Core

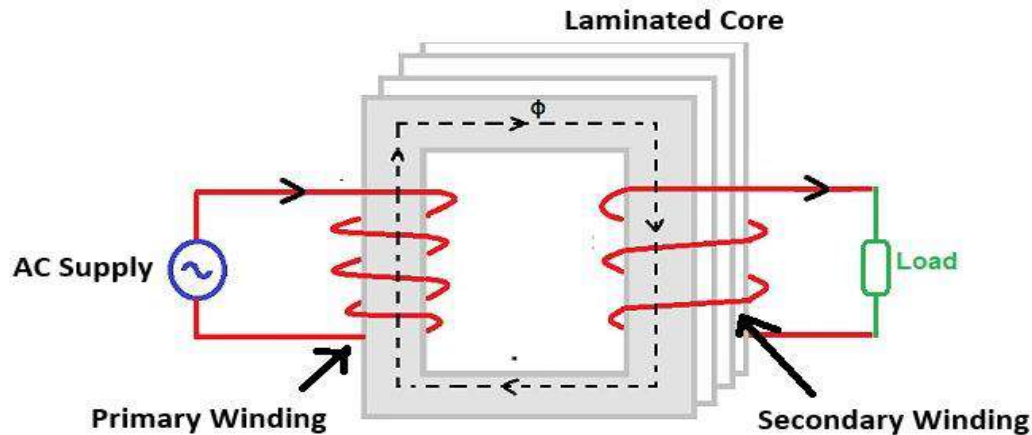
Laminated Core

7. Construction of single-phase Core type transformers

Draw the construction of single phase core type transformer

04 Marks

March-2023



Primary Winding

- Made from copper or aluminum.
- It connected to the AC voltage source.
- When current flows, it produces magnetic flux around it.

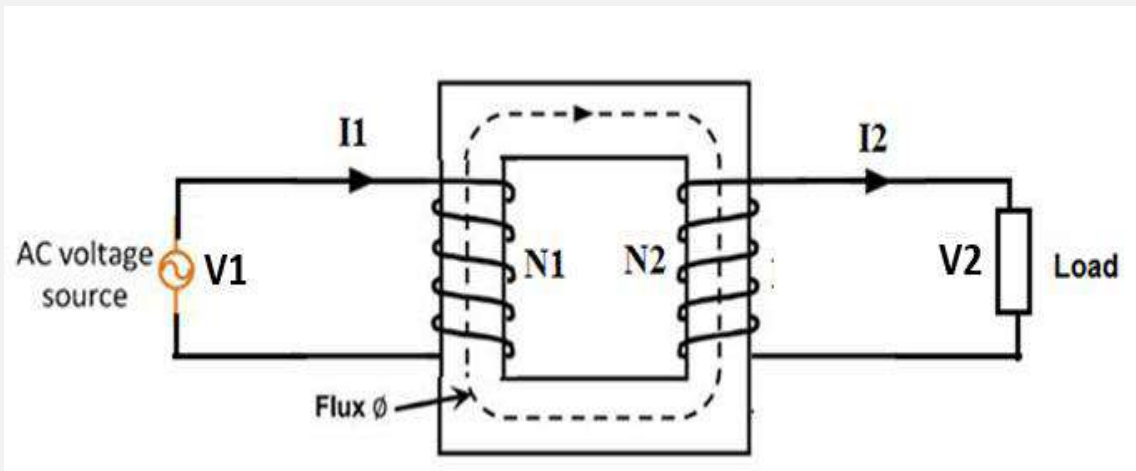
Laminated Core

- Made of thin magnetic laminations to reduce eddy current loss.
- Keeps primary and secondary windings electrically isolated.
- Carries flux from primary to secondary.

Secondary Winding

- Made from copper or aluminum.
- It connected to the load.
- When flux links with it, an EMF is induced in the winding.

Explain working principle of single-phase transformers



- AC supply to the primary winding, Current flow and produces MMF.
- MMF creates alternating magnetic flux in the core.
- Flux links to the secondary winding through the core.
- By Faraday's law, this flux induces an EMF in the secondary.
- The EMF causes current to flow in the load,
- Secondary voltage changes (step-up or step-down) based on the turns ratio with the primary.

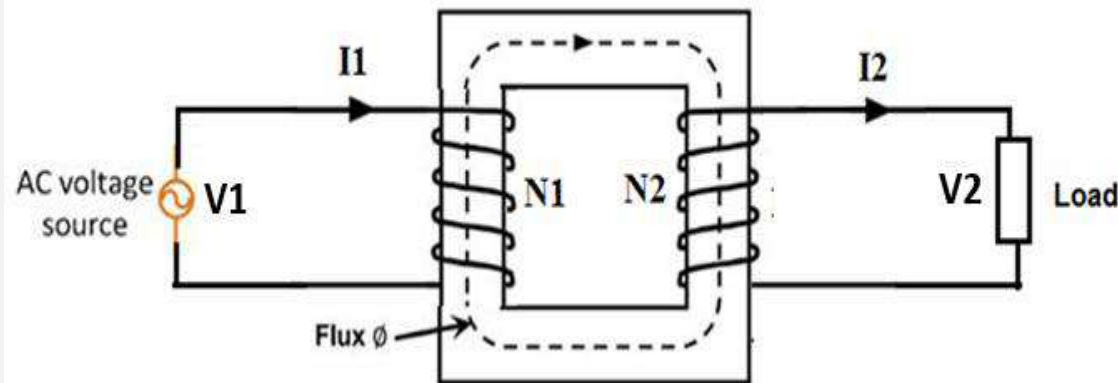
Formula:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \quad \text{and} \quad \frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Where

- V_s, V_p → Secondary & primary voltages
- N_s, N_p → Number of turns in secondary & primary
- I_s, I_p → Secondary & primary currents

[For easy Remember] Explain working of single-phase transformers

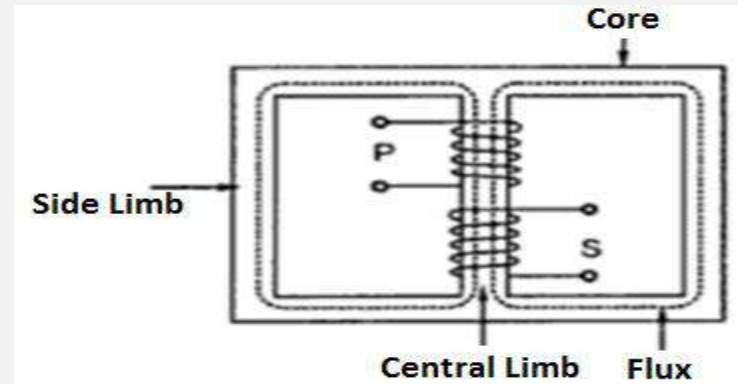
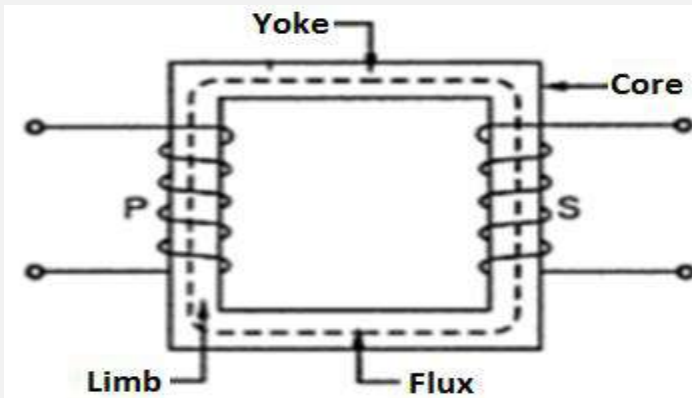


Here's the very simple, step-by-step version of the transformer working:

1. **AC supply** → Primary winding
2. Primary current → **MMF**
3. MMF → **Alternating magnetic flux** in core
4. Flux → **Links** to secondary winding
5. Flux change → **EMF induced** (Faraday's Law)
6. EMF → **Current** in load
7. Voltage changes → **Step-up or Step-down** based on turns ratio 

Explain with diagram construction of core type and shell type transformer-04 Marks

GTU - Nov-2020 May-2019, NOV-2020, July-2024, Jun-2025



Core Type

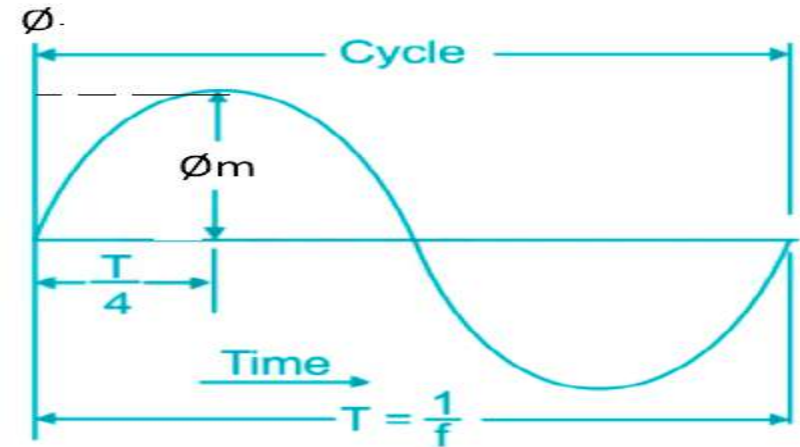
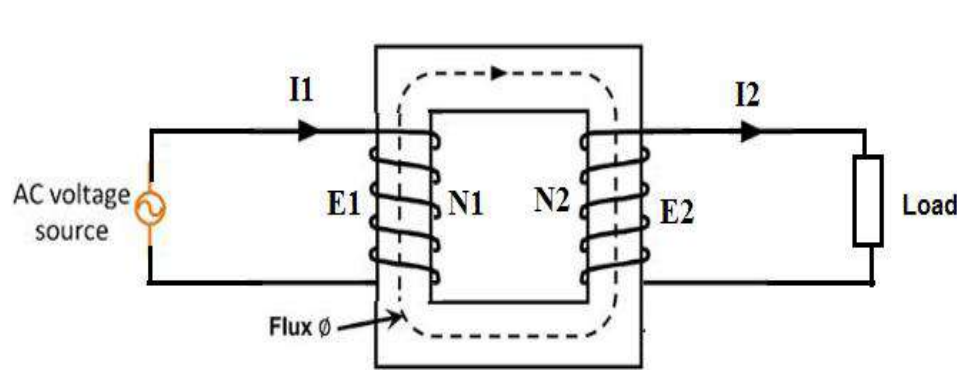
1. Winding surrounds the core
2. Two Limbs
3. Windings placed on side limbs
4. Flux shared equally in limbs.
5. Used for medium & large ratings

Shell Type

1. Core surrounds the winding
2. Three Limbs
3. Winding placed on Central limb
4. Central limb carries all flux, side limbs carry half
5. Used for small ratings

Derive the E.M.F. equation of a single phase transformer – 04 to 7 Marks

GTU- June-2019, Jan-2020, NOV-2020, March-2023, July-2024, June-2025



Let;

N_1 & N_2 windings turns, Φ_m = Maximum flux, f = Frequency, E_1 & E_2 = Induced emf

1. Average rate of change of flux over the first quarter cycle:

$$\frac{\phi_m - 0}{T/4} = \frac{\phi_m}{1/(4f)} = 4f\phi_m$$

2. Average induced emf (for 1 turn) = average rate of change of flux:

$$E_{avg} = 4f\phi_m$$

3. Form factor for a sinusoidal wave:

$$\text{Form Factor} = \frac{\text{RMS value}}{\text{Average value}} = 1.11$$

4. RMS emf for 1 turn:

$$E_{\text{rms}} = E_{\text{avg}} \times 1.11 = 4f\phi_m \times 1.11 = 4.44f\phi_m$$

5. For N turns in the winding:

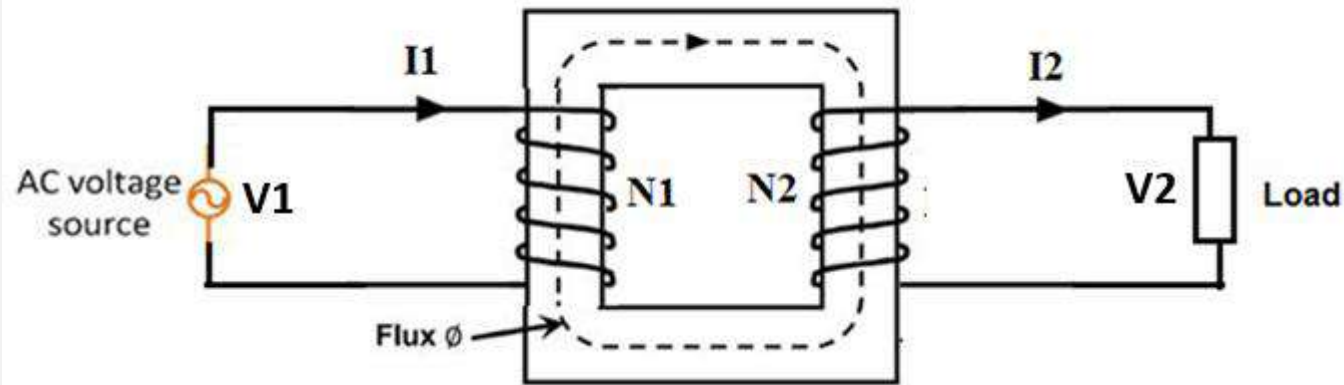
$$E_{\text{rms}} = 4.44fN\phi_m$$

Now, primary winding having N_1 turns and secondary winding having N_2 turns then final EMF equation will be:

$$E_1 = 4.44 f N_1 \phi_m$$

$$E_2 = 4.44 f N_2 \phi_m$$

8. Voltage and current ratios, Losses



The relationship between the number of turns and the voltage is given by,

$$N_1/N_2 = V_1/V_2 = I_2/I_1, \text{ Where,}$$

N_1 = Primary winding turns, N_2 = Secondary winding turns

V_1 = Voltage of primary, V_2 = Voltage at secondary

I_1 = current flow in primary winding, I_2 = current flow in secondary winding

Losses

◆ 1. Copper Loss

✓ What is it?

Loss of energy in the **transformer windings** due to their resistance.

📖 When does it happen?

Only when **current flows** through the windings.

📐 Formula:

$$\text{Copper Loss} = I^2 R$$

- I = Current in winding
- R = Resistance of winding

◆ 2. Iron Loss (also called Core Loss)

✓ What is it?

Loss of energy in the **transformer core** (iron part) due to changing magnetic field.

📖 When does it happen?

Happens **all the time**, even with **no load**, as long as the transformer is ON.

📦 Iron loss has two parts:

- **Hysteresis loss** – due to magnetizing and demagnetizing the core
- **Eddy current loss** – small currents induced inside the iron core

9. Definition of regulation and efficiency

Definition of regulation

Voltage regulation tells us how much the output voltage of a transformer drops when we connect a load (like a fan or light).

It is the percentage change in the secondary voltage from no-load to full-load, at constant primary supply voltage.

Mathematically, it is expressed as:

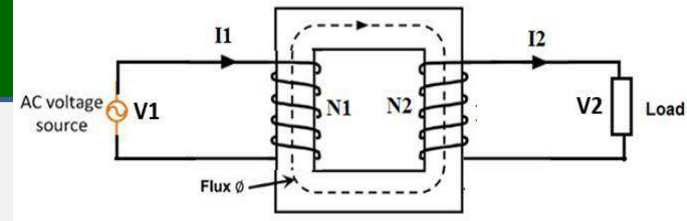
$$\text{Voltage Regulation} = \left(\frac{V_{\text{no-load}} - V_{\text{full-load}}}{V_{\text{full-load}}} \right) \times 100\%$$

Definition of Efficiency

The Efficiency of the transformer is defined as the ratio of useful output power to the input power.

$$\eta = \frac{\text{output power}}{\text{input power}} = \frac{\text{output power}}{\text{output power} + \text{losses}}$$

$$\eta = \frac{\text{output power}}{\text{output power} + \text{iron losses} + \text{copper losses}}$$



10. Ideal and practical transformer.

Ideal	practical
Windings has no resistance	Windings has resistance
No leakage flux	Leakage flux is present
No losses (no copper loss, no core loss)	Has copper loss and Iron loss
100% efficient	95-98 % efficiency
No voltage drop	Voltage drop

MCQ

An transformer work on principle of

- (a)** self-induction.
- (b)** mutual induction.
- (c)** coulomb law.
- (d)** ampere law.

.

ANSWER: b

MCQ

If ϕ_m is the maximum value of flux in the core then the average rate of change of flux is

- a. $f\phi_m$
- b. $2 f\phi_m$
- c. $4 f\phi_m$
- d. $4.44 f\phi_m$

ANSWER: C

MCQ

The primary side and secondary side of a transformer are

- (a)** electrically linked.
- (b)** magnetically linked.
- (c)** conductivity linked.
- (d)** all of the above.

ANSWER: b

MCQ

A transformer is an electric device which work on

- (a) AC.**
- (b) DC.**
- (c) both AC and DC.**
- (d) none of the above.**

ANSWER: a

MCQ

The main function of a transformer is to change

- (a) power.**
- (b) frequency.**
- (c) voltage level.**
- (d) power factor.**

ANSWER: c

4. The main purpose of transformer cores lamination is to

- (a)** reduce copper loss.
- (b)** reduce eddy current loss.
- (c)** reduce cost.
- (d)** reduce hysteresis loss.

ANSWER: b

MCQ

Emf equation of transformer is

- (a) $4\Phi fn$**
- (b) $4.11 \Phi fn$**
- (c) $4.33\Phi FN$**
- (d) $4.44 \Phi FN$**

ANSWER: d

MCQ

Transformer is a device which:_____.

- A. Can step up or step down the level of voltage.
- B. Work through on electric induction.
- C. Its Working without changing the Power.
- D. All of these

ANSWER: D

GTU Paper

Sr	Question	Mark	GTU Exam
1	Explain magnetic hysteresis OR Explain B-H Curve and Hysteresis Loop in detail	03 07	Nov-2020, June-2025 Jan-025
2	Define flux, flux density, reluctance and permeability Explain the terms:- 1. Flux density 2. Reluctance 3. Permeability 4. field strength.	04	Jan-025 June-2025
3	Describe concept of MMF.	03	Nov-2020 Jan-2025
4	Briefly describe the operating principle of a transformer. OR Explain working principle of single phase Transformer.	04	Jan-2019 Jan-2020 Aug-2022 Jun-2025
5	Derive the E.M.F. equation of a single phase transformer	04	June-2019, Jan-2020 NOV-2020, March-2023 July-2024, June-2025
6	Explain with diagram construction of core type and shell type transformer. OR Compare core type and shell type transformers.	04	May-2019 NOV-2020, July-2024 Jun-2025
7	Draw the construction of single phase core type transformer	04	March-2023
8	State comparison between practical transformer and ideal transformer	07	Jun-2025

THE END

S.G. Prajapati 

BEE-
Ch-4: Electrical Machines



Dr. S. G. Prajapati

Saturday, December 6, 2025

CONTENTS

- Fundamentals of Electrical Machines
- Generation of rotating magnetic fields,
- Construction and working of Single-phase induction motors
(Split phase, Capacitor start, Permanent split capacitor, Capacitor start/capacitor run).
- Single phase induction motor applications: pumps, refrigerators, fans, compressors, and portable drills.
- Construction and working of brushless DC motors and its application: Electrical Vehicle, washing machines, Blowers, Computers/Laptops



Basic Components

Stator – The stationary part of the machine (contains windings).

Rotor – The rotating part (connected to the shaft).

Windings – Coils of wire that carry current.

Air gap – Small clearance between stator and rotor.



Working Principles

Electromagnetic Induction (Faraday's Law): A voltage is induced in a conductor when it is exposed to a changing magnetic field.

Lorentz Force: A current-carrying conductor in a magnetic field experiences a force.

Torque Production: In motors, torque is produced by the interaction of magnetic fields in the stator and rotor.

Generation of rotating magnetic fields

Sr.	Question	Mark	Exam
1	<u>Explain the term rotating magnetic field with proper diagrams in case of a three phase induction motor.</u> OR Discuss how the rotating magnetic field is produced in three phase induction motor. OR Explain Generation of Rotating Magnetic Field in 3-phase Induction Motor	07 04 07	Jan-2019 May 2019, Mrch-2023 Jan -2020 Nov -2020

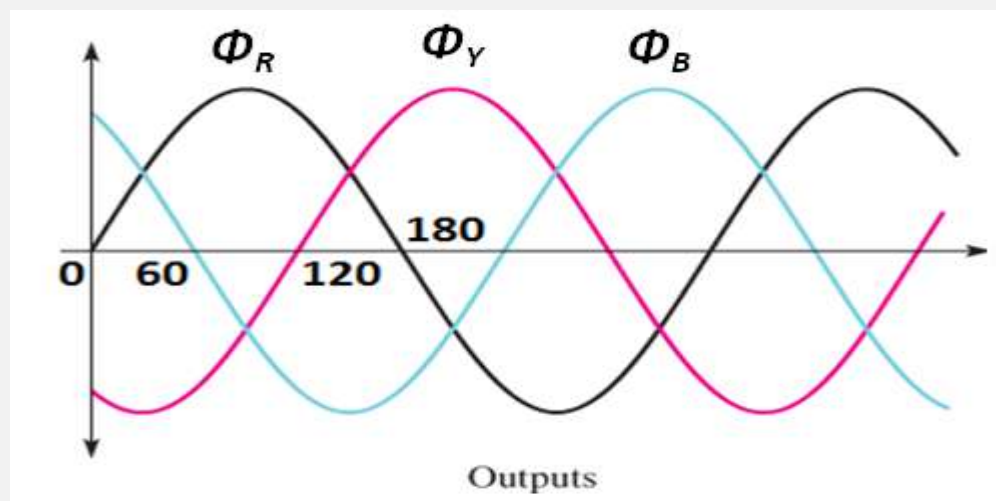
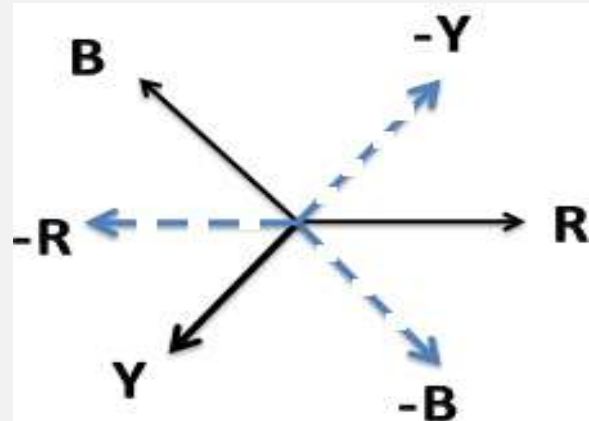
Consider 3 coils are connected at 120° apart, then the mathematical equation for the instantaneous values of the fluxes Φ_1 , Φ_2 and Φ_3 can be given as,

$$\Phi_R = \Phi_m \sin(\theta)$$

$$\Phi_Y = \Phi_m \sin(\theta - 120)$$

$$\Phi_B = \Phi_m \sin(\theta - 240)$$

The magnitude of each flux is Φ_m .



(1) when $\Theta = 0^\circ$

$$\phi_R = \phi_m \sin(0) = 0$$

$$\begin{aligned}\phi_Y &= \phi_m \sin(0 - 120^\circ) \\ &= -0.866\phi_m\end{aligned}$$

$$\begin{aligned}\phi_B &= \phi_m \sin(0 - 240^\circ) \\ &= 0.866\phi_m\end{aligned}$$

$$\Phi_R = \Phi_m \sin(\theta)$$

$$\Phi_Y = \Phi_m \sin(\theta - 120^\circ)$$

$$\Phi_B = \Phi_m \sin(\theta - 240^\circ)$$

$\Theta = 60^\circ$

$$\begin{aligned}\phi_R &= \phi_m \sin(60^\circ - 0) \\ &= 0.866\phi_m\end{aligned}$$

$$\begin{aligned}\phi_Y &= \phi_m \sin(60^\circ - 120^\circ) \\ &= -0.866\phi_m\end{aligned}$$

$$\phi_B = \phi_m \sin(60^\circ - 240^\circ) = 0$$

$$-\Phi_R$$

$\Theta = 180^\circ$

$$\phi_R = \phi_m \sin(180^\circ - 0) = 0$$

$$\begin{aligned}\phi_Y &= \phi_m \sin(180^\circ - 120^\circ) \\ &= 0.866\phi_m\end{aligned}$$

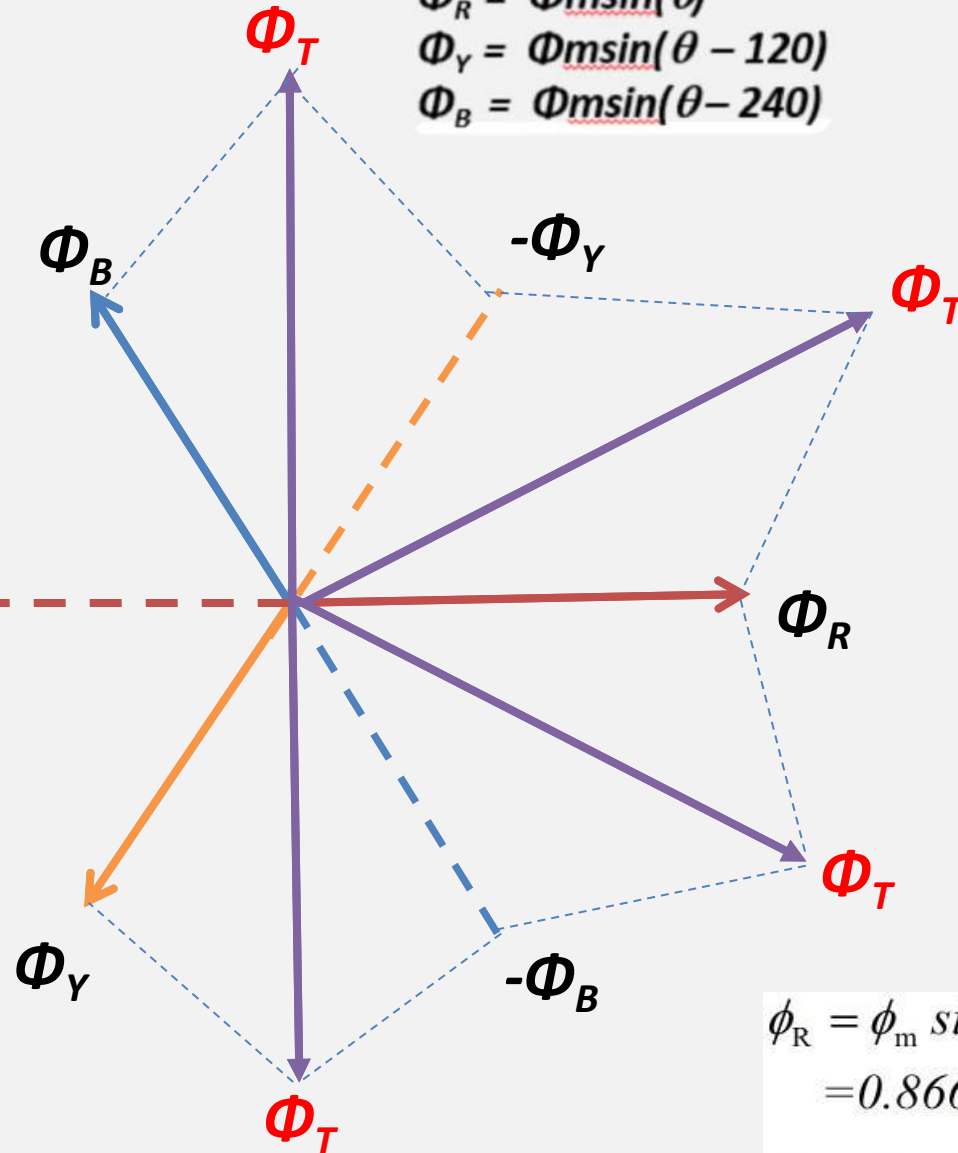
$$\begin{aligned}\phi_B &= \phi_m \sin(180^\circ - 240^\circ) \\ &= -0.866\phi_m\end{aligned}$$

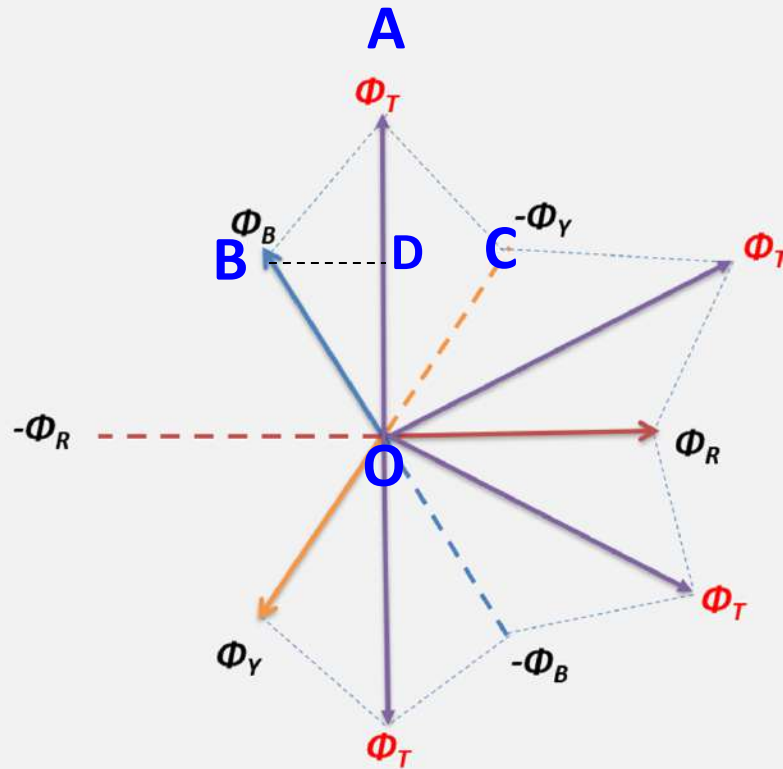
$\Theta = 120^\circ$

$$\begin{aligned}\phi_R &= \phi_m \sin(120^\circ - 0) \\ &= 0.866\phi_m\end{aligned}$$

$$\phi_Y = \phi_m \sin(120^\circ - 120^\circ) = 0$$

$$\begin{aligned}\phi_B &= \phi_m \sin(120^\circ - 240^\circ) \\ &= -0.866\phi_m\end{aligned}$$





Case-1: when $\Theta = 0^\circ$

$$\phi_R = \phi_m \sin(0) = 0$$

$$\begin{aligned}\phi_Y &= \phi_m \sin(0 - 120^\circ) \\ &= -0.866\phi_m\end{aligned}$$

$$\begin{aligned}\phi_B &= \phi_m \sin(0 - 240^\circ) \\ &= 0.866\phi_m\end{aligned}$$

To take the value of Φ_T , drawn perpendicular Line (BD) from B on ' Φ_T '.

$$OD = \frac{\phi_T}{2} \quad \& \quad OB = \phi_B = 0.866 \phi_m$$

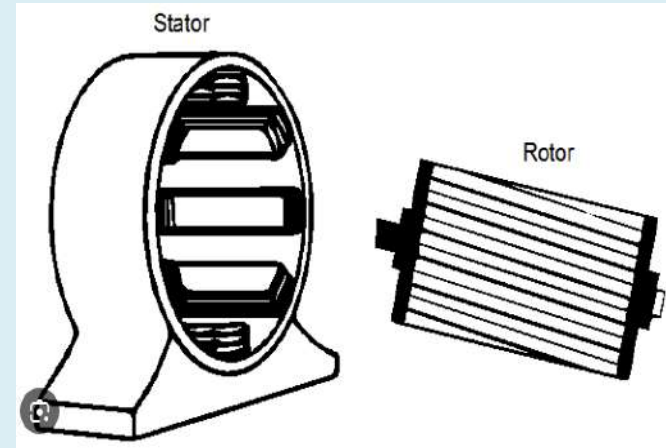
$$\text{In } \triangle OBD, \angle BOD = 30^\circ, \cos 30^\circ = \frac{OD}{OB} = \frac{\frac{\phi_T}{2}}{0.866\phi_m}$$

$$\therefore \phi_T = 2 \times 0.866 \phi_m \times \cos 30^\circ = 2 \times 0.866 \phi_m \times \frac{\sqrt{3}}{2} = 1.5 \phi_m$$

$$\therefore \phi_T = 1.5 \phi_m$$

So the magnitude of total flux is 1.5 times the maximum value of an individual flux.

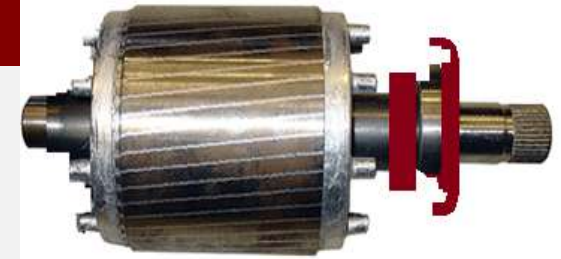
In all the cases, its value found same.



Construction and working of Single-phase induction motors



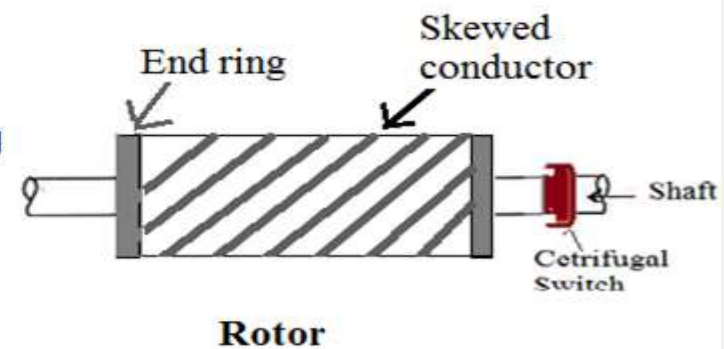
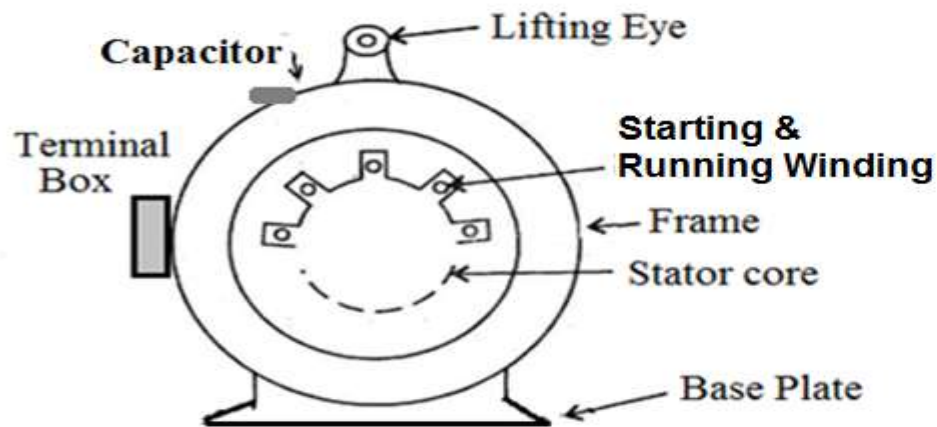
1-phase Induction motor



Outer Parts of Motor

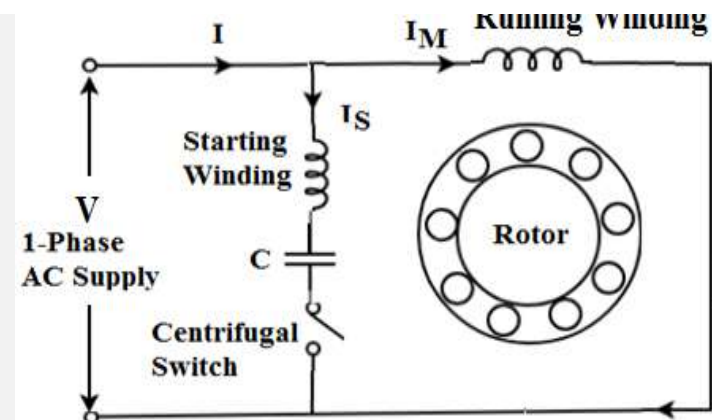
Stator Core & winding

Rotor



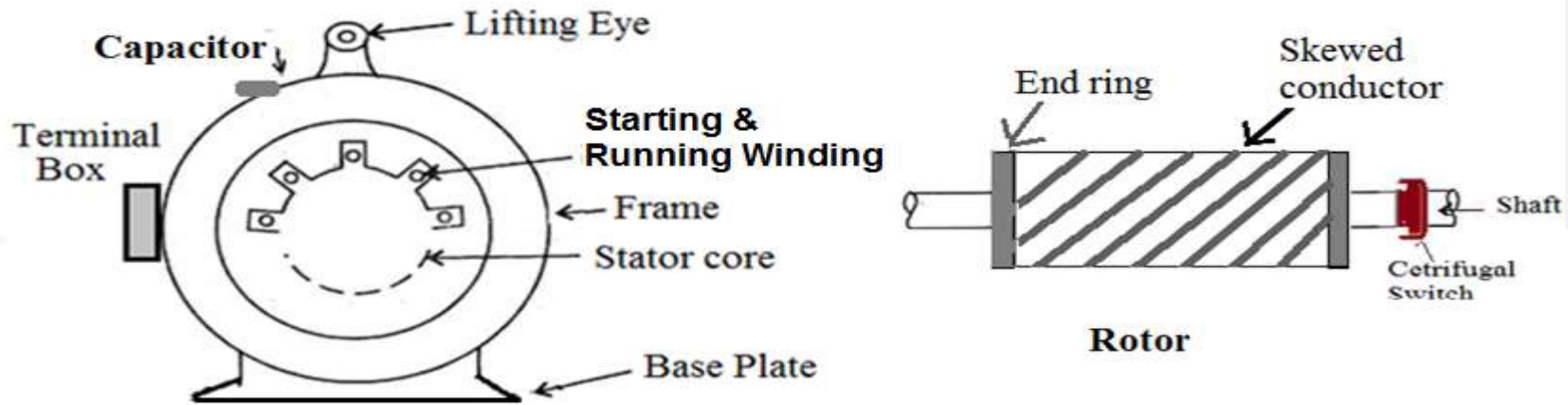
Rotor

Construction



Circuit diagram for Working

Construction of 1-Phase Induction Motor



Terminal Box: For connections

Capacitor: It use to provide starting torque and in running it use for power factor improvement.

Lifting Eye: For lift the motor

Frame:- For Support of stator core and winding. It made from cast iron material.

Base Plate: For provide base of motor for installation.

Stator:

- Stator is a Stationary Part.
- Stator core is made of large number of laminations to reduce eddy current loss.
- Starting & Running Winding is placed in stator core.

Rotor:

- Rotor is Rotating Part of motor.
- It made from skewed aluminum bare conductor & shorted on both sides by end rings.
- It slide skew to reduce humming .

Centrifugal switch : It connected with shaft of motor and Use to disconnect starting winding when motor pickup 80 % of its rated speed.

Shaft: It is rotating element use to drive mechanical machine. It is made up of steel.

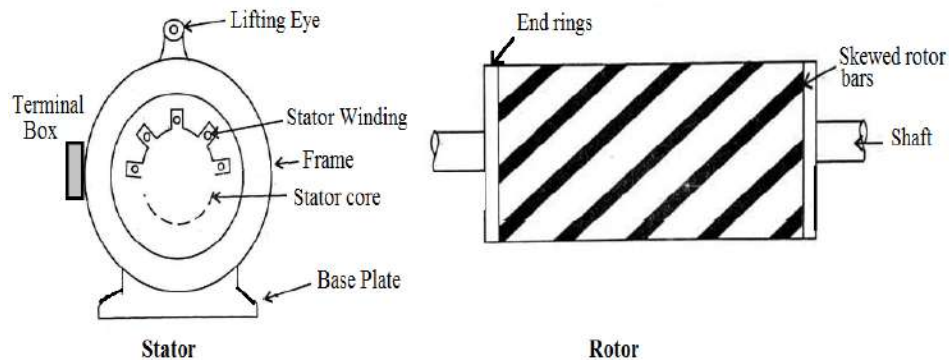
Q-3

Give a comparison between squirrel cage induction motor and wound rotor induction motor

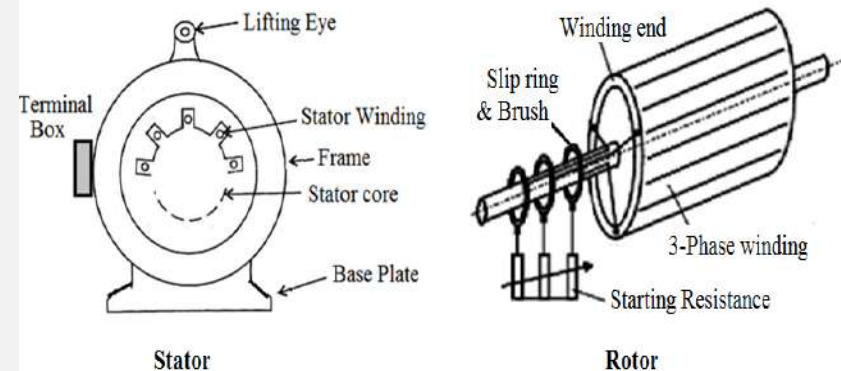
04

2019,2022,2024

Squirrel Cage Induction Motor



Slip Ring or wound rotor I.M



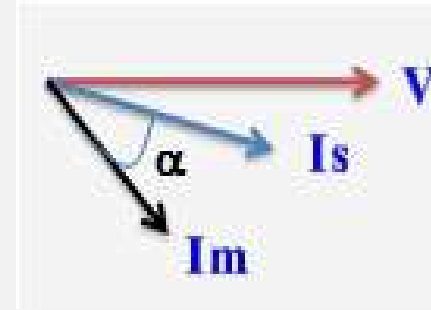
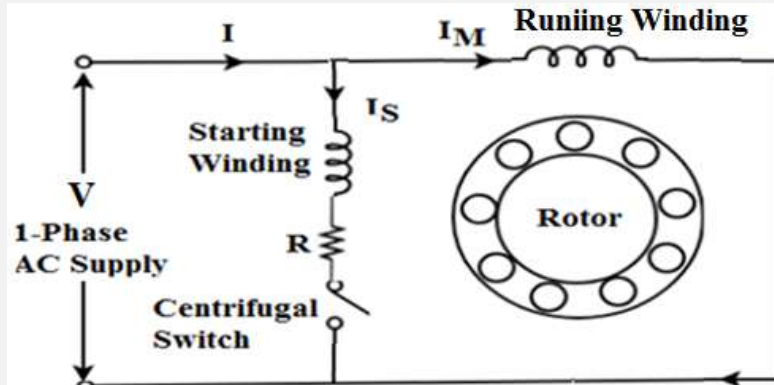
1. Stator having 3-phase winding
2. Rotor is squirrel cage type & having bar conductor shorted by end rings
3. Simple Construction
4. Low cost
5. No resistance added externally in rotor
6. Low starting torque
7. Use for fan, blower, water pumps etc.

1. Stator having 3-phase winding
2. Rotor is wound type & winding Connected with slip ring
3. Complicated Construction
4. High Cost
5. Resistance added externally in rotor
6. High starting torque
7. Use for lift, crane, compressors etc..

Q	Mention the types of single phase induction motor. Explain any one of them	04	GTU-Jan-2019
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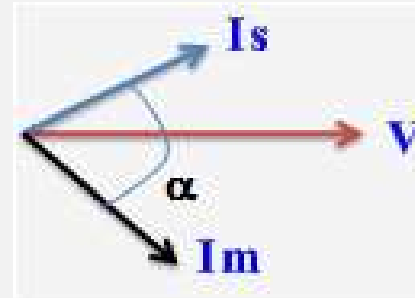
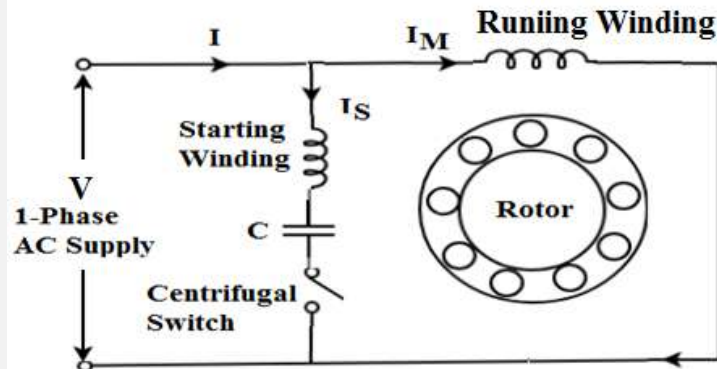
- 1. Resistance Start Motor**
- 2. Capacitor start motor**
- 3. Permanent Split Capacitor Motor (PSC)**
- 4. Capacitor start capacitor Run motor**

Working of 1-Phase Resistance Start Motor - for Exam



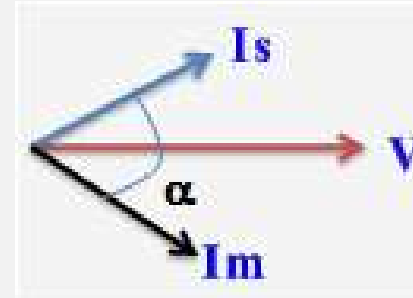
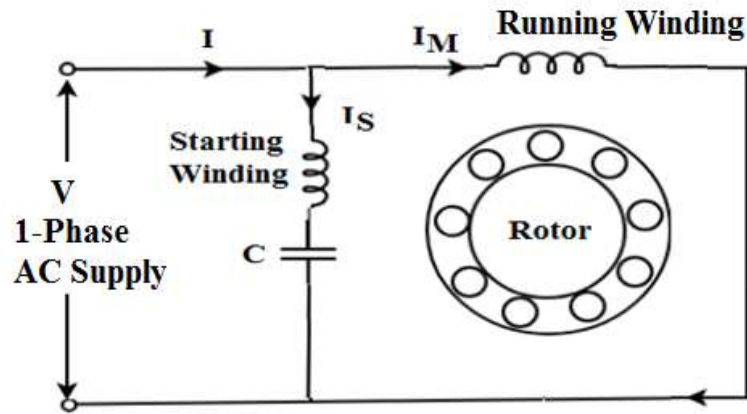
- **1-Phase supply given to stator**, rotating magnetic flux produce in stator by phase split- starting & running winding.
- This stator flux links with rotor through air gap, it induces emf in the rotor circuit, due to this, current starts flowing in the short-circuited rotor conductors, now as per principle, current carrying conductor (rotor) placed in magnetic field it experience a force, due to this force, torque produced on rotor and hence **motor starts to run**.
- When, motor pickup 70 to 80 % of its rated speed, the starting winding is automatically disconnected by centrifugal switch and now the motor run only with running winding at rated speed.
- Starting torque, **$T_s = I_m I_s \sin \alpha$** , Here, **$\alpha$** is about **$20^\circ$ to 30°** . So it has low starting torque.
- Direction of rotation of motor reversed by change in either main winding or starting winding connections. (**not both**).

Working of 1-Phase capacitor Start Motor - for Exam



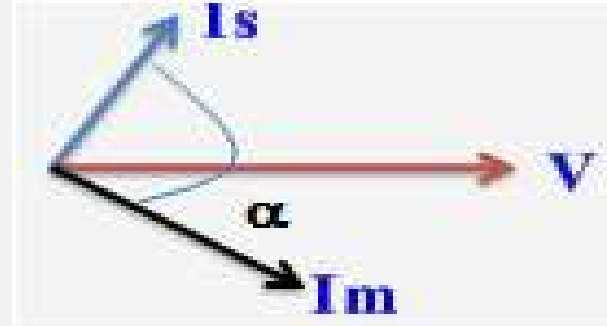
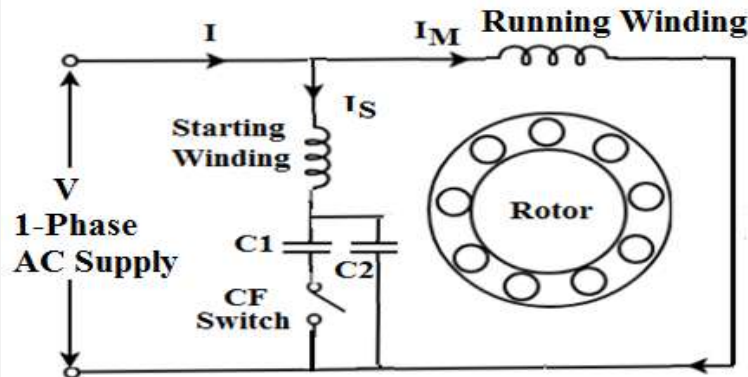
- **1-Phase supply given to stator**, rotating magnetic flux produce in stator by phase split- starting & running winding.
- This stator flux links with rotor through air gap, it induces emf in the rotor circuit, due to this, current starts flowing in the short-circuited rotor conductors, now as per principle, current carrying conductor (rotor) placed in magnetic field it experience a force, due to this force, torque produced on rotor and hence **motor starts to run**.
- When, motor pickup 70 to 80 % of its rated speed, the starting winding & capacitor automatically disconnected by centrifugal switch and now the motor run only with running winding at rated speed.
- Starting torque, **$T_s = I_m I_s \sin \alpha$** , Here, **α** is about **60° to 70°**. So it has good starting torque .
- Direction of rotation of motor reversed by change in either main winding or starting winding connections. **(not both)**.

Working of 1-Phase permanent Split capacitor motor



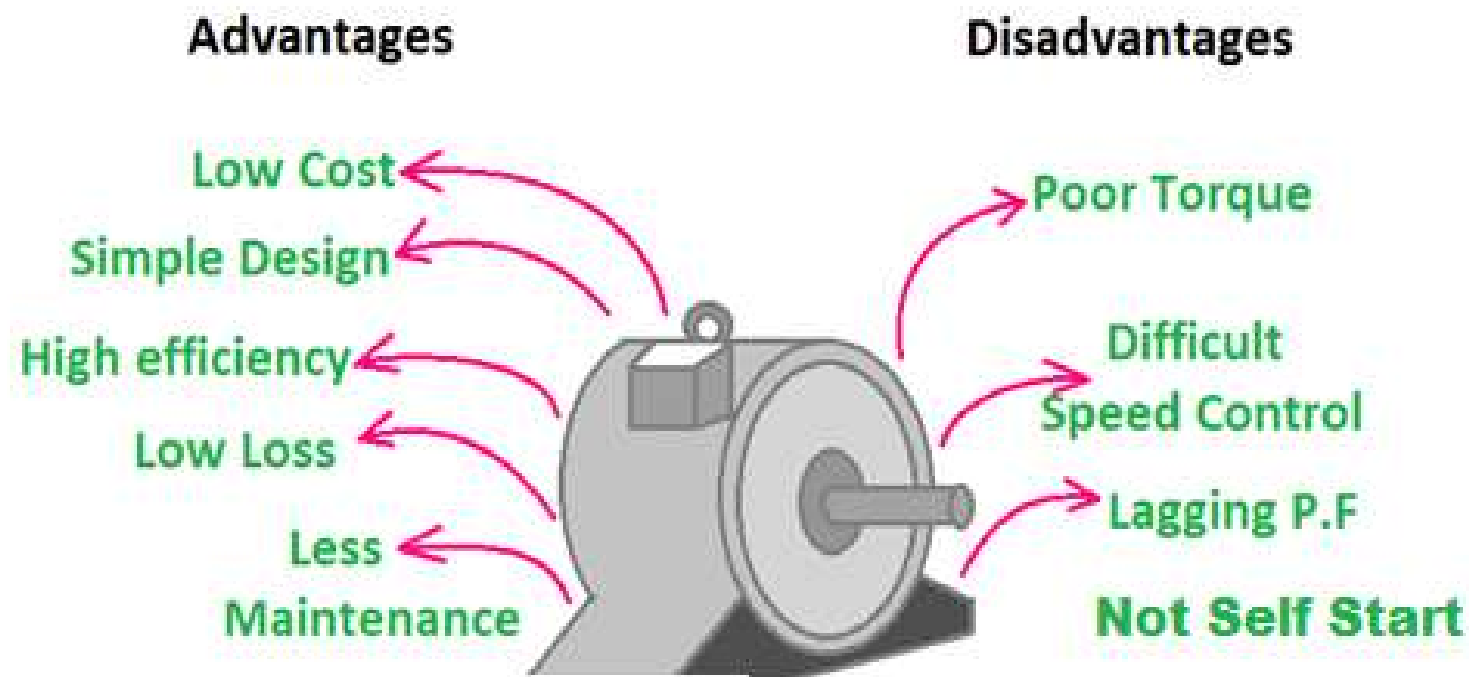
- **1-Phase supply given to stator**, rotating magnetic flux produce in stator by phase split- starting & running winding.
- This stator flux links with rotor through air gap, it induces emf in the rotor circuit, due to this, current starts flowing in the short-circuited rotor conductors, now as per principle, current carrying conductor (rotor) placed in magnetic field it experience a force, due to this force, torque produced on rotor and hence **motor starts to run** at rated speed.
- Starting torque, **$T_s = I_m I_s \sin \alpha$** , Here, α is about **50° to 60°** . So it has medium starting torque .
- Direction of rotation of motor reversed by change in either main winding or starting winding connections. **(not both)**.

Working of capacitor start capacitor run motor



- **1-Phase supply given to stator**, rotating magnetic flux produce in stator by phase split- starting & running winding.
- This stator flux links with rotor through air gap, it induces emf in the rotor circuit, due to this, current starts flowing in the short-circuited rotor conductors, now as per principle, current carrying conductor (rotor) placed in magnetic field it experience a force, due to this force, torque produced on rotor and hence **motor starts to run**.
- **When motor pickup 70 to 80 % of its rated speed, capacitor C1 is automatically disconnected by CF Switch and Now the motor run with Starting winding, capacitor C2 and running winding at rated speed.**
- Starting torque, **$T_s = I_m I_s \sin \alpha$** , Here, **$\alpha$** is about **$85^\circ$ to 90°** . So it has high starting torque .**Capacitor C1 selected high value and C2 Low value.**
- Direction of rotation of motor reversed by change in either main winding or starting winding connections. **(not both)**.

Mention Merits and Demerits of Single Phase Induction Motor



**Single phase induction motor applications:
pumps, refrigerators,
fans, compressors, and portable drills.**

✓ **Common Applications:**

1. Pumps

- Used in water pumps for domestic and agricultural purposes.

2. Refrigerators

- Powers the compressor motor inside refrigerators and freezers.

3. Fans

- Drives ceiling fans, exhaust fans, and table fans.

4. Compressors

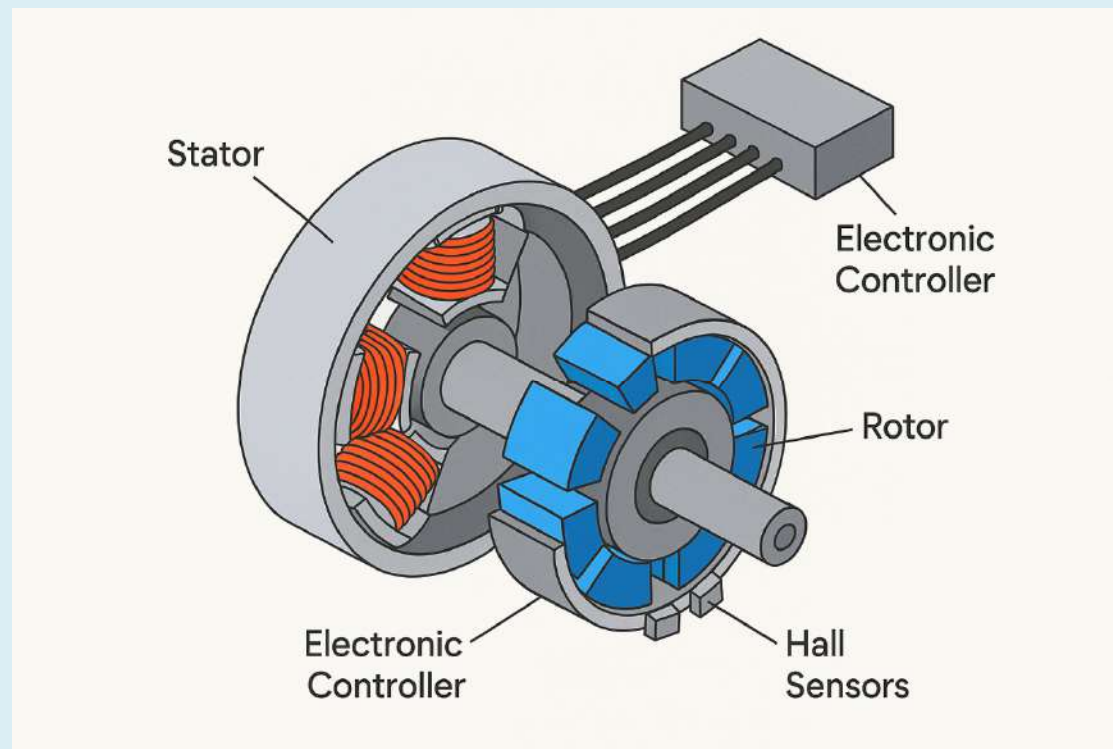
- Operates small air compressors used in garages and workshops.

5. Portable Drills

- Powers handheld electric drills and other small tools.

BLDC motor





**Construction and working of brushless DC motors and
its application:
Electrical Vehicle,
washing machines, Blowers, Computers/Laptops**

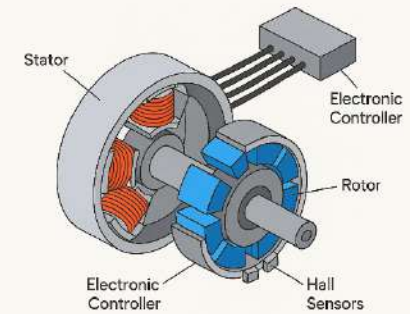
🔧 Brushless DC (BLDC) Motors

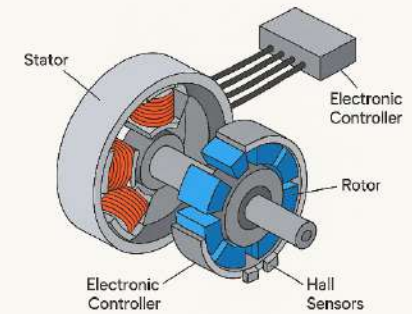
⚙️ 1. Construction of BLDC Motor

Unlike traditional brushed DC motors, **BLDC motors** eliminate brushes and commutators, resulting in higher efficiency and longer life.

Main Components:

- **Stator:**
 - Made of laminated steel and has **three-phase windings**.
 - Connected to an **electronic controller** that provides a rotating magnetic field.
- **Rotor:**
 - Contains **permanent magnets** (typically 2 to 8 poles).
 - Rotates in response to the stator's magnetic field.
- **Electronic Controller:**
 - Replaces the mechanical commutator.
 - Uses sensors (like **Hall-effect sensors**) to detect rotor position and commutate the windings electronically.



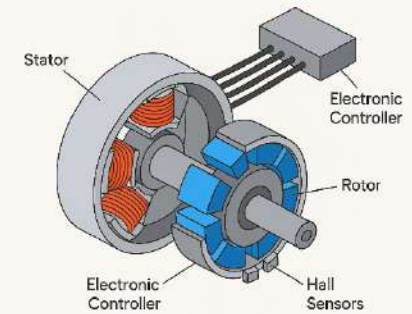


⚡ 2. Working Principle of BLDC Motor

The operation of a BLDC motor is based on the **interaction between the magnetic fields** of the rotor and stator.

Steps:

1. The **electronic controller** energizes the stator coils in a specific sequence.
2. This creates a **rotating magnetic field**.
3. The **permanent magnets on the rotor** follow this rotating field, causing rotation.
4. **Hall sensors** continuously detect the rotor position and help the controller switch windings correctly.
5. This results in **smooth, precise, and efficient rotation**.



✓ 3. Advantages of BLDC Motors

- High efficiency & reliability
- No brushes = low maintenance
- High torque-to-weight ratio
- Long lifespan
- Better speed control

BLDC Motor application: Electrical Vehicle, washing machines, Blowers, Computers/Laptops

4. Applications of BLDC Motors

1. Electric Vehicles (EVs)

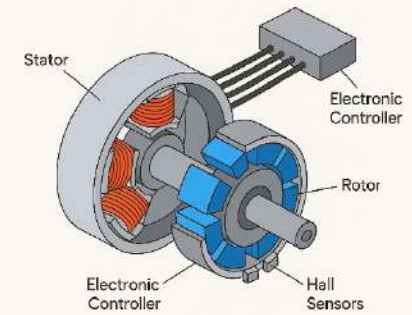
- BLDC motors are widely used in **electric scooters, e-bikes, and electric cars.**
- **Why?**
 - High efficiency = longer battery life
 - Compact and lightweight
 - Excellent speed and torque control

2. Washing Machines

- Used in **direct-drive washing machines.**
- **Why?**
 - Precise control over drum speed
 - Quiet operation
 - Energy-efficient

3. Blowers and Fans

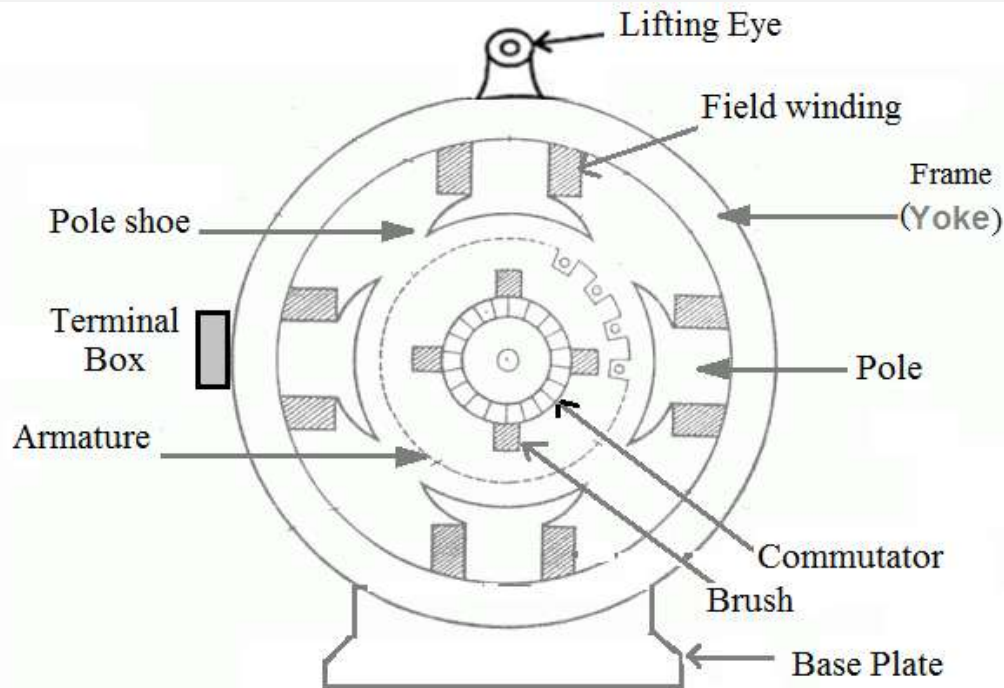
- Used in **HVAC systems, industrial blowers, and cooling fans.**
- **Why?**
 - Smooth speed variation
 - Energy savings
 - Low noise levels



4. Computers/Laptops

- Used in **cooling fans** for CPUs and GPUs.
- **Why?**
 - Small, compact
 - Long operating life
 - Quiet and efficient

Construction/Parts of DC machine



Terminal Box: For connections

Lifting Eye: For lift the motor

Frame:-For Support of stator pole and winding. It made from cast iron material.

Base Plate: For provide base of motor for installation.

Field System (Stator)

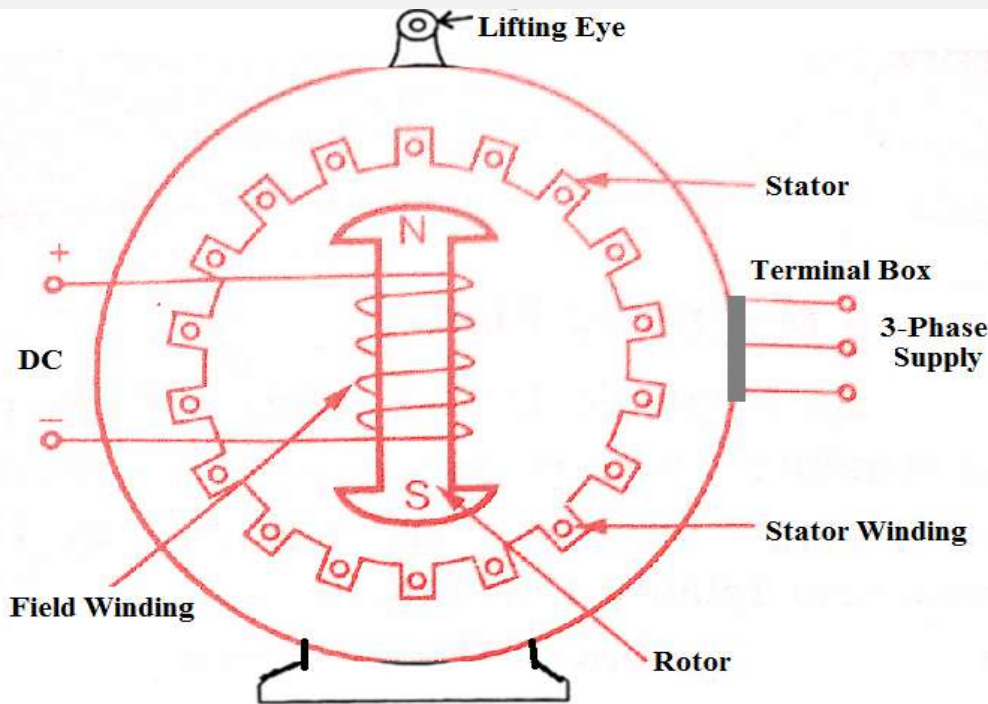
- It is a Stationary Part. It comprises Pole, Pole shoe and field winding.
- Pole is made of magnetic material to produce flux.
- Pole shoe use to spread more flux in air gap.
- Field Winding is surrounded to Pole. Use to magnetizes poles which produce [magnetic flux](#).

Armature (Rotor)

- It is Rotating Part of motor. It includes armature core, winding, brush and commutator
- **Armature Winding:** Lap winding and Wave Winding use as armature winding.
- **Commutator:** the commutator use to produce unidirectional torque.
- **Carbon Brushes:** Use to provide supply to armature.

Shaft: It is rotating element use to drive mechanical machine. It is made up of steel.

Construction/Parts of Synchronous generator



Terminal Box: For connections

Lifting Eye: For lift the machine

Frame:- For Support of stator core and winding.. It made from cast iron material.

Base Plate: For provide base for installation.

Stator

- It is a Stationary Part. It comprises of stator core and armature winding.
- The armature winding is placed in stator slots
- Three phase AC obtained from armature winding of stator by terminal Box.

Rotor

- It is Rotating Part.
- The shape of rotor is salient pole type
- Rotor Poles are excited by a D.C source through two slip rings & Brushes.
- Turbine is connected with rotor shaft to rotate pole.

GTU PAPER'S ANALYSIS

Sr	Question	Mrk	GTU Exam
General			
1	<u>Explain the term rotating magnetic field with proper diagrams in case of a three phase induction motor.</u> OR Discuss how the rotating magnetic field is produced in three phase induction motor. OR Explain Generation of Rotating Magnetic Field in 3-phase Induction Motor with diagrams and equations	07 04 07	Jan-2019 May 2019, Mrch-2023 Jan -2020 Nov -2020
3-Phase Induction Motor			
1	<u>Explain construction of 3-phases induction motor with diagram</u>	04	Mrch-2023
2	<u>Give Merits, Demerits and Applications of Induction Motor.</u>	04	2020,2024
3	<u>Compare poly phase Induction Motor and single phase Induction Motor.</u>	04	Jan-2020
4	<u>Comparison between squirrel cage induction motor and wound rotor induction motor.</u> OR Compare Squirrel cage induction motor and Slip ring Induction Motor.	03	2019, 2024, 2022, 2025
5	<u>Explain in brief working principle of Three Phase Induction Motor.</u>	04	FEB-2024
7	<u>Describe the construction of rotor for a slip ring type three phase induction motor.</u>	03	Jan-2019

Sr	Question	Mrk	GTU Exam
1-Phase Induction Motor			
1	<u>Why single-phase induction motor is not self starting while three phase induction motor is self starting. Explain in brief.</u>	04	2020
2	<u>Mention Merits and Demerits of Single Phase Induction Motor</u>	04	Aug-2022
3	<u>State types of sing phase induction motor, EXPLAIN ANY ONE</u>	03	2023, 2025
4	<u>Explain Working of capacitor start Single phase induction motor</u>	04	2023, 2025
5	<u>Describe the stator construction of a single phase induction motor.</u>	03	Jan-2019

DC Motor			
1	<u>Classify and compare various DC motor.</u>	03	2019,2024,2025
2	<u>Prepare a list of parts of a DC machine. Explain any one part in detail</u> OR <u>List the various parts of DC Motor and explain each</u>	04 07	Jan & May-2019 Mrc-2023
3	<u>Explain working principle of D.C. Motor</u>	03	Jan-2020, July-2020
4	<u>Explain construction of DC Machine.</u>	07	2024, Mr-2022,AUG-2022,2025
5	<u>State significance of the back emf in DC motor</u>	03	Jun-2019

Alternator/Synchronous Generator / Motor			
1	<u>Explain the working principles of a synchronous generator and a synchronous motor.</u>	07	Jan-2019
2	Explain working principle of synchronous motor	03	FEB-2024
3	<u>Explain in detail the construction of an Alternator</u> OR Explain construction of synchronous generator with diagram.	07	May-2019 Jan-2020, 2022, 2024 , 2025
5	<u>Explain the principle of operation of an Alternator.</u>	03	March-2022

THE END

S.G. Prajapati



BEE

CH-5: Electrical Wiring and Installation



Dr. S. G. Prajapati

Thursday, November 13, 2025

Contents



Basic Overview



FUSE, MCB, ELCB



lamps, fixtures & reflectors,



EARTHING



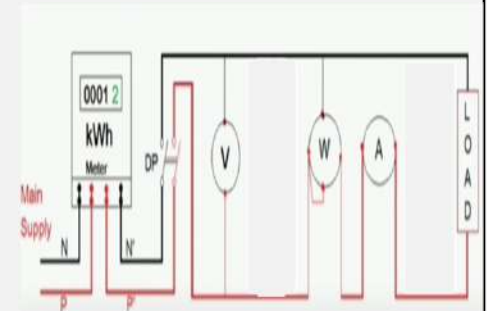
SAFETY PRECAUTIONS



BATTERIES

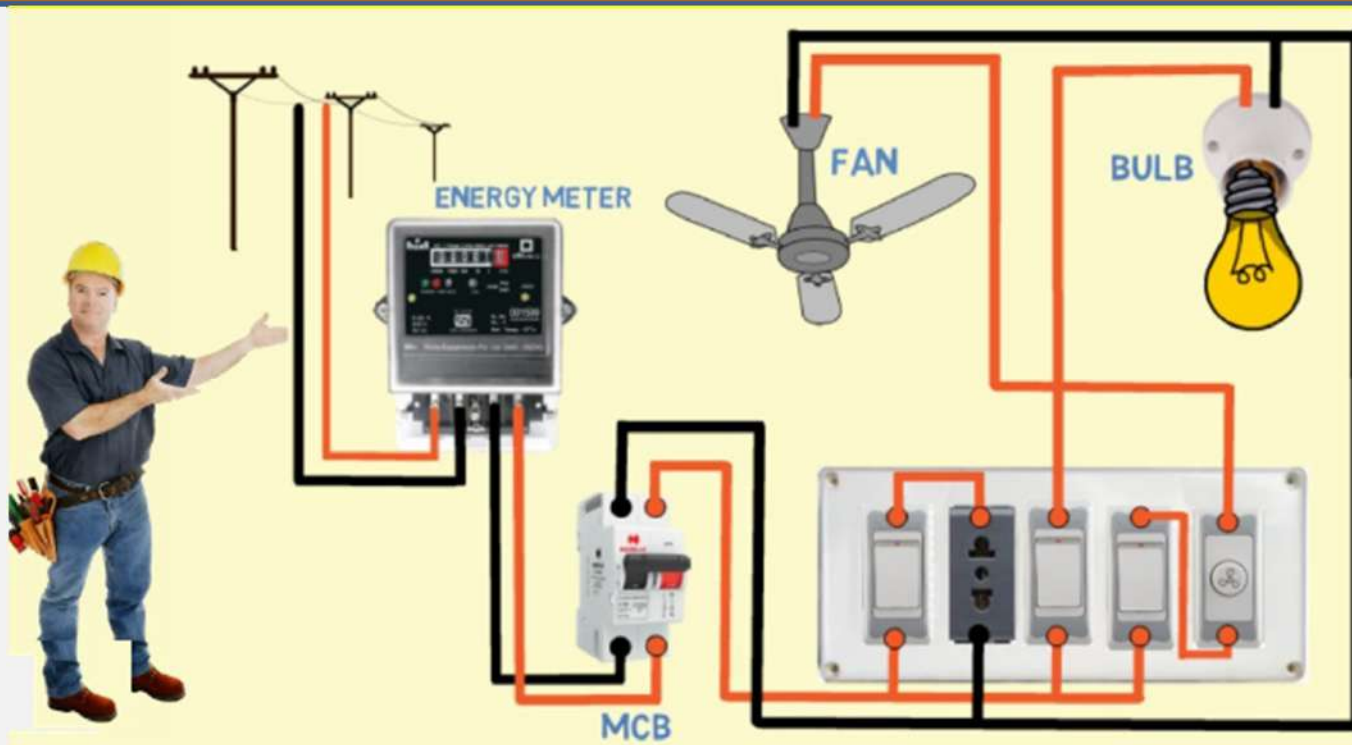


ENERGY CALULATION



Basic electrical measurements

Basic about electrical installations



Electrical installation means **the wiring and devices** used to supply electricity safely in homes, offices, or factories.

It includes **wires, switches, sockets, distribution boards, earthing, lighting, and protective devices.**

Give a list of safety devices used for home appliances.- 03 MARKS



FUSE



MCB



ELCB



EARTHING

Sr.	Question	Marks	Year
1	➤ Discuss the difference between MCB and Fuse ➤ Give the comparison of fuse and MCB	03 04	2023 2019,2018,2024

MCB



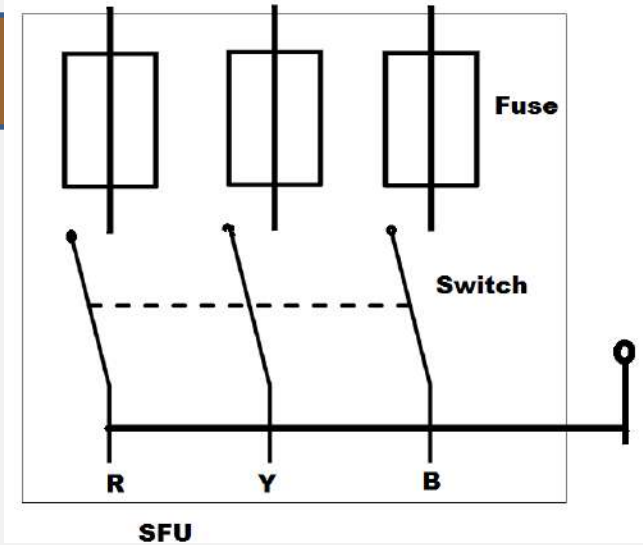
FUSE



1. It Automatic trip off circuit at the time of fault.
2. MCB can be manual ON-OFF
3. Construction is complex
4. Costly
5. Gives effective protection

1. Fuse wire melt to off circuit at the time of over current pass through it.
2. Fuse need to replace wire with new one.
3. Simple Construction.
4. Lower cost
5. Less effective protection compare to MCB

Explain the protective device Switch Fuse unit in detail



It is a **manually operated switch** that has **fuses** connected in series with the SWITCH to protect electrical equipment from overcurrent or short-circuit faults.

Here are some of the features of a switch fuse unit:

Protection

Protects against short circuits and overloads by interrupting the power supply when a fault is detected

Isolation

Allows a section of an electrical system to be safely isolated for maintenance or repairs

Manual control

Provides a way to manually control the power supply to specific areas or equipment

Ratings

Switch fuse units are available in ratings such as 30, 60, 100, 200, 400, 600, and 800 amperes

What is MCB? Where is it used?



MCB

MCB is miniature Circuit Breaker Use to protect circuit from.

- ✓ Overload and
- ✓ Short circuit

It automatic Break circuit at the time of fault.

MCB is commonly used in low voltage electrical for over current protection.

MCB are generally used for operating currents up to 63 Amps..

Question	Marks	Year
What is MCCB? Where is it used?	03	2019, 2024



MCCB

MCCB is Module case Circuit Breaker Use to protect circuit from.

- ✓ Overload and
- ✓ Short circuit

It automatic Break circuit at the time of fault.

It is similar to an MCB but designed for **higher current ratings and industrial use.**

MCCB are generally used for operating currents between 63 Amps to 800 Amps.

Commonly used in:

- **industrial and commercial electrical panels.**
- **Factories and large buildings**
- **Main distribution boards**
- **Generators and heavy motors**

What is ELCB? Where is it used?



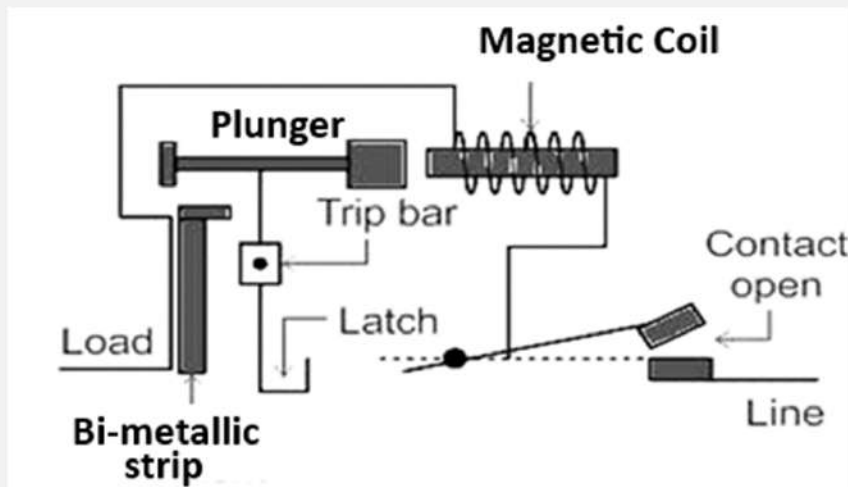
ELCB

- ✓ ELCB is Earth Leakage Circuit Breaker Use to protect circuit from Earth leakage.
- ✓ It automatic Break circuit at the time of earth leakage fault.
- ✓ ELCB is commonly used in LV & HV voltage electrical for EARTH LEAKAGE protection.



Fuse	MCB	ELCB
1. Fuse wire melts and breaks the circuit 2. Protects from overload and short circuit 3. Fuse need to replace wire with new one. 4. Simple Construction. 5. Lower cost 6. Less effective protection compare to MCB	1. It Automatic trip off circuit at the time of fault. 2. Protects from overload and short circuit 3. MCB can be manual ON-OFF 4. Construction is complex 5. Costly 6. Gives effective protection	1. It Automatic trip off circuit at the time of Leakage current. 2. Protects from earth leakage and electric shock 3. ELCB can be manual ON-OFF 4. Construction is complex 5. High Costly 6. Gives effective protection

	Question	Marks	Year
4	Write a short note on Miniature Circuit Breaker (MCB)	04	2019

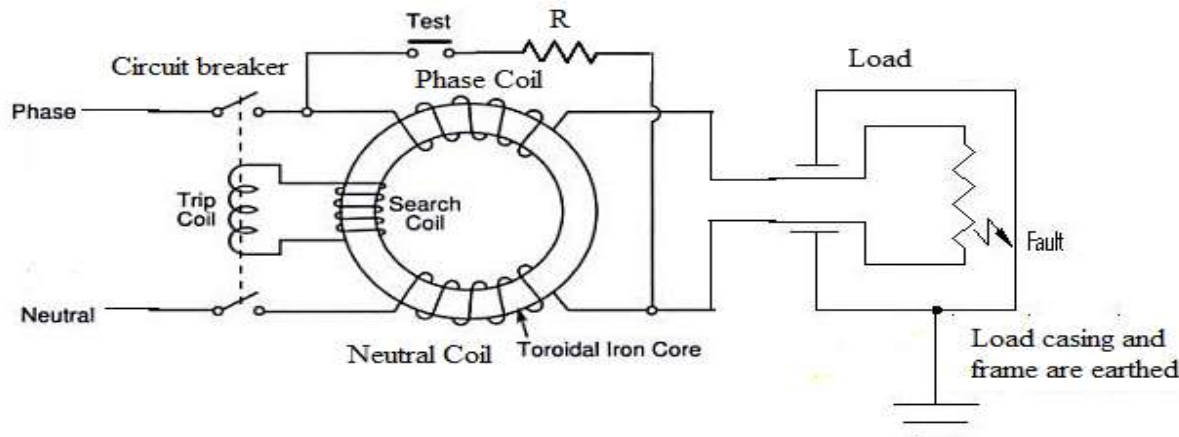


The Miniature Circuit Breaker (MCB) Use to protect circuit from.

- ✓ Overload and
- ✓ Short circuit

- In normal condition, the current pass through MCB is under limit, Hence, MCB work as a normal switch.
- At overload, current passing through bi-metallic strip increases, Hence bi-metal strip heat up, it will bend towards the trip bar and disconnects the contacts.
- At short circuit, Current increase very high and very fast→ so A solenoid coil produces a strong magnetic field, so plunger attracts towards the trip bar and disconnects the contacts.

Question	Marks	Year
➤ Explain Working of Earth Leakage Circuit Breaker (ELCB)	07	2023
➤ Write a short note on Earth Leakage Circuit Breaker (ELCB)	04	2019 2018



It is a device that **protects humans from electric shocks** caused by **leakage currents** to the earth.

- **Circuit description:** ELCB has phase coil, neutral coil and the search coil; all the coils wound on a common transformer core.
- **At normal conditions**, same current pass through phase and neutral coil, so the total sum of the opposing magnetic flux on core is zero no emf induce in search coil trip circuit not operate and Contact remain close.
- **At Earth leakage condition**, current passing through neutral coil will reduced, So the total sum of the opposing magnetic flux on core is now present, so emf induce in Search coil, it gives signal to trip circuit and disconnects the contacts.

Question	Marks	Year
Explain the following protective devices in detail: (i) SFU (ii) MCB (iii) ELCB	07	2020

(i) SFU – Switch Fuse Unit

It is a **manually operated switch** that has **fuses** connected in series with the SWITCH to protect electrical equipment from overcurrent or short-circuit faults.

Applications:

- Used in **distribution boards** for domestic and industrial circuits.
 - Commonly used up to **100 A or more**.
- Protects **motors, lighting circuits, and cables**.

(ii) MCB – Miniature Circuit Breaker

MCB automatically switches OFF the electrical circuit during **overload** or **short-circuit** conditions. Magnetic coil & bi-metallic strip are main part of MCB.

Applications:

- Used in **domestic wiring, distribution boards, and industrial panels**.
- Commonly rated from **6 A to 125 A**.

(iii) ELCB – Earth Leakage Circuit Breaker

It is a device that **protects humans from electric shocks** caused by **leakage currents** to the earth.

Applications:

- Installed in **residential and commercial** electrical systems.
- Used to **protect people from electric shock** and prevent **fire hazards** caused by current leakage.

Question	Marks	Year
Write advantages and disadvantages of ELCB.	03	2025

✓ Advantages of ELCB

1. Protects against **electric shock** due to earth leakage.
2. Prevents **fire hazards** caused by current leakage.
3. **Highly sensitive** and reacts quickly (in milliseconds).
4. **Automatically disconnects** the power during faults.

✗ Disadvantages of ELCB

1. **Cannot protect** against overload or short circuit.
2. May **trip unnecessarily** due to small leakage or wiring faults.
3. Needs **proper earthing** to work correctly.
4. More **expensive** than normal fuses or MCBs.



Question	Marks	Year
Compare MCB and ELCB	03	2022, 20

MCB	ELCB
	
1. It Automatic trip off circuit on short circuit and overload event. 2. MCB can be manual ON-OFF 3. No Test Button is given 4. Use for protection of Electrical devices 5. Lower Cost	1. It Automatic trip off circuit on earth leakage event. 2. ELCB can be manual ON-OFF 3. Test Button is given to test ELCB 4. Use for personal safety 5. Higher Cost

Question	Marks	Year
List the various safety devices used for domestic purpose	03	2024

Device	Function	Protection Against
Fuse	Melts to break circuit	Overload, short-circuit
MCB	Trips automatically	Overload, short-circuit
ELCB/RCCB	Detects leakage current	Electric shock
SFU	Manual switch with fuse	Overload, short-circuit
Earthing	Provides low-resistance path to ground	Electric shock
Main Switch	Disconnects power manually	ISOLATION

Describe the necessity of using electrical protective devices.	04	2025
--	----	------

Main Reasons (Necessity):

1. Protection from Electric Shock:

Devices like earthing and ELCB prevent electric shock to humans.

2. Protection of Equipment:

Fuses and circuit breakers protect electrical appliances from overload or short circuit.

3. Fire Prevention:

Protective devices stop excessive current that could heat wires and cause fires.

4. System Safety and Stability:

They help maintain safe voltage and current levels in the circuit.

5. Automatic Disconnection:

They cut off power automatically during faults, preventing further damage.

6. Prevention of Power Loss:

By quickly isolating faulty parts, they keep the rest of the system running safely.

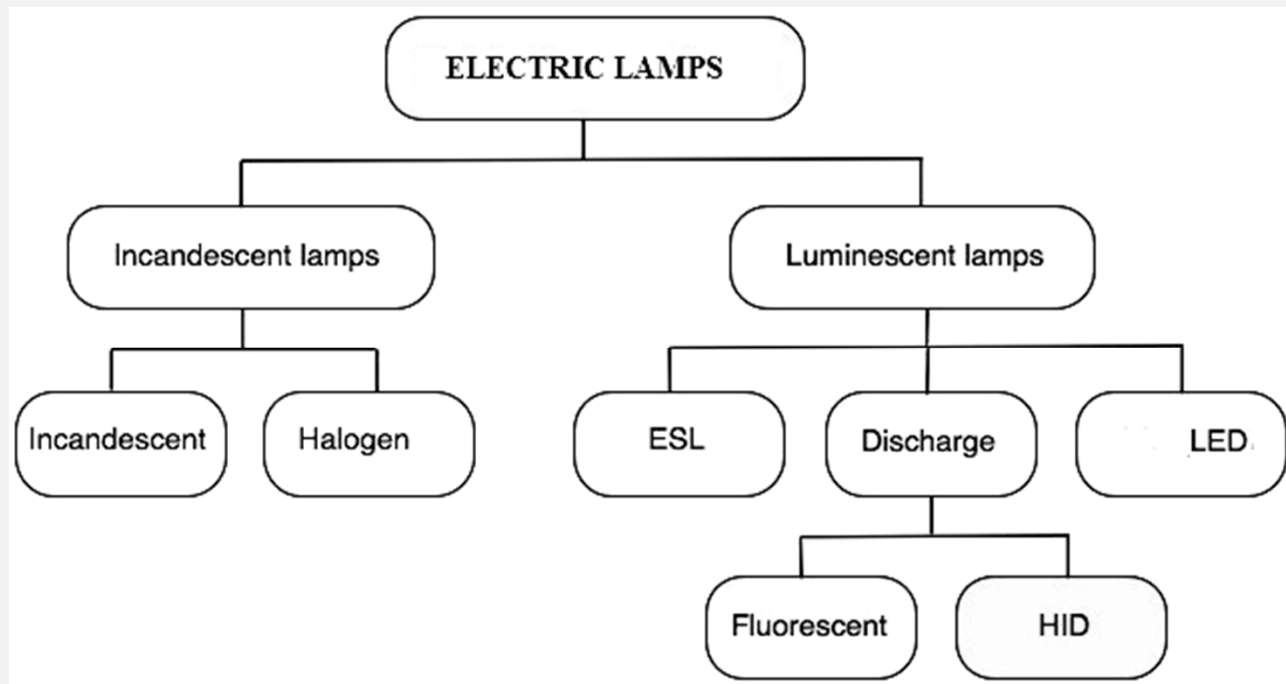
GTU PAPERS QUESTIONS-MCB & ELCB

Sr.	Question	Marks	Year
1	➤ Discuss the difference between MCB and Fuse ➤ Give the comparison of fuse and MCB	03 04	2023 2019,2018,2024
2	Compare MCB and ELCB	03	2022, 20
3	Explain the following protective devices in detail: (i) SFU (ii) MCB (iii) ELCB	07	2020
4	Write a short note on Miniature Circuit Breaker (MCB)	04	2019
5	What is MCCB? Where is it used?	03	2019, 2024
6	Write the comparison of Fuse, MCB and ELCB.	04	2025
7	List the various safety devices used for domestic purpose OR Give a list of safety devices used for home appliances	03 03	2024, 2025
8	➤ Explain Working of Earth Leakage Circuit Breaker (ELCB) ➤ Write a short note on Earth Leakage Circuit Breaker (ELCB)	07 04	2023 2019 2018
9	Write advantages and disadvantages of ELCB.	03	2025
10	Describe the necessity of using electrical protective devices.	04	2025

2. Types of lamps, fixtures & reflectors, illumination schemes for domestic, industrial & commercial premises, Lumen requirements for different categories

Types of lamps

1	Incandescent Lamps • Traditional filament-based lamps • Low efficiency, short lifespan • Used in homes, decorative lighting	
2	Fluorescent Lamps • Tube Lights– Energy-efficient, long lifespan • Compact Fluorescent Lamps (CFLs) – Used in homes, offices	
3	LED Lamps • Highly energy-efficient and long-lasting • Available in various wattages and color temperatures • Used in residential, commercial, and industrial lighting	
4	Halogen Lamps • Bright white light, slightly more efficient than incandescent • Used in spotlights, projectors, and stage lighting	
5	High-Intensity Discharge (HID) Lamps • Includes Metal Halide (MH), High-Pressure Sodium (HPS), and Mercury Vapor Lamps • Used in industrial areas, stadiums	
6	Smart Lamps • LED-based lamps with IoT control • Used in smart homes, offices	



**fixtures &
reflectors**

a) Ceiling Fixtures		
Recessed Lighting – Recessed lights, also known as downlights are a type of lighting fixture installed into the ceiling or wall.		
Surface-Mounted – Fixed on the ceiling, used in warehouses, homes		
b) Wall-Mounted Fixtures		
Sconces – Decorative, used in hotels, corridors		
Bulkhead Lights – Industrial, outdoor lighting		
c) Hanging Fixtures		
Pendant Lights – Used in dining areas, offices		
Chandeliers – Decorative, used in hotels, reception areas		
d) Track & Spot Lighting Fixtures		
Used in museums, showrooms, commercial stores		

reflectors control the **focus** of light.

e) Reflectors

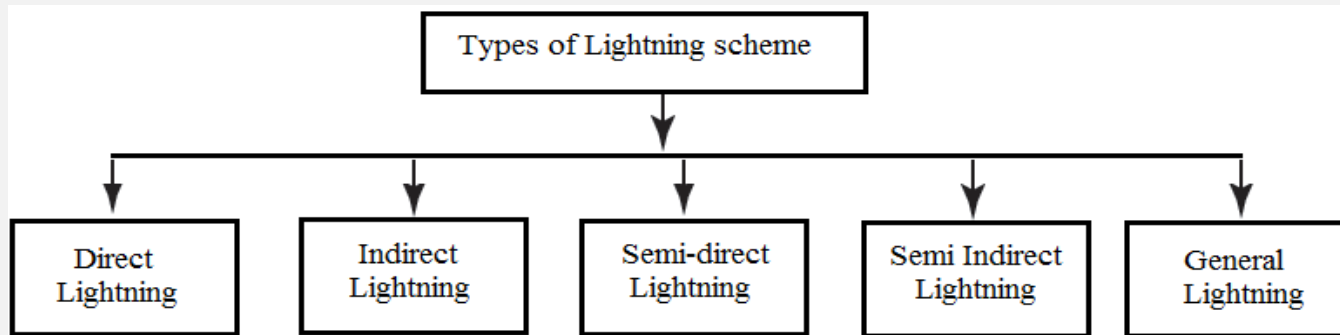
- **Parabolic Reflectors (PAR)** – Used in stadiums, floodlights



- **Elliptical Reflectors** – Stage lighting, spotlights

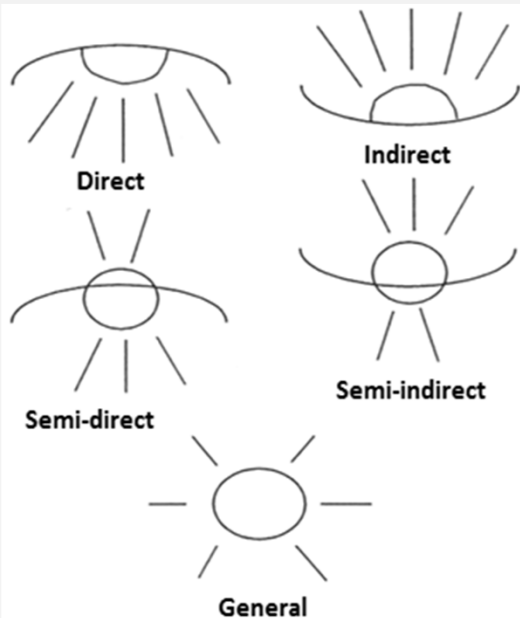


Illumination Schemes for Different Premises



(1) Direct lighting:

In direct lighting schemes, 100% of light has fallen on the area to be lighted required, not on the ceiling and walls, example:- Table lamp.



(2) Indirect lighting:

In indirect lighting schemes, first light is falling on the ceiling and it reflected in the area to be lighted required.

(3) Semi directs lightning:

In semi direct lightning schemes, 80% of light has fallen on the area to be lighted required while 20% of light is falling on the ceiling and walls.

(4) Semi indirect lighting:

In semi indirect lightning schemes, 80% of light is falling on up-Wards and 20% of light is falling on downwards.

(5) General lightning:

In general lightning schemes all the light is equally distributed upwards and downwards.

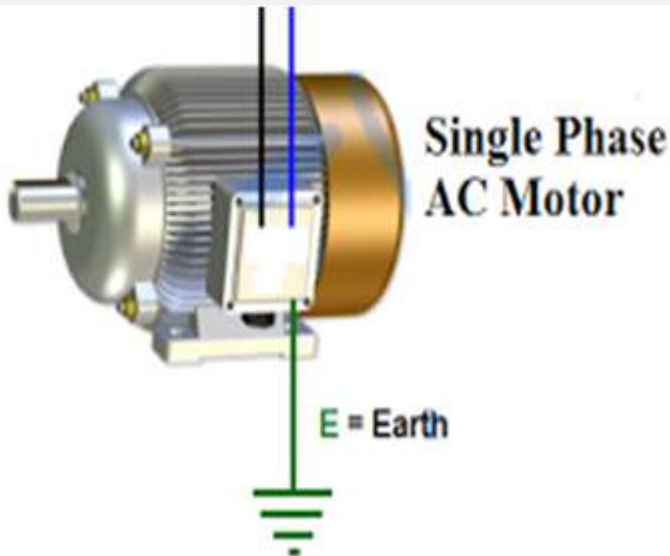
Lumen Requirements for Different Categories

Lumen (lm) measures the total light output. Different spaces require specific levels: Lumen is measured by lux meter,

Sr. No.	Location	Preferred Lumens/m ²
1	Entrance	150
2	Hall	300
3	Bed room	200
4	Kitchen	150
5	Dining Room	125
6	Bathroom	75
7	Study Room	300
8	Stairs	150
9	Verandah	100
10	Gallery	150

3. EARTHING

About Earthing



To connect the metallic (conductive) Parts of an Electric appliance or installations to the earth (ground) is called **Earthing** or **Grounding**.

- ✓ Earth is a conductor that carries no current in normal operation.
- ✓ Earth or ground wire is a safety protective wire for the human body and electrical equipments.
- ✓ Earth is a conductor that connect with equipment body and it grounded which carries current under faulty condition.

Earthing – Types of Earthing and its importance



Need (Importance) of Earthing

1. Safety of Human Beings:

Protects people from electric shock in case of insulation failure or leakage.

2. Protection of Equipment:

Prevents damage to electrical appliances

3. Voltage Stabilization:

Maintains the voltage of all parts of the electrical system at a safe level.

4. Path for Fault Current:

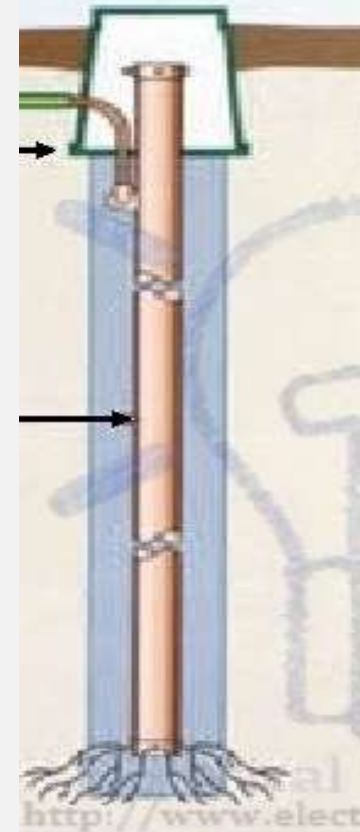
Provides a low-resistance path for fault current to flow safely into the ground.

5. Protection Against Lightning:

Carries lightning discharge safely to the earth, avoiding fire or damage.

Types/Methods of grounding

(1) Plate earthing and (2) Pipe earthing



(1) Plate earthing

CONVENTIONAL COPPER PLATE EARTHING

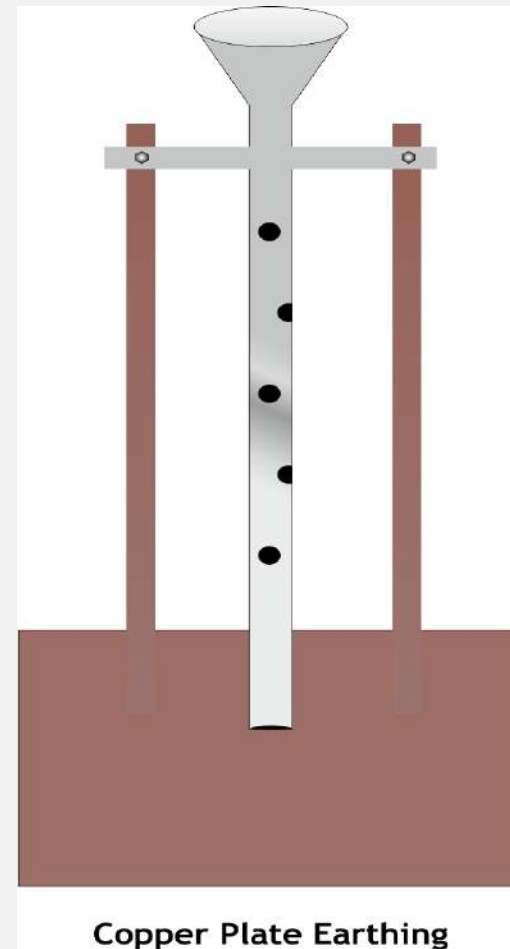
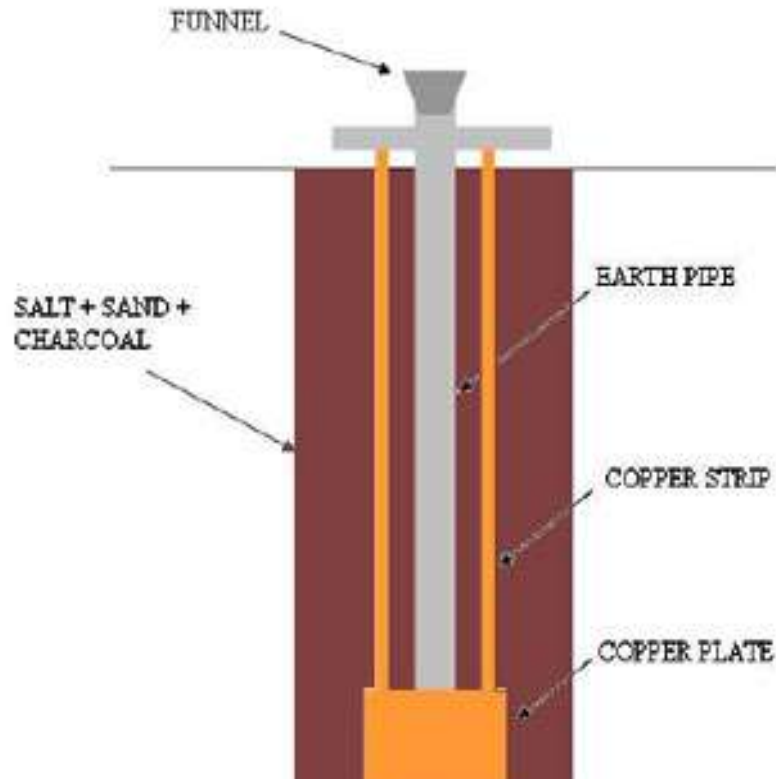


Plate Earthing

- Following are the plate dimensions commonly used:

For Copper plate: 60cm × 60cm × 3.18mm and

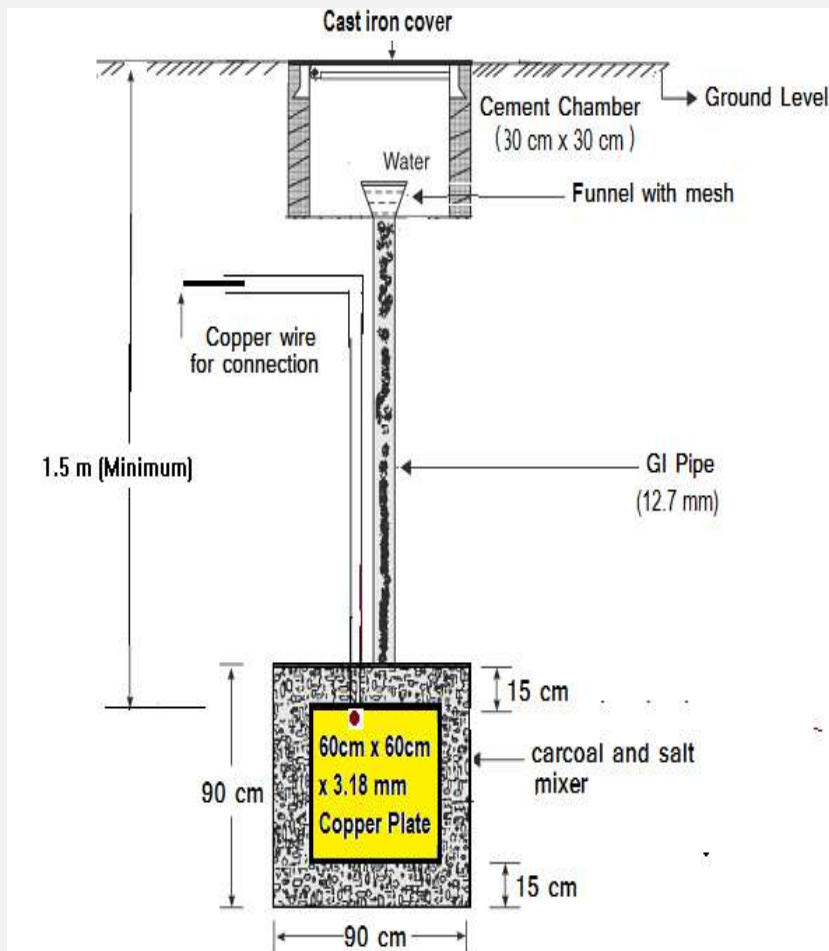
For G.I plate: 60cm × 60cm × 6.35mm

Classify different types Earthing and explain any one in detail	07 04 07	2023 2022 2019, 2025
Explain plate earthing with diagram		

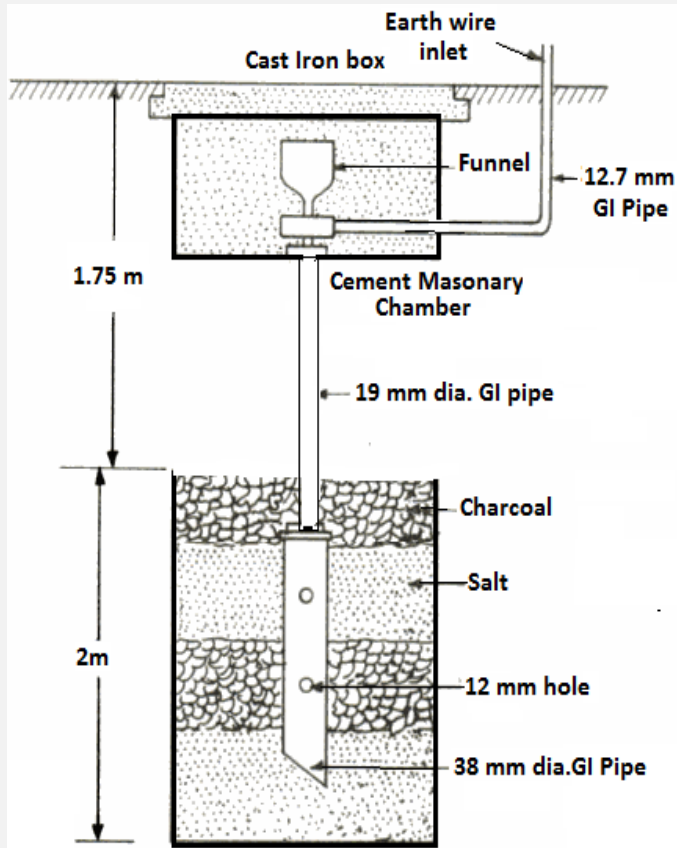
There are mainly **three types of earthing** used in electrical systems:

1. Pipe Earthing
2. Plate Earthing
3. Rod (or Strip) Earthing

Plate Earthing



- First, the selected plate is placed vertically inside the ground at a depth of 1.5 m (minimum) and is embedded in alternate layers of coal and salt to a thickness of 15 cm.
- The earth wire is drawn through G.I pipe and is perfectly bolted to the earth plate.
- One thing you must keep in your mind that; the nuts and bolts must be copper for copper plate and must be galvanized iron for G.I plate.
- The G.I pipe is fitted with a funnel on the top of the ground; from that pipe, water is poured to maintain plate area at wet condition to keep low resistance.
- A cement masonry chamber is built with a cast iron cover for easy regular maintenance.



Pipe earthing is a **common and effective method** of earthing (grounding) used in homes and industries to safely discharge **fault currents** into the ground.

1. First, the pipe is placed upright at a depth of 1.75 m in a permanently wet ground and the area (15 cms) surrounding the GI pipe is filled with an alternate layers of charcoal and salt to increase the effective area of the earth and to reduce the earth resistance respectively.
2. **The efficiency of the earthing system is improved by pouring water through the funnel periodically.**
3. The GI earth wires of sufficient cross-sectional area are run through a 12.7 mm diameter pipe and from the 19 mm diameter pipe it secured tightly by screw with 38 mm diameter GI pipe.
4. **A cement masonry chamber is built with a cast iron cover for easy regular maintenance.**
5. Compared to the plate earth system the pipe earth system can carry larger leakage currents as a much larger surface area is in contact with the soil for a given electrode size.

GTU PAPERS QUESTIONS-Earthing

Sr.	Question	Marks	Year
1	Briefly describe pipe earthing	07	2023
2	Classify different types Earthing and explain any one in detail Explain plate earthing with diagram	07 04 07	2023 2022 2019, 2025
3	Explain different types earthing and its importance in electrical utility system in detail.	07	2020
	What is need of earthing ? Explain the different method of earthing.	04	2025

4. SAFETY PRECAUTIONS

Write safety precautions for Electrical Appliances	04	2022, 2020
State safety precautions of electrical appliances	04	2025

1. Always **switch off** the appliance when not in use.
2. **Do not touch** electrical items with **wet hands**.
3. Use **good quality wires** and plugs.
4. **Do not overload** sockets with many appliances.
5. Make sure the appliance is **properly earthed**.
6. Keep electrical items **away from water**.
7. Use **fuses or circuit breakers** for protection.
8. Check for **damaged wires** and replace them.
9. **Follow instructions** given by the manufacturer.
10. Use **insulated tools** when repairing.

GTU PAPERS ANALYSIS

Sr. No	Question	Marks	GTU Exam Y.
Safety			
1	Write safety precautions for Electrical Appliances	04	2022, 2020
	State safety precautions of electrical appliances	04	2025

5. BATTERIES

Types of Batteries

1. Primary Batteries (Not Chargeable)

EX.

Alkaline and zinc carbon batteries



2. Secondary Batteries (Rechargeable)

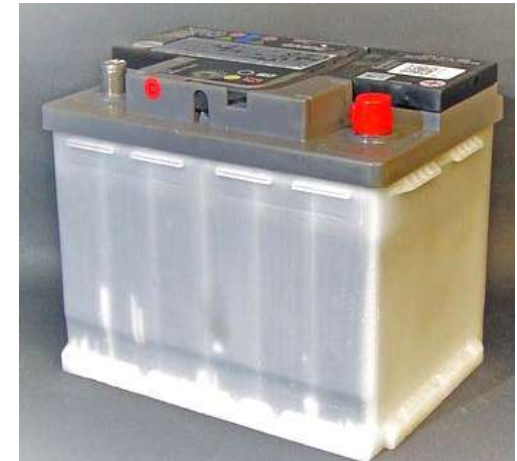
EX.

Lithium-ion(Li-ion)

Nickel Cadmium(Ni-Cd)

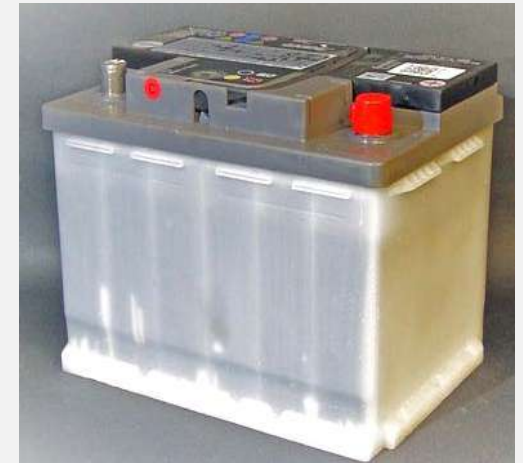
Nickel-Metal Hydride(Ni-MH)

Lead-Acid



2. Secondary Batteries

1. Lithium-ion(Li-ion)
2. Nickel Cadmium(Ni-Cd)
3. Nickel-Metal Hydride(Ni-MH)
4. Lead-Acid



Various Batteries & its Applications

Alkaline Button/Coin Cell Batteries (1.5 Volts) Silver oxide		Use in Computers, Calculators, watches and cameras
Lithium ion (Li-ion)		Mobile phones, watches and cameras
Lead acid batteries		for vehicles, including cars, trucks, boats, tractors and motorcycles, emergency lighting and uninterruptible power supplies
Zinc carbon		Torches, clocks, shavers, radios, toys and smoke alarms

Types of Batteries

Alkaline



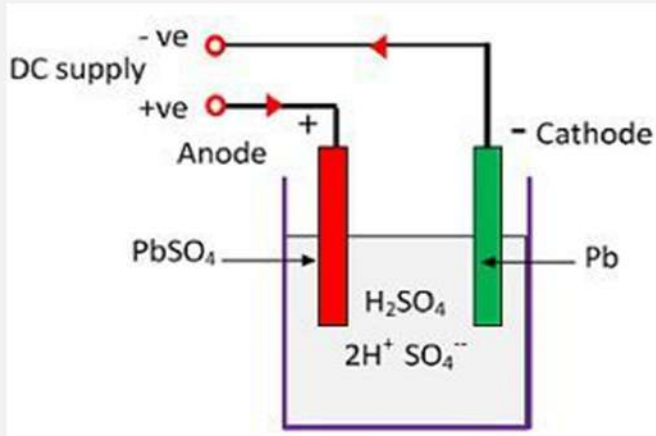
Wireless Mike

Nickel cadmium (NiCd)



Cordless power tools,
laptop computers,
shavers, motorized toy
's, and personal
stereos

Explain the construction of the lead-acid battery with neat diagram	04	2020
What is battery? Explain the construction and working of any battery.	04	2025



Battery is a source of DC Electricity, that produces electricity by chemical reaction. It consists of the following main parts:

1. Container: Made of hard rubber or plastic.

2. Plates (Electrodes):

Positive Plate: Made of **lead dioxide (PbO_2)**.

Negative Plate: Made of **lead (Pb)**.

3. Electrolyte:

A mixture of **sulphuric acid (H_2SO_4)** and water. It allows ions to move between plates.

4. Separators:

Thin sheets of insulating material between the plates. Prevents short circuiting but allows ion flow.

5. Terminals:

Positive (+) and **Negative (-)** terminal for external connection.

The lead-acid battery works on **electrochemical reactions** between **lead (Pb)**, **lead dioxide (PbO_2)**, and **sulphuric acid (H_2SO_4)**.

During Discharging (Battery in Use)

- The battery **supplies current** to an external circuit.
- **Chemical energy → Electrical energy**

During Charging (Connected to DC Supply)

- The battery is **recharged** by passing **electric current** from an external DC source.
- **Electrical energy → Chemical energy**

Question	Marks	Year
State and explain in brief important electrical characteristics of battery.	04	2019
OR State important characteristics of batteries.	04	2025

Here are some **important characteristics of a battery** explained in an **easy and simple way** 🖐

1. 🔋 **Voltage (V):**

1. The electric pressure a battery provides.
2. Example: A AA cell has **1.5 volts**.

2. 🔋 **Capacity (Ah or mAh):**

1. How much charge the battery can store.
2. Tells how long it can power a device.
3. Example: 2000 mAh means it can supply 2000 mA for 1 hour.

3. ⚡ **Energy Density:**

1. How much energy it stores per unit weight or volume.
2. Higher energy density = longer usage time for the same size battery.

4. 🔁 **Cycle Life:**

1. Number of times a battery can be charged and discharged before it weakens.
2. Example: A Li-ion battery may last **500–1000 cycles**.

5. 🌡 **Temperature Range:**

1. The range of temperatures where the battery works properly.
2. Too hot or cold can reduce performance or damage it.

6. ⌚ **Shelf Life:**

1. How long a battery can be stored without losing much charge.
2. Important for backup or emergency use.

7. ⚙ **Efficiency:**

1. How much of the stored energy can be used effectively.
2. Higher efficiency = less energy wasted as heat.

8. 🔧 **Internal Resistance:**

1. Lower resistance means better performance and less heat during use.

GTU PAPERS QUESTIONS

Sr.	Question	Marks	Year
1	State and explain in brief important electrical characteristics of battery.	04	2019
	OR State important characteristics of batteries.	04	2025
2	Explain the construction of the lead-acid battery with neat diagram	04	2020
3	What is battery? Explain the construction and working of any battery.	04	2025

6. ENERGY CALULATION

6. Compute the monthly energy charges for an air conditioner having consumption of 2 kW. Daily usage of the air conditioner is 10 hours. Energy charges are Rs 8 per unit.

04 MARKS

$$\text{Daily Unit} = 2 \times 10 = 20 \text{ KWh}$$

$$\begin{aligned} \text{Monthly Unit Consumption} &= 20 \times 30 (\text{Day of month}) \\ &= 600 \text{ Unit} \end{aligned}$$

$$\text{Monthly Energy Charges} = 600 \times 8 = \text{Rs.4800}$$

6. Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts . **07 MARKS**

Appliances	Rating in W	Qty.	Daily Use (Hr.)	Unit Consume (KWh)/Day	Unit Consume / Month (KWH)	Cost (KWh x 3.50)
Bulb	40	4	5 h	$800/1000=0.8$	$0.8 \times 30=24$	84
Tube light	60	4	5 h	$1200/1000=1.2$	36	126
TV	100	1	6 h	$600/1000 = 0.6$	18	63
Washing M/c	400	1	3 h	$1200/1000 = 1.2$	36	126
Water Pump	373	1	15 mint = $\frac{1}{4}$ h	$93.25/1000=0.09$	2.7	9.45

Rs. 408.45

Calculate the energy charge for a month of march, if 3 bulbs of 20 W for 5 h, 2 tube lights of 40 W for 5 h, a TV of 120 W for 9 h, a washing machine of 400 W for 3 h, a water pump of 1 HP for 30 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts .

Appliances	Rating in W	Qty.	Daily Use (Hr.)	Unit Consume (KWh)/Day	Unit Consume / Month (KWH)	Cost (KWh x 3.50)
Bulb	20	3	5 h	$300/1000=0.3$	$0.3 \times 31=9.3$	32.55
Tube light	40	2	5 h	$400/1000=0.4$	12.4	43.4
TV	120	1	9 h	$1080/1000 = 1.08$	33.48	117.18
Washing M/c	400	1	3 h	$1200/1000 = 1.2$	37.2	130.2
Water Pump	746	1	30 mint =1/2 h	$373/1000=0.373$	11.56	40.46

Rs. 363.79

A home utilize 230 unit for two month, calculate it energy charge as per below slab. For first 50 unit 3.05, second 50 unit 3.50 and third 100 unit is 4.15 Rupees per month.



1. Monthly Unit consumption $230/2=115$

$50 \times 3.05=152.5$

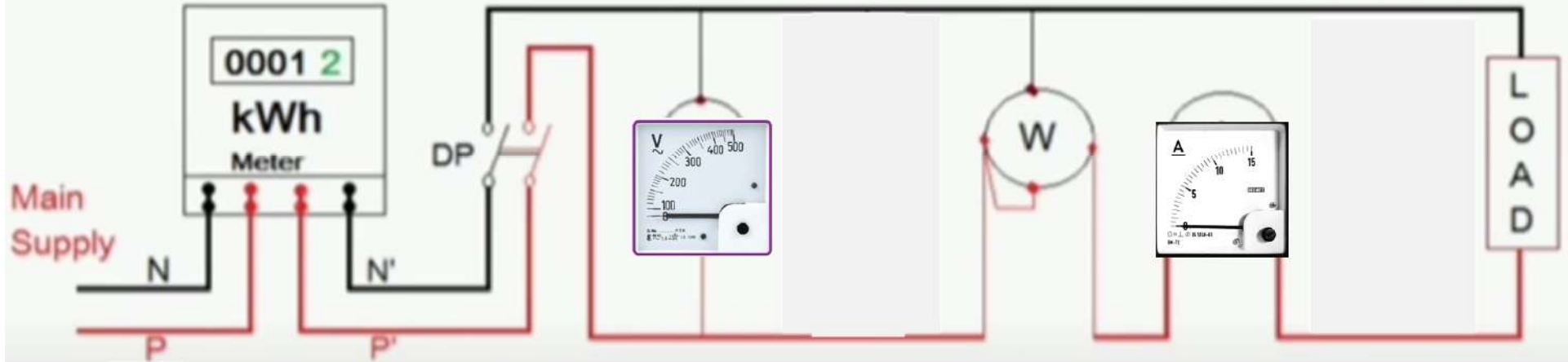
$50 \times 3.50=175$

$15 \times 4.15=62.25$

Total: 389.75

2. Energy charge for two month is $389.75 \times 2 = \text{Rs.}779.5$

Basic electrical measurements with Ammeter, Voltmeter, Wattmeter and Energy meter



Ammeter → Measures current

Voltmeter → Measures voltage

Wattmeter → Measures power

Energy Meter → Measures total energy used

GTU PAPERS ANALYSIS

Sr. No	Question	Marks	GTU Exam Y.
Example			
1	Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watt	07	2019
2	Compute the monthly energy charges for an air conditioner having Power consumption of 3 kW and daily uses 10 hours. Energy charges Rs. 5 per unit	04	2023 2019

THE END

S.G. Prajapati

